Lesson 16

PRODUCER BEHAVIOR: COST ANALYSIS

Economists argue that sunk cost should not be included in a rational person's decision-making process while opportunity cost should be included.

VARIABLE COST (VC)

Costs, which vary with the level of activity (or output), are called variable costs. Variable cost is a cost of labor, material or overhead that changes according to the change in the volume of production units. Combined with fixed costs, variable costs make up the total cost of production. While the total variable cost changes with increased production, the total fixed cost stays the same.

Fixed Cost (FC)

Costs, which do not vary with the level of activity or output, are called fixed costs. In long run, there are no fixed costs. Fixed cost does not vary depending on production or sales levels, such as rent, property tax, insurance, or interest expense.

Total Cost (TC)

Total cost (TC) is the sum of all fixed and variable costs. It plot as a vertical summation of the horizontal line total fixed cost (TFC) curve and the upward sloping total variable cost (TVC) curve.

$$TC = FC + VC$$

Average Cost or Average total cost (AC or ATC)

Total cost per unit of output, found by dividing total cost by the quantity of output. Average total cost, usually abbreviated ATC, can be found in two ways. Because average total cost is total cost per unit of output, it can be found by dividing total cost by the quantity of output. Alternatively, because total cost is the sum of total variable cost and total fixed cost, average total cost can be derived by summing average variable cost and average fixed cost. Average cost (AC) is the vertical summation of the AFC & AVC. Average variable cost plus average fixed cost equals average total cost.

$$\mathbf{AVC + AFC = ATC \text{ or } AC}$$

Average variable cost (AVC)

AVC is an economics term to describe the total cost a firm can vary (labor, etc.) divided by the total units of output.

$$\mathbf{AVC = TVC/Q}$$

Average fixed cost (AFC)

AFC is total; fixed cost divided by the total units of output.

$$\mathbf{AFC = TFC/Q}$$

AC = AFC + AVC, where average fixed cost (AFC) is a downward sloping line as you are dividing a fixed number by an increasing number of output units. By contrast, average variable cost (AVC) first falls as output increases and then rises.

Study of AC is necessary for firms to be able to set the price or (average revenue) at which they will sell. Also they will be interested in knowing how AC is broken down into AFC & AVC.

MARGINAL COST (MC)

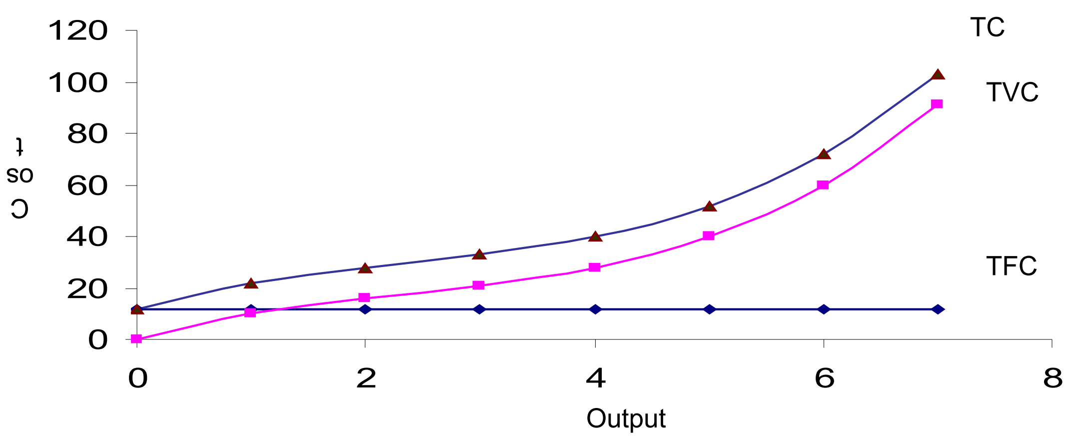
The change in total cost (or total variable cost) resulting from a change in the quantity of output produced by a firm in the short run. Marginal cost indicates how much total cost changes for a give change in the quantity of output. Because changes in total cost are matched by changes in total variable cost in the short run (remember total fixed cost is fixed), marginal cost is the change in either total cost or total variable cost. Marginal cost, usually abbreviated MC, is found by dividing the change in total cost (or total variable cost) by the change in output.

Marginal cost is the addition to TC caused by a unit increase in output. More generally:

$$\mathbf{MC = \Delta TC/\Delta Q}$$

The secret of the shape of the MC curve lies in the law of diminishing marginal returns. The relationship between MC and AC is a reflection of the relationship between MPP & APP. That is: both MC and AC fall in the beginning, then MC starts to rise, cutting AC from below at the latter's turning point (minima).
In the long run, the law of diminishing marginal returns does not apply to the extent that it does in short run.

TC = TVC + TFC			
Output (Q)	**TFC**	**TVC**	**TC**
0	12	0	12
1	12	10	22
2	12	16	28
3	12	21	33
4	12	28	40
5	12	40	52
6	12	60	72
7	12	91	103



RELATIONSHIP BETWEEN AC AND AVC
Initially, AC falls more rapidly than AVC because AC is a summation of AFC & AVC and since both are falling the effect of two falling curves is greater than the effect of one falling curve. After the turning point in AVC, both AC and AVC rise but the gap between them narrows because of same reasoning as given above.
There is an **inverse** relationship between costs and productivity, i.e. as productivity rises, costs fall and
vice versa.
The equivalent of constant, increasing and decreasing returns to scale in terms of costs are economies of scale, diseconomies of scale and constant costs (or constant returns to scale).

i. In the case of economies of scale, long run total cost (LRTC) is an upward sloping curve but with falling slope. Note that the slope can never become zero or negative, though.
ii. In diseconomies of scale, LRTC is an upward sloping curve with an increasing slope.
iii. In constant costs, LRTC is a positively sloped straight line.

THE LONG-RUN AVERAGE COST CURVE (LRAC)
The long-run average cost (LRAC) curve for a typical firm is U shaped.

i. As a firm expands, it initially experiences economies of scale (due to productive efficiency, better utilization of resources etc.); in other words, it faces a downward sloping LRAC curve.
ii. After the scale of operation is increased further, however, the firm achieve constant costs i.e., LRAC become flat.
iii. If the firm further increases its scale of operation, diseconomies of scale set in (due to problems with managing a very large organization etc.) and the LRAC assumes a positive slope.

The following assumptions are made while deriving LRAC curves:

Price of factors are constant, technology is fixed, firms choose that combination of factors at which the MPP of the last dollar spent on each input is equal.

Long-run marginal cost (LRMC):

In case a firm is enjoying economies of scale, each incremental unit will cost less than the preceding one i.e., LRMC will be falling. The opposite will be true for diseconomies of scale. In case of constant costs, each incremental unit will cost the same, i.e., the LRMC will be constant.

Relation between SRAC and LRAC curves:

The LRAC curve for a firm is actually derived from its SRAC curves. The exact shape of the LRAC is a wave connecting the least cost parts of the SRAC curves. In practice however, LRAC is shown as a smooth U-shaped curve drawn tangent to the SRAC. This is also called an envelope curve.

Lesson 17

REVENUE & PROFIT MAXIMIZATION ANALYSIS

REVENUES

Revenues are the sale proceeds that accrue to a firm when it sells the goods it produces; in other words, they are the cash inflows that the firm received by way of selling its products.

Total Revenue (TR), Average Revenue (AR) and Marginal Revenue (MR):

Total revenue (TR), average revenue (AR) and marginal revenue (MR) concepts apply in the same way as they did to TC, AC and MC.

i. **TR = P x Q.**
ii. **AR = TR/Q;** AR is usually equal to price unless the firm is engaged in price discrimination.
iii. **MR = ΔTR/ΔQ.**

PRICE-TAKING FIRM

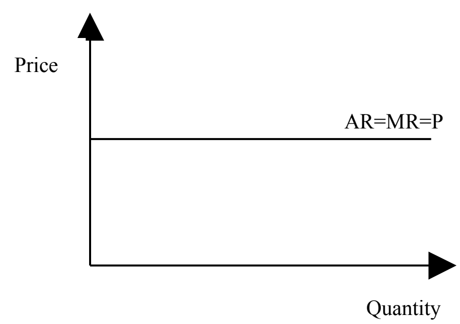
A firm that does not have the ability to influence market price is a price-taker. In perfect competition, the firm is price taker. There are large number of buyers and sellers and firm can not influence on the market price. Price is set by the forces of demand and supply.

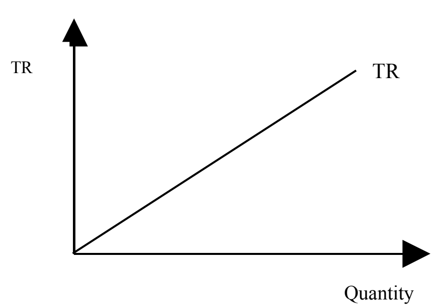
PRICE-MAKING FIRM

A firm that influences the market price by how much it produces can be called a price-maker or price-setter. In Monopoly, firm is price maker. A monopoly or a firm within monopolistic competition that has the power to influence the price it charges as the good it produces does not have perfect substitutes. A monopoly is a price maker as it holds a large amount of power over the price it charges.

DERIVING A FIRM'S AR & MR CURVES FOR PRICE TAKING FIRM

For a price taker, **AR=MR=P**. In this case, the demand (or AR) curve the firm faces is a horizontal line. TR for a price-taking firm is a straight line from the origin.

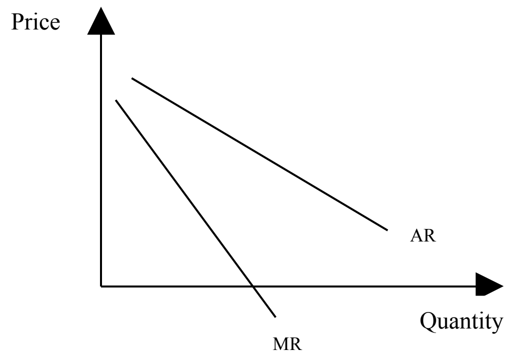


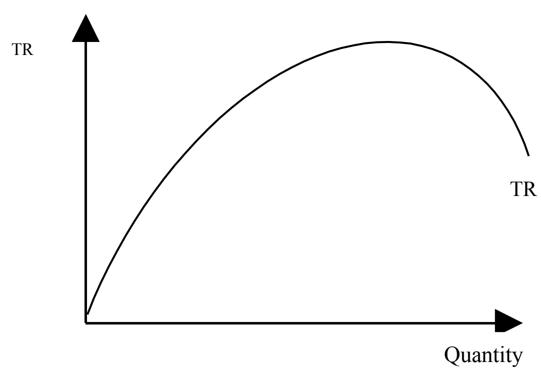


DERIVING A FIRM'S AR & MR CURVES FOR PRICE MAKING FIRM

Q (ships)	P = AR Rs. Crores	TR= P×Q Rs. Crores	MR
1	8	8	
2	7	14	6
3	6	18	4
4	5	20	2
5	4	20	0
6	3	18	-2
7	2	14	-4

A price maker faces a downward sloping demand (or AR) curve i.e., it cannot sell more without reducing price. But this means lowering the price for all units, not just the extra units it hopes to sell. The demand faced by a price maker is elastic, when MR is positive and therefore TR increases due to a decrease in price. Demand is inelastic when MR is negative, and therefore TR falls due to a decrease in price.





PROFIT MAXIMIZATION

Firms are interested in profit maximization. Profit is the difference between total revenue & total cost. Higher the difference, higher is the level of profit. Economists say that when firms earn zero accounting profits, they actually earn normal economic profits because TC already includes the normal profits that owners of the firms need for themselves to stay in the business. Positive profits are, for this reason, called supernormal profits as they are over and above what the owners normally require as a return for their entrepreneurship.

$$\text{Profit} = TR - TC$$

APPROACHES OF PROFIT MAXIMIZATION

Profit maximization can be studied using the TR-TC approach and the MR-MC approach.

i. In the TR-TC approach, it is assumed that firm is price maker and firm is operating in short run. Total profit is the vertical distance between TR and TC.
ii. In the MR-MC approach, two steps are followed to identify maximum profit. First: the profit-maximizing output is identified – this is the point where MR cuts MC. Second: the size of maximum profit is calculated using AC and AR curves.

Assumptions:

1. Demand curve is downward sloping
2. Firm is operating in the short run

TR & TC APPROACH

According to this approach, profit is maximized at that point where the difference between total revenue & total cost is maximum. In this table, profit is maximized at quantity of 3, where profit is at its maximum of 4.

Q(units)	TR	TC	T π
0	0	6	-6
1	8	10	-2
2	14	12	2
3	**18**	**14**	**4**
4	20	18	2
5	20	25	-5
6	18	36	-18
7	14	56	-42

MR & MC APPROACH

According to this approach, profit is maximized at the point where MC=MR. In this table, profit is maximized at quantity of 4 where MR=MC=2

Q	P=AR	TR	MR	TC	AC	MC	T π	A π
0	9	0	-----	6	----	----	-6	----
1	8	8	8	10	10	4	-2	-2
2	7	14	6	12	6	2	2	1
3	6	18	4	14	4 2/3	2	4	1 1/3
4	**5**	**20**	**2**	**18**	**4 1/2**	**4**	**2**	**1/2**
5	4	20	0	25	5	7	-5	-1
6	3	18	-2	36	?	11	-18	-3
7	2	14	-4	56	8	20	-42	-6

If MR & AR remain same over the long run, then the profit maximizing output will be obtained where MR intersects LRMC.
If AC is always above AR, then firms will never be able to make a profit. In this case, the point where MR=MC, represents the loss-minimizing point.
When MC and MR intersect at two points, not one, then Firms should produce at that point of intersection of MR and MC beyond which, MC exceeds MR.
If a firm's AR is below its AVC, it will shut down since it is not covering any part of its fixed costs.

Note: Graphical illustration of these two approaches is discussed in detail in the video lectures.

EXERCISES

How will the length of the short run for a shipping company depend on the state of the shipbuilding industry?
If the shipbuilding industry is in recession, the short run (and the long run) may be shorter. It will take less time to acquire a new ship if there is no waiting list, or if there are already ships available to purchase (with perhaps only minimal modifications necessary).
Up to roughly how long is the short run in the following cases?
(a) A mobile ice-cream firm. (b) A small grocery. (c) Electricity power generation.

a) 2-3 days: the time necessary to acquire new bicycles, equipment and workers.
b) Several weeks: the time taken to acquire additional premises.
c) 3-5 years: the time taken to plan and build a new power station.

How would you advise the naanwaala (bread-maker) next door as to whether he should (a) employ an extra assistant on a Sunday (which is a high demand day); (b) extend his shop, thereby allowing more customers to be served on a Sunday?

a) If maximizing profit is the sole aim, then he should employ an additional assistant if the extra revenue from the extra customers that the assistant can serve is greater than the costs of employing the assistant.
b) Only if the extra revenue from the extra customers will more than cover the costs of the extension plus the extra staffing.

Given that there is a fixed supply of land in the world, what implications can you draw from about the effects of an increase in world population for food output per head?
Other things being equal, diminishing returns would cause food output per head to decline (a declining MPP and APP of labour). This, however, would be offset (partly, completely or more than completely) by improvements in agricultural technology and by increased amounts of capital devoted to agriculture: this would have the effect of shifting the APP curve upwards.
The following are some costs incurred by a shoe manufacturer. Decide whether each one is a fixed cost or a variable cost or has some element of both.
(a) The cost of leather. (b) The fee paid to an advertising agency. (c) Wear and tear on machinery. (d) Business rates on the factory. (e) Electricity for heating and lighting. (f) Electricity for running the machines. (g) Basic minimum wages agreed with the union. (h) Overtime pay. (i) Depreciation of machines as a result purely of their age (irrespective of their condition).
(a) Variable. (b) Fixed (unless the fee negotiated depends on the success of the campaign). (c) Variable (the more that is produced, the more the wear and tear). (d) Fixed. (e) Fixed if the factory will be heated and lit to the same extent irrespective of output, but variable if the amount of heating and lighting depends on the amount of the factory in operation, which in turn depends on output. (f) Variable. (g) Variable (although the basic wage is fixed per worker, the cost will still be variable because the total cost will increase with output if the number of workers is increased). (h) Variable. (i) Fixed (because it does not depend on output).
Assume that a firm has 5 identical machines, each operating independently. Assume that with all 5 machines operating normally, 100 units of output are produced each day. Below what level of output will AVC and MC rise?
20 units. Below this level, the one remaining machine left in operation will begin to operate at a level below its optimum. (Note that with 5 machines producing 100 units of output, minimum AVC could be achieved at 100, 80, 60, 40 and 20 units of output, but between these levels some machines may be working at less than their optimum and some at more than their optimum. Thus if the optimum level for a machine is critical, then the AVC curve may look ‘wavy’ rather than a smooth line.
Why is the minimum point of the AVC curve (y) at a lower level of output than the minimum point of the AC curve (z)?
Because between points y and z marginal cost is above AVC (and thus AVC must be past the minimum point) but below AC (and thus AC cannot yet have reached the minimum point).

Even though AVC is rising beyond point y, the fall in AFC initially more than offsets the rise in AVC and thus AC still falls.

What economies of scale is a large department store likely to experience?

Specialized staff for each department (saving on training costs and providing a more efficient service for customers); being able to reallocate space as demand shifts from one product to another and thereby reducing the overall amount of space required; full use of large delivery lorries which would be able to carry a range of different products; bulk purchasing discounts; reduced administrative overheads as a proportion of total costs.

Why are firms likely to experience economies of scale up to a certain size and then diseconomies of scale after some point beyond that?

Because economies of scale, given that most arise from increasing returns to scale, will be fully realized after a certain level of output, whereas diseconomies of scale, given that they largely arise from the managerial problems of running large organizations, are only likely to set in beyond a certain level of output.

How is the opening up of trade and investment between, say eastern and western Europe, likely to affect the location of industries within Europe that have (a) substantial economies of scale; (b) little or no economies of scale?

a) Given that production will take place in only one or two plants, new plants will tend to be located near to the centre of the new enlarged European market.
b) Plants will still tend to be scattered round Europe, given that the customers are scattered.

These effects will be the result of attempts to minimize transport costs and thus will be more significant the higher are transport costs per kilometer.

Name some industries where external economies of scale are gained. What are the specific external economies in each case?

Two examples are:

- Financial services: pool of qualified and experienced labour, access to specialist software, one firm providing specialist services to another.
- Various parts of the engineering industry: pool of qualified and experienced labour, access to specialist suppliers, possible joint research, specialized banking services.

Would you expect external economies to be associated with the concentration of an industry in a particular region?

Yes. There may be a common transport and communications infrastructure that can be used; there is likely to be a pool of trained and experienced labour in the area; joint demand may be high enough to allow economies of scale to be experienced in the supply of some locally extracted raw material.

If factor X costs twice as much as factor Y (Px/Py = 2), what can be said about the relationship between the MPPs of the two factors if the optimum combination of factors is used?

MPPx/MPPy = 2. The reason is that if MPPx/Px = MPPy/Py, then, by rearranging the terms of the equation, MPPx/MPPy must equal Px/Py (= 2).

Could isoquants ever cross?

Not for a given state of technology, otherwise it would mean that at one side of the intersection the higher output isoquant would be ‘south-west’ of the lower output isoquant. This would mean that a higher output could be achieved by using less of both factors of production!

Could they ever slope upward to the right?

Yes. It would mean that one of the two factors had a negative marginal productivity that was greater than the positive marginal productivity of the other: i.e. that MPPa/MPPb (or MPPb/MPPa) was negative (a negative marginal rate of factor substitution).

This situation will occur when so much is used of one factor that diminishing returns have become so great as to produce substantial negative marginal productivity: isoquants will bend back on themselves beyond the points where they become vertical or horizontal. The firm, however, will not produce along this portion of an isoquant, because the price ratio (Pa/Pb) will (virtually) never be negative.

What will happen to an isocost if the prices of both factors rise by the same percentage?
It will shift inwards parallel to the old isocost.
Why do the prices of cattle and sheep prices fall so drastically "on", or just "after" the first day of Eid-ul-Azha?
The supply curve for cattle and sheep is fixed in the short-run, i.e. a vertical supply curve, therefore price will be determined by demand. Since demand for "cattle for sacrifice" falls drastically after or on the first day of Eid-ul-Azha, the price has to come down drastically as well for the market to clear.
Explain the shape of the LRMC curve for a firm with a typical U-shaped LRAC curve.
At first economies of scale cause the LRMC to fall. Then because of (marginal) diseconomies of scale, additional units of production begin to cost more to produce than previous units: the LRMC begins to slope upwards. But the LRAC is still falling because the LRMC is below it pulling it down. It is not until the LRMC crosses the LRAC that the firm will experience a rising LRAC and hence average diseconomies of scale.
Will the "envelope curve" be tangential to the bottom of each of the short-run average cost curves? Explain why it should or should not be.
No. At the tangency points the two curves must have the same slope. Thus the slope at the tangency point is not zero (the slope at the turning point or minima of the SRAC curves).
What would the isoquant map look like if there were (a) continuously increasing returns to scale; (b) continuously decreasing returns to scale?

a) The isoquants would get progressively closer and closer together.
b) The isoquants would get progressively further and further apart.

What can we say about the slope of the TR and TC curves at the maximum profit point? What does this tell us about marginal revenue and marginal cost?
The slopes are the same. But given that the slope of the total curve gives the respective marginal, this means that marginal revenue will be equal to marginal cost.
Fill in the missing figures in the table below.

Q	***P = AR***	***TR***	***MR***	***TC***	***AC***	***MC***	***Tπ***	***Aπ***
0	9			6				
1	8			10				
2	7			12				
3	6			14				
4	5			18				
5	4			25				
6	3			36				
7	2			56				

Q	***P = AR***	***TR***	***MR***	***TC***	***AC***	***MC***	***Tπ***	***Aπ***
0	9	0		6	–		–6	–

			8			4		
1	8	8		10	10		–2	–2
			6			2		
2	7	14		12	6		2	1
			4			2		
3	6	18		14	4.3		4	1.3
			2			4		
4	5	20		18	4.5		2	0.5
			0			6		
5	4	20		25	5		–5	–1
			–2			9		
6	3	18		36	6		–18	–3
			–4			16		
7	2	14		56	8		–42	–6

Why should the figures for MR and MC be entered in the spaces between the lines?

Because marginal revenue (or cost) is the extra revenue (or cost) from moving from one quantity to another.

You are given the following information for a firm.

Q	0	1	2	3	4	5	6	7
P	12	11	10	9	8	7	6	5
TC	2	6	9	12	16	21	28	38

Construct a detailed table like the one you constructed in the earlier question with TR, AC, MR, TC, AC, MC, TΠ and AΠ. Use your table to draw "two" diagrams (one with the marginal revenue and cost curves, and one with the total (or average) revenue and cost curves) and use them to show the "profit-maximizing output" and the "level of maximum profit", respectively. Confirm your findings by reference to the table you construct.

Q	***P = AR***	***TR***	***MR***	***TC***	***AC***	***MC***	***Tπ***	***Aπ***
0	12	0		2	–		–2	–
			11			4		
1	11	11		6	6		5	5
			9			3		
2	10	20		9	4.5		11	5.5
			7			3		
3	9	27		12	4		15	5
			5			4		
4	8	32		16	4		16	4
			3			5		
5	7	35		21	4.2		14	2.8
			1			7		
6	6	36		28	4.7		8	1.3
			–1			10		
7	5	35		38	5.4		–3	–0.4

The curves will be a similar shape to those discussed in the lecture, and included in the slides handout. The peak of the TΠ curve will be at Q = 4. This will be the output where MR and MC intersect.

Will the size of normal 'profit' vary with the general state of the economy?

Yes. Normal profit is the rate of profit that can be earned elsewhere (in industries involving similar level of risk). When the economy is booming, profits will normally be higher than when the economy is in recession. Thus the 'normal' profit that must be earned in any one industry must be higher to prevent capital being attracted to other industries.

Given the following equations:

$$\mathbf{TR = 72Q - 2Q^2;\ TC = 10 + 12Q + 4Q^2}$$

Calculate the maximum profit output and the amount of profit at that output using both methods.

(a) $T\Pi = 72Q - 2Q^2 - 10 - 12Q - 4Q$

$= -10 + 60Q - 6Q^2 \qquad (1)$

$\therefore\ dT\Pi\,/dQ = 60 - 12Q$

Setting this equal to zero gives:

$60 - 12Q = 0$

$\therefore\ 12Q = 60$

$\therefore\ \mathbf{Q = 5}$

(b) $MR = dTR/dQ = 72 - 4Q$

$MC = dTC/dQ = 12 + 8Q$

Setting MR equal to MC gives:

$72 - 4Q = 12 + 8Q$

$\therefore\ 12Q = 60$

$\therefore\ \mathbf{Q = 5}$

To find the level of maximum profit, we must substitute Q = 5 into equation (1). This gives:

$T\Pi = -10 + (60 \times 5) - (6 \times 5^2)$

$= -10 + 300 - 150$

= Rs. 140

Lesson 18

PROFIT MAXIMIZATION ANALYSIS (CONTINUED) & MARKET STRUCTURES

PROFIT MAXIMIZATION USING CALCULUS

If total revenue (TR) and total cost equation are given as follows:

$$\mathbf{TR = 48q - q^2}$$
$$\mathbf{TC = 12 + 16q + 3Q^2}$$

Q	TR	TC	Tπ = TR –TC
0	0	12	-12
1	47	31	16
2	92	56	36
3	135	87	48
4	176	124	52
5	215	167	48
6	252	216	36
7	287	271	16

Profit is maximized at the point where:

$$\mathbf{MC = MR}$$

MC function can be found by taking derivative of total cost function. i.e.:

$$MC = d\,TC / dQ$$
$$MC = 16 + 6Q$$

MR function can be found by taking derivative of total revenue (TR) function i.e.:

$$MR = d\,TR / dQ$$
$$= 48 - 2Q$$

As profit is maximized at the point where MR = MC, so by equating values of MC and MR function, we get,

$$\mathbf{MR = MC}$$
$$16 + 6Q = 48 - 2Q$$
$$6Q + 2Q = 48 - 16$$
$$8Q = 32$$
$$\underline{\mathbf{Q = 4}}$$

The equation for total profit is,

$$\mathbf{T\pi = TR - TC}$$
$$= 48Q - Q^2 - (12 + 16Q + 3Q^2)$$
$$= 48Q - Q^2 - 12 - 16Q - 3Q^2$$
$$= -4Q^2 + 32Q - 12$$

Putting Q = 4, we get,

$$T\pi = -4(4)^2 + 32(4) - 12$$
$$= -64 + 128 - 12$$
$$\underline{\mathbf{T\pi = 52}}$$

So profit is maximized where output is 4 and the maximum profit is 52.

We can also calculate AR from this information:

$$\mathbf{TR=48Q - Q2}$$
$$AR=TR/Q= 48Q/Q-Q2/Q$$
$$= 48-Q$$
$$= 48- 4$$
$$\underline{\mathbf{AR = 44}}$$

Slope of MR curve is twice as the slope of AR curve:

Slope of AR is obtained by differentiating AR function with respect to Q.

$$\textbf{Slope of AR} = dAR / dQ$$
$$= d/dQ\ (48-Q)$$
$$= -1$$

Slope of MR is obtained by differentiating MR function with respect to Q.

$$\textbf{Slope of MR} = dMR / dQ$$
$$= d/dQ\ (48-2Q)$$
$$= -2$$

MARKET STRUCTURES

Economists have identified four broad market structures:

- Perfect competition
- Monopoly
- Monopolistic competition
- Oligopoly

Type of market	Number of firms	Freedom of entry	Nature of product	Examples	Implication for demand curve of firm
Perfect competition	Very many	Unrestricted	Homogenous (undifferentiated)	Grains (wheat) or vegetables	Horizontal; firm is a price taker
Monopolistic competition	Many / Several	Unrestricted	Differentiated	Plumbers, restaurants	Downward sloping but relatively elastic; firm has some control over prices.
Oligopoly or Cartel	Few	Restricted	1. Undifferentiated or 2. Differentiated	Cement, cars, electrical appliance, oil.	Downward sloping relatively inelastic but depends on reactions of rivals to a price change
Monopoly	One	Restricted or completely blocked	Unique	WAPDA, or KESC	Downward sloping more inelastic than oligopoly; firm has considerable control over price

Market structure refers to how an industry (broadly called market) that a firm is operating in is structured or organized.

The key ingredients of any market structure are:

- Number of firms in the market/industry
- Extent of barriers to entry
- Nature of product
- Degree of control over price

Knowledge about market structure can help answer four questions:

ii. How much profit a firm will make (normal or supernormal)
iii. How much quantity it will produce at its profit-maximization point (i.e. whether it will be a large level of output or a small one relative to the market)
iv. Whether or not a higher level of output would increase the cost or productive efficiency of the firm or allocative efficiency for society (see the summary on monopoly for details)
v. Are the prices set too high, too low, or just right?

PERFECT COMPETITION

The main assumptions of perfect competition are:

i. Large number of buyers and sellers, therefore firms price-takers.
ii. No barriers to entry (also implies free mobility of factors of production).
iii. Identical/homogeneous products
iv. Perfect information/knowledge

The word **perfect** in perfect competition is not used its normative sense. Rather it means that competition in the industry is of an extreme nature. It is used as a benchmark with which to compare other types of market structures.
Perfect competition can be thought of as an **extreme form of capitalism**, i.e. all the firms are fully subject to the market forces of demand and supply.
Concentration ratio is used to assess the level of competition in an industry. It is simply the percentage of total industry output that is produced by the five largest firms in the industry.

PROFIT MAXIMIZATION UNDER PERFECT COMPETITION IN THE SHORT RUN
The short run is the period where at least one factor of production is fixed. In perfect competition, it also means that no new firms can enter the market. Equilibrium analysis can help us answer questions about the market-clearing price and quantity; where the profits are maximized and how much are these profits; how individual firms make their short run supply decisions and how these translate into the long-run industry supply curve.
In the short run, a perfectly competitive firm can settle at equilibrium where it is making super normal profits, normal profits, loss, or where it decides to shut down.
In the short run, the firm's supply curve is identical to the positive part of MC. The short run industry supply curve is simply the horizontal summation of the supply curves of individual firms.

The demand (or AR) curve for the industry is downward sloping but for any individual perfectly competitive firm, is horizontal. Thus, the firm can sell as much at the given market price. For this reason, the AR and MR curves align under perfect competition.

Lesson 19

MARKET STRUCTURES (CONTINUED)

PROFIT MAXIMIZATION UNDER PERFECT COMPETITION IN THE LONG RUN

In the long run, all the factors of production are variable. In the long run, any firm can enter or leave the industry. If there are supernormal profits in the short run, more firms will be attracted to the market and the increase in supply will push prices down to eliminate supernormal profit possibilities in the long run. By contrast, if firms are making losses in the short run, they will leave the industry in the long run causing supply to fall, prices to rise and normal profitability to be restored. In the long run, therefore, perfectly competitive firms can only earn normal profits.

ALLOCATIVE EFFICIENCY AND PRODUCTIVE EFFICIENCY

Public interest is concerned with both Allocative efficiency and productive efficiency.

a. **Allocative efficiency:** The optimal point of production for any individual firm is where MR=MC. The optimal point of production for any society is where price is equal to marginal cost. This is called the point of maximum Allocative efficiency and is achieved in perfect competition (because MR=MC, and MR=AR=P for a perfectly competitive price taking firm, therefore P=MC).
b. **Productive efficiency:** This is attained when firms produce at the bottom of their AC curves, that is, goods are produced in the most cost efficient manner. Perfectly competitive firms also achieve this in the long run because they produce at P=MC and this intersection point also happens to be the point of tangency with the lowest part of the AC curve. Thus P= AC minimum.

MONOPOLY

Monopoly defines the other pole or extreme of the market structure spectrum. Usually refers to a situation where there is a single producer in the market. However, it actually depends upon how narrowly you define the industry.

MONOPOLY POWER

Economists are often interested in how much monopoly power any firm (not necessarily a monopoly) has. Here monopoly stands for the extent to which the firm can raise prices without driving away all it customers. In other words, monopoly power and price elasticity of demand are inversely related.

Firms whose customers are more have more monopoly power. A monopolistic firm faces inelastic demand of the product & its demand curve is negatively sloped. While in perfect competition, demand curve has infinitely elasticity.

PROFIT MAXIMIZATION UNDER MONOPOLY

In monopoly, firm earns profit when MC=MR and MC curve cuts the MR curve from below. MC curve is not the supply curve of the firm as it was in the perfect competition. This is also the major difference between monopoly & perfect competition.

i. The profit maximizing or best level of output is given where MR=MC. Price is then read off the demand curve which is downward sloping. Note however, the difference with perfect competition, where the firm's demand curve was horizontal and not downward sloping like the industry. In a monopoly, however, the firm "is" the industry and therefore faces the same demand curve as the industry (a downward sloping one).
ii. Depending upon the level of AC at the point where MR=MC, the monopolist might be earn supernormal profits, breaking even or minimizing short run losses.
iii. Price is greater than MR in equilibrium. Therefore, price is not equal to MC. As such, therefore, the supply curve for the firm is not the rising part of the MC curve.

A monopolist can make supernormal profits even in long run because there is no easy entry for other firms as in the case of perfect competition. Therefore, a monopolist can maintain her high price even in the long run.

HOW CAN A MONOPOLIST RETAIN ITS MONOPOLY?

- **i.** These can be due to “natural” reasons or “active policies” pursued by the monopolist.
- **ii.** Large initial fixed costs may be involved, which makes it prohibitive for others to enter.
- **iii.** Natural monopoly experiences economies of scale as its operation becomes bigger and bigger and therefore it is cost-effective for only one single firm producing for the entire economy, rather than two or more firms.
- **iv.** Product differentiation or brand loyalty.
- **v.** Active pricing strategies (limit pricing: charging a price below a potential entrant’s AC to drive him out or discourage him from entering).
- **vi.** The “threat of takeover” by the monopolist sometimes prevents other firms from entering.
- **vii.** The monopolist controls the supply of key factors of production.
- **viii.** The monopolist produces a product, which no one else can imitate, i.e. is protected by patents or copyrights.

Lesson 21

MARKET STRUCTURES (CONTINUED)

PRICE DISCRIMINATION (PD) happens when a producer charges different prices for the same product to different customers. A seller with a degree of monopoly power has the ability to price discriminate. This means being able to charge a different price to different customers.

TYPES OF PRICE DISCRIMINATION

PD can be of three types:

- **i.** 1st degree PD
- **ii.** 2nd degree PD
- **iii.** 3rd degree PD

1ST DEGREE PD

In this type, everyone charged according to what he can pay. Seller can charge the highest price of any product from customers. First-degree price discrimination occurs when identical goods are sold at different prices to each individual consumer. Obviously, the seller is not always going to be able to identify who is willing to pay more for certain items, but when he or she can, his profit increases.

For example, this type of price discrimination can be observed in the sale of both new and used cars. People will pay different prices for cars with identical features, and the salesperson must attempt to estimate the maximum price at which the car can be sold. This type of price discrimination often includes a bargaining aspect, where the consumer attempts to negotiate a lower price.

2ND DEGREE PD

In this type, different prices charged to customers who purchase different quantities.

Examples of this can often be found in the hotel and airline industries where spare rooms and seats are sold on a last minute standby basis. In these types of industry, the fixed costs of production are high. At the same time the marginal or variable costs are small and predictable. If there are unsold airline tickets or hotel rooms, it is often in the businesses best interest to offload any spare capacity at a discount prices, always providing that the cheaper price that adds to revenue at least covers the marginal cost of each unit.

In retail stores, second-degree price discrimination also exists. A reduced price may be offered if you buy two t-shirts instead of just one. This form helps to get rid of merchandise and generate more revenue for a company.

3RD DEGREE PD

In this type, seller charge different prices to different customers in different markets.

For example, exporters may charge a higher price in overseas markets if demand is estimated to be more inelastic than it is in home markets. In Pakistan, there is food chain like Mc Donald's, pizza hut, KFC etc. They sell their products at different prices in different countries. Moreover, senior citizens are considered a group, and are often offered discounts at movie theaters, for transportation, in restaurants, and even in retail stores where seniors may have a "senior day" each week that allows them to take a discount on merchandise. "Students" are another segmented group that may be offered lower prices. Both seniors and students have a higher elasticity of demand and can generally afford to pay less than the average worker.

Consequences of PD:

PD can allow firms making losses to make profits, firms to increase their supernormal profits if make supernormal profits; allow goods to be produced that would otherwise not be produced.

PRE-REQUISITES / CONDITIONS OF PRICE DISCRIMINATION

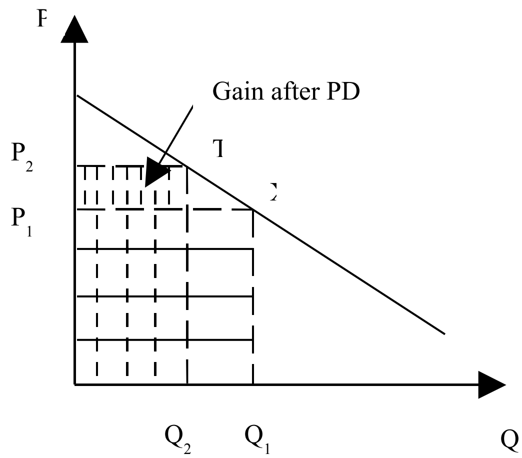
- **i.** That markets should be independent (it should not be possible for the different customers to arbitrage the price differences in the market).

ii. Firms should have the flexibility to price discriminate (i.e. should have some control over prices, so perfect competition ruled out).

iii. Price elasticity of demand for different customers should be different.

BENEFITS OF PRICE DISCRIMINATION

Price discrimination can be both, **beneficial or harmful** for public interest depending on a number of factors (equity or fairness concerns, the production of goods otherwise not produced, the use to which price-discriminating firms put their supernormal profits to, etc.).



MONOPOLISTIC COMPETITION

Monopolistic competition is also characterized by a large number of buyers and sellers and absence of entry barriers. In these two respects it is like perfect competition. Firms are price-takers but not in the extreme sense of perfect competition. Products are differentiated and in this respect, it is different from perfect competition.

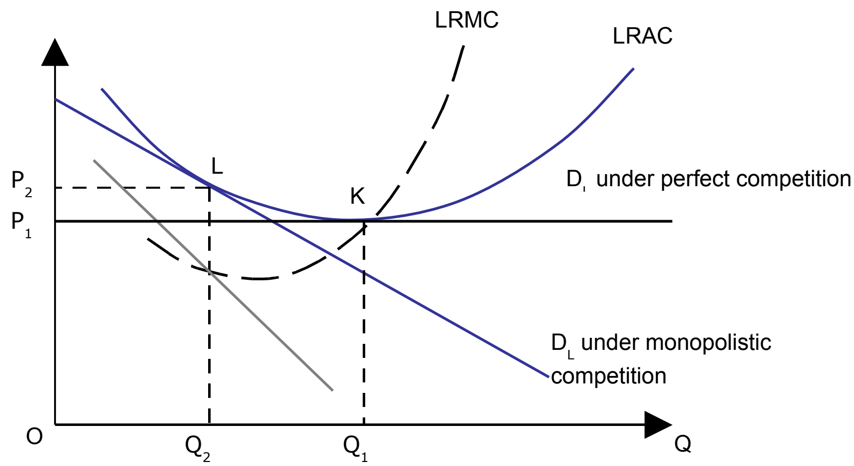
Thus the characteristics of a monopolistically competitive market are almost the same as in perfect competition, with the exception of heterogeneous products, and that monopolistic competition involves a great deal of non-price competition (based on subtle product differentiation). A firm making profits in the short run will break even in the long run because demand will decrease and average total cost will increase. This means in the long run, a monopolistically competitive firm will make zero economic profit. This gives the company a certain amount of influence over the market; because of brand loyalty, it can raise its prices without losing all of its customers. This means that an individual firm's demand curve is downward sloping, in contrast to perfect competition, which has a perfectly elastic demand schedule.

SHORT RUN AND LONG RUN UNDER MONOPOLISTIC COMPETITION:

In the short run, super normal profits are possible, but, in long run only normal profits can be earned. Equilibrium obtains where the AR curve becomes tangent to the AC curve. Public interest depends upon the position of AC at the point of tangency. If the AR curve is steep then the point of tangency will produce an output that will be well to the left of right the point where P= MC or P=AC $_{\text{minimum}}$.

Since products are differentiated, there is room and rationale for advertising and product promotion.

Monopolistic competition & public interest



Lesson 22

MARKET STRUCTURES (CONTINUED)

OLIGOPOLY

Similar to monopoly in the sense that there are a small number of firms (about 2-20) in the market and, as such, barriers to entry exist. It is similar to perfect competition in the sense that firms compete with each other, often feverishly, which may result in prices very similar to those that would obtain under perfect competition. It is similar to monopolistic competition since there is a possibility of having differentiated products.

DIFFERENCE OF OLIGOPOLY WITH OTHER MARKET STRUCTURES

An oligopoly is a market form in which a market or industry is dominated by a small number of sellers (oligopolists). The word is derived from the Greek for few sellers.

- Because there are few participants in this type of market, each oligopolist is aware of the actions of the others. The decisions of one firm influence, and are influenced by the decisions of other firms. Strategic planning by oligopolists always involves taking into account the likely responses of the other market participants. This causes oligopolistic markets and industries to be at the highest risk for collusion.
- It is not possible to identify any single equilibrium in oligopoly. Theory of firm is not clearly discussed & established as the theory of firm in the other three market structures. Reason for that is the firms are interdependent.

COLLUSION

Collusion occurs when two or more firms decide to cooperate with each other in the setting of prices and/or quantities. Firms collude in order to maximize the profits of the industry as a whole by behaving like a single firm. In doing so, they try to increase their individual profits. In the study of economics and market competition, collusion takes place within an industry when rival companies cooperate for their mutual benefit. Collusion most often takes place within the market form of oligopoly, where the decision of a few firms to collude can significantly impact the market as a whole. Cartels are a special case of explicit collusion. Collusion which is not overt, on the other hand, is known as tacit collusion.

At one time, all the firms sit together and combine their decisions in order to maximize profits & behave like monopoly. But at the same time, since all these firms have separate identity, they have the desire also to maximize their own individual profits as well. They might behave like single firm but they can also try to maximize their individual profits. This opposing situation creates tension. This tension can lead to collusion to break down.

TWO POSSIBLE SCENARIOS OF OLIGOPOLY

This tension between collusion & competition give rise to two possible scenarios that the oligopolist firms can have:

1. Collusive oligopoly
2. Non-collusive oligopoly

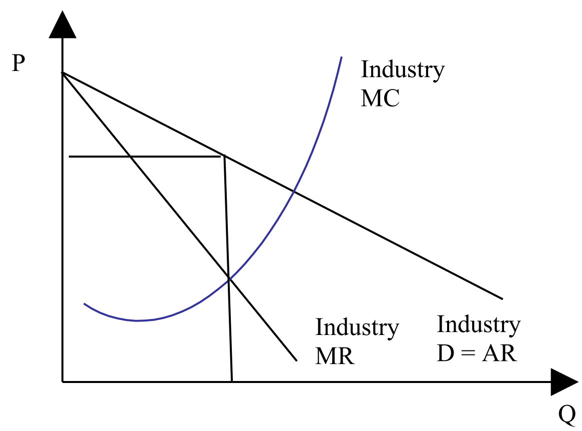
1- COLLUSIVE OLIGOPOLY (CARTEL)

A collusive oligopoly (or cartel) can be formed by deciding upon market shares, advertising expenses, prices to be charged (identical or different) or production quotas, such as OPEC, are collusive oligopolies. A firm can collude in many different ways. For example, they can collude on the market share in total profits. Collusion can also be done in terms of how much advertising expenditures each firm would have to put. They can also set the prices and quotas. If firms are not of equal size, then quotas can be allocated according to the MC of each firm. Cost of the cartel firm is minimized if the MC of each of the firm is equal. But the problem with this quota system is that firms which have higher MC will get lower quotas and the firms which have lower MC will get higher quotas.

Cartel

A cartel is a formal (explicit) agreement among firms. Cartels usually occur in an oligopolistic industry, where there are a small number of sellers and usually involve homogeneous products.
A cartel is most likely to survive when the number of firms is small, there is openness among firms regarding their production processes; the product is homogeneous; there is a large firm which acts as price leader; industry is stable; government's strictness in implementing anti-trust (or anti-collusion) laws. Govt regulations are helpless against internationally operational cartels or when collusion is tacit (or hidden) not explicit.

Profit maximizing cartel



2- NON- COLLUSIVE OLIGOPOLY
If different firms in the oligopolistic structures do not cooperate with each other is known as non collusive oligopoly. In this case, collusion breaks down because the incentive to cheat is very high. This can arise, for instance, in a situation where there is a lure of very high profits so that individual firms cheat on their quota and try to increase output and profits. But this causes everyone else to do the same and therefore supply soars and prices tumble producing in effect a non-collusive oligopoly.
The incentive to collude becomes strong for members of a non-collusive oligopoly when firms are not making good profits. Thus oligopolies usually oscillate between collusive and non-collusive equilibria.

COLLUSION & GAME THEORY
The Prisoner's Dilemma Situation
Consider about the two prisoners who have committed a crime together. Both have been arrested by the police and kept in separate cells. They have been interrogated separately about their crime. The dilemma is this if both confess to a crime they each have to face the punishment of 5 years in jail. If they do not confess then police has no evidence to keep them and police will let them go free. Their punishment will be very minor or might be zero in this case.

If one testifies for the prosecution against the other and the other remains silent, the betrayer goes free and the silent accomplice receives the full 10-year sentence. If both remain silent, both prisoners are sentenced to only six months in jail for a minor charge. If each betrays the other, each receives a five-year sentence. Each prisoner must make the choice of whether to betray the other or to remain silent. However, neither prisoner knows for sure what choice the other prisoner will make. So this dilemma poses the question: How should the prisoners act?
The dilemma arises when one assumes that both prisoners only care about minimizing their own jail terms. Each prisoner has two and only two options: either to co-operate with his accomplice and stay quiet, or to defect from their implied pact and betray his accomplice in return for a lighter sentence. The outcome of each choice depends on the choice of the accomplice, but each prisoner must choose without knowing what his accomplice has chosen.
In deciding what to do in strategic situations, it is normally important to predict what others will do. This is not the case here. If one prisoner knew the other prisoner would stay silent, his best move is to betray as he then walk free instead of receiving the minor sentence. If he knew the other prisoner would betray, his best move is still to betray, as he receive a lesser sentence than by silence. Betraying is a dominant strategy. The other prisoner reasons similarly, and therefore also chooses to betray. Yet by both defecting they get a lower payoff than they would get by staying silent. So rational, self-interested play results in each prisoner being worse off than if they had stayed silent.
A prisoner's dilemma situation for oligopolistic firms arises when 2 or more firms by attempting independently to choose the best strategy anticipation of whatever the others are likely to do, all end up in a worse position than if they had cooperated in the first place.

Payoff of a matrix for firm X& Y
(profit in Rs: at different prices)

		X's price	
		Rs.2	**Rs.1.80**
Y's price	**Rs.2**	**A.** 10m each	**B.** 5m for Y 12m for X
	Rs.1.80	**C.** 12m for Y 5m for X	**D.** 8m each

There are four points to be noted in this table:

- If both X & Y firms charge same price of Rs.2 then they get same profit of Rs.10 million as shown by option A.
- If both firm independently thought about reducing the price to Rs.1.8 then they have to take into account the decision of other firm. They have to think about what their rival will do? Their rival can do two things either to lower the price or kept the same price level. Now if X kept his price at Rs.2 the worst thing for X would be that its rival Y cuts its price to Rs.1.8. X's profit will now fall to Rs.5 million and Y's profit will increase to Rs.12 million due to lower price. This is shown in option C.
- If however, X cuts its price to Rs.1.8 the worst outcome still would be Y to cut its price too to Rs.1.8. but this time X's profit will only fall to Rs.8 million and Y's profit will also fall to Rs.8 million. This is shown in option D.
- However if X think optimistically and cuts its price to Rs.1.8 with his optimistic assumption that Y will leave with its price at Rs.2. if X is right in his assumption then he will earn the maximum profit of Rs.12 million and Y will earn Rs.5 million. This option is shown in option B.

Maximin strategy

Maximin strategy is a cautious (pessimistic) approach in which firms try to maximize the worst payoff they can make. It is the policy of adopting the safer side. It means the firm is trying to maximize the minimum profit that it will make.

Maximax strategy

A Maximax strategy involves choosing the strategy which maximizes the maximum payoff (optimistic). This policy arises from the optimistic approach that your rival will react most favorable to you. It means firm is going for the maximum possible profit.

Dominant strategy game:

Both these strategies leads towards the same strategy that is cutting down of price to Rs.1.8. this type of game is called the dominant strategy game. Given that both X & Y are tempted to lower price, they both end up tempting the lower profit i-e Rs.8 million each. If they collude and charge the same price, they will get profit of Rs.10 million each. Thus collusion rather than price war would be beneficial for both.

EXERCISES

Give two examples of markets which fall into each of the following g categories.
Perfect competition: grains; foreign exchange.
Monopolistic competition: taxis; van hire, restaurants.
Oligopoly: (homogeneous) white sugar; (differentiated) soap; banks.
Monopoly: WAPDA (electricity transmission); local bus company on specific routes.
Would you expect general building contractors and restaurant owners to have the same degree of control over price?
Other things being equal, restaurant owners are likely to produce a more differentiated product/service than general builders (as opposed to specialist builders), and are thus likely to face a less elastic demand. This gives them more control over price. Note, however, that the control over price depends on the degree of competition a firm faces. If, therefore, there were only a few builders in a given town, but many restaurants, the above arguments may not hold.
It is sometimes claimed that the market for the stocks/ shares of a company is perfectly competitive, or nearly so. Go through the four main perfect competition assumptions you have been taught about (large no. of price taking firms, no entry barriers, homogenous product, and perfect information) and see if they apply to HUBCO shares.

- **a.** Most aspects of the four assumptions of perfect competition apply.
- **b.** There is a very large number of shareholders (although there are some large institutional shareholders.)
- **c.** People are free to buy HUBCO shares (though, in reality, this depends on how liquid, i.e. accessible/available for sale the HUBCO shares are).
- **d.** All HUBCO shares are the same.
- **e.** Buyers and sellers know the current HUBCO share price, but they have imperfect knowledge of future share prices.

Is the market for gold perfectly competitive?
It is almost similar to the market for HUBCO shares. There are many buyers and sellers of gold, who are thus price takers, but who have imperfect knowledge of future gold prices. Also, countries with large gold stocks (e.g. the USA) could influence the price by large-scale selling (or buying). [Note also that the 'price' would have to refer to a weighted average of the price in all major currencies to take account of exchange rate fluctuations.]
What are the advantages and disadvantages of using a 5-firm "concentration ratio" rather than a 10-firm, 3-firm or even a 1-firm ratio?
The fewer the number of firms used in the ratio, the more useful it is for seeing just how powerful the largest firms are. The problem with only including one or two firms in the ratio, however, is that it will not pick up the significance of the medium-to-large firms. For example, if we look at the 3-firm ratio for two industries, and if in both cases the three largest firms have a 50 per cent market share, but in one industry the next largest three firms have 45 per cent of the market (a highly concentrated industry), but in the other industry the next three largest firms have only 5 per cent of the market (an industry with many competing firms), the 3-firm ratio will not pick up this difference. Clearly, this problem is more acute when using a 2-firm or a 1-firm ratio.
The more the firms used in the ratio, the more useful it is for seeing whether the industry is moderately competitive or very competitive. It will not, however, show whether the industry is dominated by just one or two firms. For example, the 10-firm ratio for two industries may be 90 per cent. But if in one case there are 10 firms of roughly equal size, all with a market share of approximately 9 per cent, then this will be a much more competitive industry than the other one, if that other one is dominated by one large firm which has an 85 per cent market share.
A more complete picture would be given of an industry if more than one ratio were used: perhaps a 1-firm, a 2-firm, a 5-firm and a 10-firm ratio.
Why do economists treat normal profit as a cost of production?
Because it is part of the opportunity cost of production. It is the profit sacrificed by not using the capital in some alternative use.

What determines (a) the level and (b) the rate of normal profit for a particular firm?

It is easier to answer this in the reverse order.

- **a.** The level of normal profit depends on the total amount of capital employed.
- **b.** The rate of normal profit is the rate of profit on capital that could be earned by the owner in some alternative industry (involving the same level of risks).

Will the industry supply be zero when the price of a firm A falls below P1 , where P1 < AVC for the firm?

Once the price dips below a firm's AVC curve, it will stop production. But only if "all" firms have the same AVC curve will the "entire industry" stop production. If some firms have a lower AVC curve than firm A, then industry supply will not be zero at P1.

Why is perfect competition so rare?

- Information on revenue and costs, especially future revenue and costs, is imperfect.
- Producers usually produce differentiated products.
- There are frequently barriers to entry for new firms.

Why does the market for fresh vegetables approximate to perfect competition, whereas that for aircraft does not?

There are limited economies of scale in the production of fresh vegetables and therefore there are many producers. There are such substantial economies of scale in aircraft production, however, that the market is only large enough for a very limited number of producers, each of which, therefore, will have considerable market power.

What advantages might a large established retailer have over a new e-commerce rival to suggest that the new e-commerce business will face difficulties establishing a market for internet shopping?

- Customers are familiar with the retailer's products and services and may trust their quality.
- Consumers may prefer to be able to ask advice from a sales assistant, something they can't do when buying over the internet.
- The retailer may have sufficient market strength to match any lower prices offered by the e-commerce firm.
- The retailer may have sufficient market strength to force down prices from its suppliers.
- Consumers may prefer to see and/or touch the products on display to assess their quality.
- Consumers may prefer the 'retail experience' of going shopping.

As an illustration of the difficulty in identifying monopolies, try to decide which of the following are monopolies: Pakistan Telecommunications Corporation Limited PTCL); your local morning newspaper; the village post office; ice cream seller inside the cinema hall; food sold in a university cafeteria; the board game 'Monopoly'.

In some cases there is more obvious competition than in others. For example, with the growth of mobile phones supplying phone services too, PTCL has lost some of its monopoly status for a section of the population. In other cases, such as ice creams in the cinema, village post offices and university cafeterias, there is likely to be a local monopoly. In all cases, the closeness of substitutes will very much depend on consumers' perceptions.

A monopoly would be expected to face an inelastic demand. And yet, if it produces where MR = MC, MR must be positive, demand must therefore be elastic. Therefore the monopolist must face an elastic demand! Can you solve this puzzle?

Demand is elastic at the point where MR = MC. The reason is that MC must be positive and therefore MR must also be positive. But if MR is positive, demand must be elastic. Nevertheless, at any given price a monopoly will face a less elastic demand curve than a firm producing the same good under monopolistic competition or oligopoly. This enables it to raise price further before demand becomes elastic (and before the point is reached where MR = MC).

If the shares in a monopoly (such as a water supply company in a European country) were very widely distributed among the population, would the shareholders necessarily want the firm to use its monopoly power to make larger profits?

If the water company raised its charges and thereby made a larger profit, shareholders would gain from larger dividends, but as consumers of water would lose from having to pay the higher charges. Except in the case of shareholders with only a few water shares, however, the gain is likely to outweigh the loss. Nevertheless, with shares very widely distributed, the average net gain would be only very small, and the wider the distribution, the more shareholders there would be who would suffer a net loss from the higher charges.

In what respects might the behaviour of Microsoft, increasingly becoming a monopoly in the software and operating systems market, be deemed to be: (a) against the public interest; (b) in the public interest?

- **a)** Prices are likely to be higher, given the lack of competition; there may be less product development, because potential competitors fear Microsoft's power to block their entry to the market, or drive them from it if they do succeed in entering; less choice for consumers.
- **b)** By developing products that are in general use round the world, it is more convenient for businesses and their employees, who do not have to learn different sets of programmes or have problems with incompatibility of programmes and operating systems; monopoly profits can lead to high levels of investment and product development, which can help to reduce prices over the longer term.

In which of the following industries are exit costs likely to be low: (a) steel production; (b) market gardening; (c) nuclear power generation; (d) specialist financial advisory services; (e) production of a new medicine. Are these exit costs dependent on how narrowly the industry is defined?

- **a)** High. The plant cannot be used for other purposes.
- **b)** Relatively low. The industry is not very capital intensive, and the various tools and equipment could be sold or transferred to producing other crops.
- **c)** Very high. The plant cannot be used for other purposes and decommissioning costs are very high.
- **d)** Low. The capital costs are low and offices can be sold.
- **e)** Low to moderate. It is likely that a pharmaceutical company can relatively easily switch to producing alternative drugs. Substantial exit costs are only likely to arise if the company is committed to a long-term research and development programme or if equipment is not transferable to producing alternative drugs.

Give some other examples of monopolistic competition.

Examples include: taxis, car hire, hotels and restaurants, insurance agents, estate agents, office equipment suppliers, antique dealers, computer systems.

Why may a food shop charge higher prices than wholesale markets (or supermarkets) for 'essential items' and yet very similar prices for "delicacy" items?

Because the demand for such essential items from a local food shop is likely to be less price-elastic than the demand for the delicacy items: if people run out of basic items, they will want to obtain them straight away from the nearest shop rather than waiting until they visit the supermarket. Also the supermarkets may obtain bulk discount from their suppliers on basic items, but not on delicacy items, where the sales turnover is much lower.

Which of these two items is a PSO or Shell petrol station more likely to sell at a discount: (a) petrol; (b) sweets? Why?

Petrol. The reason is that demand is more price elastic. People will be tempted to buy now, rather than waiting, if they see a reasonable discount. In the case of sweets, these are often an impulse buy and the price is very low anyway relative to the amount already spent on petrol. A penny or two price reductions will probably make very little difference to sales.

In monopolistic competition, why does the LRMC curve cross the MRL curve directly below the tangency point of the LRAC and ARL curves?

One way of answering the question is to note that long-run profits are maximized where long-run MR equals long-run MC (let's call it QL). But at QL, long-run AR equals long-run AC, whilst at any other output long-run AR is below long-run AC. Thus profits must be maximized at QL.

Assuming that supernormal profits can be made in the short run in a monopolistically competitive industry; will there be any difference in the long-run and short-run elasticity of demand? Explain.

Yes. The entry of new firms, attracted by the supernormal profits, will make the long-run demand for the firm more elastic: there are now more alternatives for consumers to choose from.

Why would you expect additional advertising dollars spent by a firm to cause smaller and smaller increases in sales? In other words why should advertising suffer from "diminishing returns"?

Because fewer and fewer additional people will see each extra advertisement (i.e. many of the people will have seen the adverts already and thus there will be little additional effect on their demand).

Which would you rather have: five restaurants to choose from, each with very different menus and each having spare tables so that you could always guarantee getting one; or just two restaurants, charging a bit less but with less choice and where you have to book quite a long time in advance?

Many people would choose the first, but clearly it is a question of personal preference.

How will advertising affect a cartel's MC and AR curves? How will this affect the profit-maximizing output?

If advertising increases total cartel sales, the cartel's AR curve will shift to the right and possibly become less elastic. The MC curve will only shift if the advertising varies with output. Given that the amount that member firms will advertise might not be known and, even if it were, the exact effects of any amount of advertising on AR are impossible to identify and compute, it would become difficult for the cartel to identify the profit-maximizing price with any degree of precision.

You have been taught about the conditions that facilitate the formation of a cartel? Which of these conditions were to found in the oil market in (a) the early 1970s; (b) the mid-1980s; (c) 2000?

- There are <u>relatively few</u> oil producing countries (but more in the 1980s than in the 1970s).
- The OPEC members <u>meet openly</u> to discuss pricing and quotas (in all three periods)
- <u>Production methods</u> are relatively similar, although costs vary according to the accessibility of the oil.
- <u>The (final) product is very similar</u> and there is an international price for each type of crude.
- <u>Saudi Arabia is the dominant member of</u> OPEC: its dominance over the world market, however, waned from the mid-1980s as non-OPEC production increased and there was a world glut of oil. With a growing world economy in the late 1990s, Saudi Arabia's influence grew again.
- <u>Entry barriers,</u> however, have "not" been significant. This has allowed several non-OPEC members (e.g. Mexico, Norway and the UK) to break into the market.
- <u>The market is relatively stable</u> in the short run (given the price and income inelasticity of demand). There has been a problem, however, of a decline in demand over the longer term.
- <u>Governments</u> round the world have been relatively powerless to curb OPEC's collusion, although from time to time (e.g. during the Gulf War) the USA has released oil from its huge stock piles to prevent excessive price increases.

Could OPEC have done anything to prevent the long-term decline in real oil prices since 1981?

Very little, given that the supply of substitutes (both oil and non-oil) for OPEC oil has increased substantially. Perhaps, with hindsight, if OPEC had not raised prices so much in 1973/74 and 1979 there would have been less incentive to develop substitutes and to break the power of the cartel.

Many oil analysts are predicting a rapid decline in world oil output in 10 to 20 years as world reserves are depleted. What effect is this likely to have on OPEC's behaviour?

The fall in output will drive up prices. Provided that OPEC can prevent its members from pumping oil more rapidly to take advantage of the rising price, OPEC's power could increase. It could demonstrate to its members the rising trend in oil prices and attempt to persuade them of the benefit of reducing production even further. It could 'sell' this policy to the world as one of being prudent with dwindling oil stocks.

In which of the following industries is collusion likely to occur: bricks, margarine, cement, crisps, washing powder, blank audio or video cassettes, and carpets?

In all cases collusion is quite likely: check out the factors favouring collusion discussed in the lecture and also above. In some cases it is more likely than others: for example, in the case of cement, where there is little product differentiation and a limited number of producers, collusion is more likely than in the case of carpets, where there is much more product differentiation.

Assume that there are two major oil companies operating filling stations in an area. The first promises to match the other's prices. The other promises that it will always sell at Re.1 per liter cheaper than the first. Describe the likely sequence of events in this 'game' and the likely eventual outcome. Could the promise of the second company be seen as credible (i.e. you will believe)?

Prices would be driven down, and hence profits reduced, until one of the companies could no longer stick to its promise – either the first accepting that its price will be Re. 1 above the second, or the second accepting the same price as the first. Alternatively both companies simultaneously may decide to abandon their policy and collude to raise prices. This may involve a secret meeting between them, or simply 'letting it be known' that they would be willing to raise prices, providing that the other company did the same.

The promise of the second company could be seen as credible if it had lower costs or greater financial backing than the first company. In such circumstances, the first company may be forced to give up its policy first. If they have similar costs and financial strength, then the threat is not credible.

Consider a train company which charges different prices for first and standard class, for traveling on different days in the week or different times in the day etc. Are these examples of price discrimination?

Price discrimination occurs when the same product or service (with the same marginal cost) is sold at different prices to different customers. Thus, strictly speaking, charging a different price for first and standard class, for travel on different times of day, or on different days of the week, or at different times of the year are not the purest examples of price discrimination, since (a) the service is different and (b) the marginal cost is not the same. On the other hand, charging a different price for children, students, old people, people traveling on single rather than return tickets etc. are examples of price discrimination since they allow travel on the same seat on the same train to different classes of people.

Are these various forms of price discrimination in the traveler's interest?

If the lower-price fares are making travel possible for people who could otherwise not afford it, then clearly they are benefiting. For the people paying the higher-priced fares, then there are advantages and disadvantages. Clearly, they will not like paying more than they would in the absence of price discrimination, but given that at peak times some lines are operating to full capacity, the higher price may be necessary to prevent queuing or grossly overcrowded trains (though note, as explained in the answer to the last question, charging higher prices at peak times to everyone is strictly speaking not a form of price discrimination).

If, over time, consumers are encouraged to switch their use of dial-up internet connections to off-peak periods, what will happen to peak and off-peak prices?

The difference between the prices will narrow.

To what extent is peak-load pricing (i.e. charging the highest price for a product/service when the loan of demand for it is highest; e.g. charging a high rate for dial-up internet connection in the day rather than after midnight) in the interests of consumers?

It may help to keep the average price down, if it spreads the use of fixed factors (like bandwidth or telephone lines) more evenly. It may also help to ease congestion (e.g. on trains) at peak times for those who have no alternative but to use the service at that time. Peak users may prefer a higher priced journey to a more congested journey or having to queue, and possibly running the risk of not getting the service (e.g. not getting on the train or bus because it is full).

Is total consumption likely to be higher or lower with a system of peak and off-peak prices as opposed to a uniform price at all times?

Higher, since some people would only be prepared to buy the product at off-peak prices.

Which type of price discrimination do cinemas pursue when they charge different prices for adults and children? First, second or third degree? Would it be possible for the cinema to pursue either of the other two types?

It is third-degree price discrimination. It groups cinema goers into two types: adults and children. It could not practice first-degree discrimination: it would not be possible to negotiate a separate ticket price with each customer! It could possibly practice a form of second-degree price discrimination, however, if it gave tokens to people each time they purchased a ticket and then sold tickets at reduced prices to people with tokens.

If all cinema seats could be sold to adults in the evenings at the end of the week, but only a few on Mondays and Tuesdays, what price discrimination policy would you recommend to the cinema in order for it to maximize its weekly revenue?

Offer reduced-price tickets to children in the evenings as well as in the afternoon for the first part of the week, but not for the end of the week.

Would the cinema make more profit if it could charge adults a different price in the afternoon and the evenings?

Possibly. The danger for the cinema, however, is that adults who would have gone to the cinema anyway may now choose to go in the afternoon, thereby losing the cinema revenue. Ideally the cinema would like to discriminate in such a way as to encourage people to go in the afternoon at a reduced price who would not have gone at all (whether in the afternoon or the evening), like old people for e.g., if they had to pay the higher price.

Why is the Prisoners' Dilemma game discussed in the lecture a dominant strategy 'game'?

Because, whatever assumption is made about the other's behavior, each prisoner is likely to confess.

How would each prisoner's strategy change if there were five prisoners (who committed the joint crime) and not two, and if all five all of them had been caught?

The more people there were involved in the crime, the greater would be the likelihood of one of them confessing and therefore the greater the temptation for any individual prisoner to confess.

Can you think of any other non-economic examples of the prisoners' dilemma?

Children in a class agreeing not to do homework, but parents keeping them apart after school so that they can persuade their children to do their homework, telling them, 'The other children will also be doing theirs and you will not want to show up by doing badly compared with them.' What should the children do? Do their homework in the fear that everybody else would do the homework (the equivalent of "confessing" in the fear of the other prisoners confessing) or not do the homework hoping that the others won't do it as well (the equivalent of "not confessing" in the hope that the others won't do it either).

Lesson 24

WELFARE ECONOMICS (CONTINUED)

THE MARKET FOR FACTORS OF PRODUCTION

The circular flow of income and expenditure shows the flow of goods and factors between households and firms. Firms are the demanders of the factors and households are the suppliers of the factors.

The Demand for Factor of Production:

The demand for factors of production (like labor) is a derived demand, because it is "derived" from the goods market. For e.g., the demand of labor increases when the demand for a labor-intensive good rises, and as firms try to produce more of that good by employing more labor.

Leisure:

Leisure is the time not used for working, or earning wages. It is usually the time that a laborer uses for relaxation and all activities other than work or necessary sleep.

The Marginal Disutility of Work (MDU_W):

As the supply of hours of labor increases, wage rate should also be increased. The relationship between hours provided by labor and wage rate is positive. And the curve for labor supply curve is positively sloped. But this curve can also be negatively sloped due to the marginal disutility of working hours. The marginal disutility of work (MDU_W) means the negative impact on the laborer of working for one additional unit of time. The MDU_W curve defines the supply curve for labor.

The Opportunity Cost:

The opportunity cost of working is leisure (and vice versa) that the worker could have enjoyed during that time had he not been working.

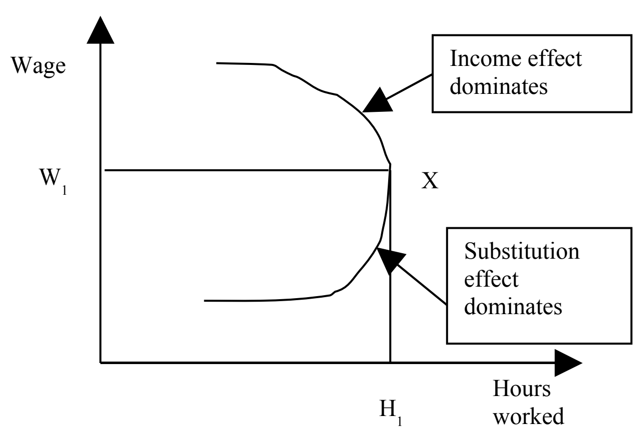
Supply and Demand Curve for Labor:

The labour supply curve may bend backwards above a certain wage rate as the income effect of higher income dominates the substitution effect of higher wages.

The wage rate is the marginal cost of labour to the firm and is directly proportional to the hours worked.

The demand curve for labour can be derived from the intersection of the wage rate lines (horizontal parallel lines) and the marginal revenue product of labour (a downward sloping concave function) given by $MRP_L = MPP_L \times MR_i$, where subscript "L" stands for labour and subscript "i" stands for the good which the laborer helps produce.

Back ward bending labor supply curve



The Value of Marginal Product of Labor (VMP_L):

The value of marginal productivity of labor can be represented by the following formula:

The value of marginal product of labour (VMP_L) = $MPP_L \times P_i$

It is equal to MRP_L when P = MC (as in perfect competition), but otherwise $VMP_L > MRP_L$.

One important difference between labour and land, capital is that the latter two can be purchased but labour can only be rented.

A **rent** is a periodic payment as a reward for hiring the factor of production for that period, where the purchase price of capital is the "value" of owning that capital for its entire life.

The Net Present Value (NPV) and Discounting:

Decisions about purchasing capital or land are often made on the basis of the net present value (NPV) associated with the decision. The NPV of an asset is the discounted value of the net returns that the asset generates over a period of time plus the discounted value of its disposal value at the end of the period minus the initial purchase cost.

Discounting is the process of converting a stream of future incomes and expenses into a present value. The discount rate is the rate at which the future incomes are discounted.

EXAMPLE:

- **Rate of discount = 10%**
- **The formula for discounting is as follows:**

$$PV = \sum \frac{X_i}{(1+r)^i}$$

Where **PV** is present value

Xi is earnings from the investment in the year i

r is the rate of discount

Σ is the sum over i, of the discounted earnings.

Present value of machine that generates Rs. 1,000 for four years and then sold as scrap for Rs. 1,000 at the end of year 4?

Year 1+ Year 2 + Year 3 + Year 4

$= 1000/(1.1)^1 + 1000/(1.1)^2 + 1000/(1.1)^3 + 2000/(1.1)^4$

$= 909 + 826 + 751 + 1366$

= Rs. 3852

If machine costs less than this then Buy, otherwise don't Buy.

Net present value = PV – Purchase cost

THE ECONOMICS OF INFORMATION PRODUCTS

The economics of information products or (internet products) involves studying how economic principles apply to the production, distribution and consumption of these products. Information economics or the economics of information is a branch of microeconomic theory that studies how information affects an economy and economic decisions.

The internet has reduced the **marginal cost of distributing information** to zero, as once a product is launched on the web, any number of potential customers can access/view it without any additional cost to the producer of the information.

Since the average cost of information product is falling over the entire range of output the market structure most consistent with such a product is (natural) monopoly.

Buying and selling information is not the same as buying and selling most other goods. First of all, information is non-rivalrous, which means that consuming information doesn't mean that someone else cannot also consume it. Obviously this does not apply to normal goods, like food, in which one's consumption precludes another's.

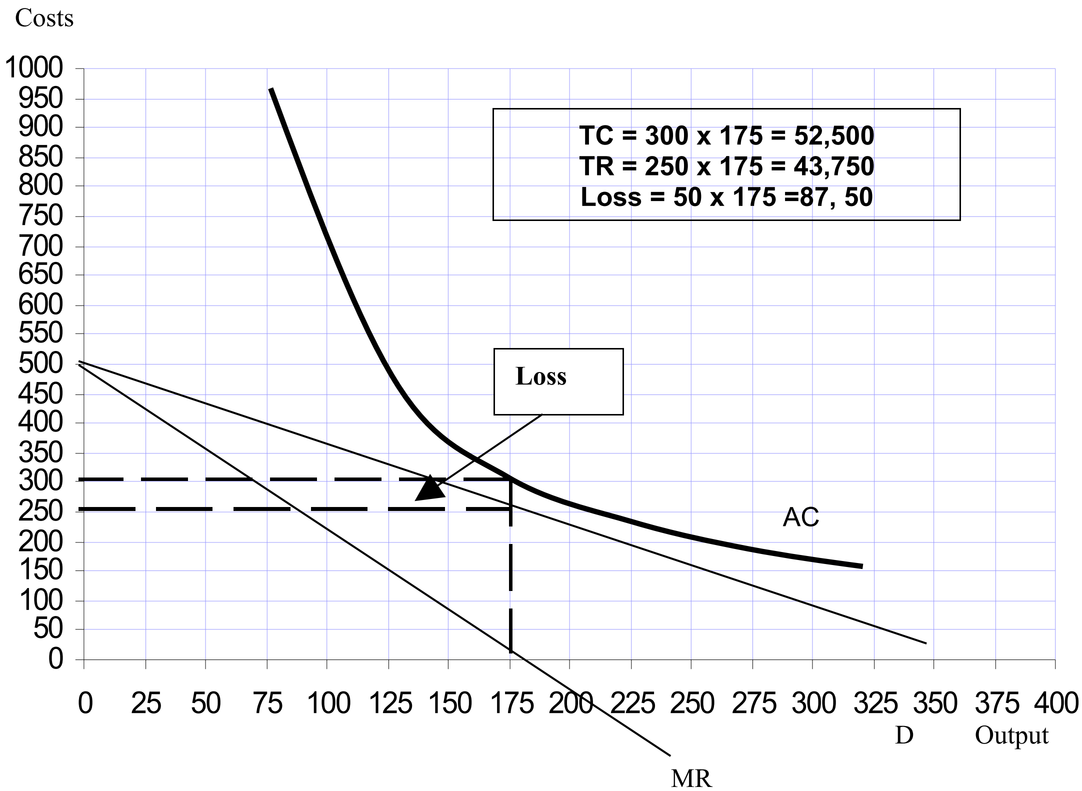
A related characteristic that alters information markets is that information has almost zero marginal cost. This means that once the first copy exists, it cost nothing or almost nothing to make a second copy. This makes it easy to sell over and over. However, it makes classic marginal cost pricing completely infeasible.

Experience goods are goods that people must get a flavor of before they can consider buying them. In economics, an experience good is a product or service where product characteristics such as quality or price are difficult to observe in advance, but these characteristics can be ascertained upon consumption.

A typical information product

Online economics course			
Q	**TC**	**MC**	**AC**
50	50,000		1000
100	50,000	0	500
150	50,000	0	330
200	50,000	0	250
250	50,000	0	200
300	50,000	0	166

A typical information product
Profit or loss



EXERCISES
WELFARE ECONOMICS AND EXTERNALITIES

If monopoly power existed in an industry, would production be above or below the socially efficient level (assuming no externalities)? Which would be greater, MSB or P?

A firm with monopoly power produces where MR = MC. But this is below the socially efficient level of output which obtains where P = MC.

Assuming no externalities, and perfect competition show that social efficiency is achieved in the factor markets, where: $MSB_f = MRP_f = P_f = MDU_f = MSC_f$ (where MRP is the marginal product of a factor, MDU is the marginal disutility of supplying it and f is any factor).

The MRP_f is the marginal benefit to the employer from employing a factor. Profits will be maximised where MRP = P_f. The MDU_f is the marginal cost to the factor supplier from supplying a factor. The factor supplier's 'surplus' will be maximised where $P_f = MDU_f$. Under perfect competition, since $MRP_f = P_f$ for each producer, and $P_f = MDU_f$ for each factor supplier, then, since the market price for the factor (P_f) is the same for all firms and factor suppliers, then $MRP_f = MDU_f$ for all firms and factor suppliers. In the absence of externalities in the factor market, $MSB_f = MRP_f$, and $MDU_f = MSC_f$. Thus: $MSB_f = MRP_f = P_f = MDU_f = MSC_f$, i.e. $MSB_f = MSC_f$

Note: The concepts in this question will not be tested in the exam.

Trace through the effects in both factor and goods markets of the following:

a) **An increase in the productivity of a particular type of labour.**
b) **An increase in the supply of a particular factor.**

The following charts illustrate the effects:

(a) *1. Labour demand*

$MRPl\uparrow$ (i.e. $MSBl\uparrow$) $\rightarrow$ $MRPl > W$ $\rightarrow$ employment of labour$\uparrow$ $\rightarrow$ $W\uparrow$

2. Labour supply

$W\uparrow$ $\rightarrow$ $W > MDUl$ (i.e. $W > MSCl$) $\rightarrow$ supply of labour$\uparrow$ $\rightarrow$ $MDUl\uparrow$ (movement up along *MDUl* and hence *MSCl* curve) .

These adjustments would continue until $MSBl = MSCl$.

3. Producer supply

$MRPl\uparrow$ $\rightarrow$ $MC\downarrow$ $\rightarrow$ $P > MC$ (i.e. $P > MSC$) $\rightarrow$ production$\uparrow$ $\rightarrow$ $P\downarrow$

4. Consumer demand

$P\downarrow$ $\rightarrow$ $MU > P$ $\rightarrow$ consumption$\uparrow$ $\rightarrow$ $MU\downarrow$ (movement down along *MU* and hence *MSB* curve)

These adjustments would continue until $MSB = MSC$ (in goods markets).

(b) *1. Factor supply*

$Sf\uparrow$ (i.e. $MSCf\downarrow$) $\rightarrow$ $Sf > Df$ $\rightarrow$ $Pf\downarrow$

2. Factor demand

$Pf\downarrow$ $\rightarrow$ $MRPf > Pf$ $\rightarrow$ employment of factor$\uparrow$ (movement down along *MRPf* and hence *MSBf* curve)

These adjustments would continue until $MSBf = MSCf$.

3. Producer supply

$Pf\downarrow$ $\rightarrow$ $MC\downarrow$ $\rightarrow$ $P > MC$ (i.e. $P > MSC$) $\rightarrow$ production$\uparrow$ $\rightarrow$ $P\downarrow$

4. Consumer demand

$P\downarrow$ $\rightarrow$ $MU > P$ $\rightarrow$ consumption$\uparrow$ $\rightarrow$ $MU\downarrow$ (movement down along *MU* and hence *MSB* curve)

These adjustments would continue until $MSB = MSC$ (in goods markets).

Note: The concepts in this question will not be tested in the exam.

If MU_X/MU_Y were greater than P_X/P_Y, how would consumers behave? What would bring consumption back to equilibrium where $MU_X/MU_Y = P_X/P_Y$?

Consumers would buy relatively more of X and relatively less of Y. But as they did this, MU_X/MU_Y would fall (because of diminishing marginal utility) until $MU_X/MU_Y = P_X/P_Y$.

Note: The concepts in this question will not be tested in the exam.

If MC_X/MC_Y were greater than P_X/P_Y how would firms behave? What would bring production back into equilibrium where $MC_X/MC_Y = P_X/P_Y$?

Firms would produce relatively more of good Y and relatively less of good X. But as they did this, MC_X/MC_Y would fall (because of the law of diminishing returns and hence increasing marginal costs as production increases) until $MC_X/MC_Y = P_X/P_Y$.

What are marginal external costs?

The different between marginal social cost and marginal private costs is marginal external cost. In other words, it is the marginal cost of the externality to society.

Is it likely that the MSB curve will be parallel to the MU curve? Explain your reasoning.

No. It is likely that the marginal external costs of consumption will increase as more is consumed, and thus the curves will get further apart (making the MSB curve steeper than the MU = MB curve). For example, the marginal pollution costs of cars gets progressively greater as more and more cars come onto the roads and the environment becomes less and less able to absorb the additional quantities of pollutants.

Give other examples of each of the four types of externality.

a) External costs of production (MSC > MC): The pollution of rivers and streams by slurry and nitrate run-off from farms; road congestion near a factory.

b) External benefits of production (MSC < MC): Beneficial spin-offs from the development of new products (for example, the various space programmes in the USA, the USSR and Europe have contributed to advances in medicine, materials technology, etc.); where the opening of a new environmentally friendly factory results in less output from factories that pollute.

c) External costs of consumption (MSB < MB): The effect of CFC aerosols on the ozone layer; the unpleasant sight of kites stuck in trees and wires.

d) External benefits of consumption (MSB > MB): People decorating the outside of their houses or making their gardens look attractive benefits neighbours and passers-by; car owners getting their cars properly serviced so as to reduce the smoke emitted and the pollution associated with it.

Some roads could be regarded as a public good, but some could be provided by the market. Which types of road could be provided by the market? Why? Would it be a good idea?

Roads where there are relatively few access points and where therefore it would be practical to charge tolls. Charges could be regarded as a useful means of restricting use of the roads in question, or, by charging more at peak times, of encouraging people to travel at off-peak times. Such as system, however, could be regarded as unfair by those using the toll roads, and might merely divert congestion onto the non-toll roads.

Which of the following have the property of non-rivalry: (a) a can of drink; (b) public transport; (c) a commercial radio broadcast; (d) the sight of flowers in a public park?

(a) No. (b) No (passengers take up seats). (c) Yes. (d) Yes (unless I get in your way!).

Give some other examples of public goods. Does the provider of these goods (the government or local authority) charge for their use? If so is the method of charging based on the amount of the good that people use? Is it a good method of charging? Could you suggest a better method?

Two examples are: national defence; urban roads. In both cases the user is not directly charged. The funding comes from taxation. In the case of roads, part of the funding comes from road users generally (in the form of taxes on petrol and road fund licenses) and part from general or local taxation. Only in the case of petrol tax is the charging related to the amount that people use the public good. It encourages people to use the roads less, and thus takes into account the marginal cost (i.e. repairs and maintenance) of road provision. In this sense, however, roads are not a pure public good because using them does create a small amount of wear and tear on them (although a significant portion of road maintenance costs are due simply to deterioration through time).

If the marginal cost of provision is zero (as is the case with a pure public good) then charging people according to how much they use it will not cause an efficient allocation of resources: with a zero marginal cost, the price should be zero. Charging people according to how much they use it, however, could be regarded as fair according to the benefit principle – but not according to the principle of vertical equity.

Name some goods or services provided by the government or local authorities that are not public goods.

Education, health, libraries, parks. Only limited people can use these facilities.

If health care is provided free, the demand is likely to be high. How is this high demand dealt with? Is this a good way of dealing with it?

It is dealt with by a system of queuing. Emergency cases are usually dealt with immediately, or at least very quickly, but non-emergency cases may have to wait weeks, months or even years for treatment.

Many people would argue that for reasons of equity, and the special nature of health, it is better to solve the problem of waiting lists by diverting more resources into health care, rather than by using a system of charging people. Except where there are initially idle resources or inefficiencies, this approach will result in a lower provision of other publicly provided goods or services, or higher taxes.

How would you attempt to value time that you yourself save (a) getting to work; (b) going on holiday; (c) going out in the evening?

The approach is to ask what the opportunity cost to you of that time is. What else could you have done with the time and what, as a result, would you have been prepared to pay to save time? Thus in the case of (b), if you have a long holiday and the travel is seen as an enjoyable part of it, then there may be no time cost to you, or it may even be seen as a benefit; whereas if you only have a week off work and want to get to the sun as quickly as possible, then you will want to minimize travel time and thus may be prepared to pay quite a lot more to fly rather than go over land.

Imagine that a public project yields a return of 13 per cent (after taking into account all social costs and benefits), whereas a 15 per cent private return could typically be earned in the private sector. How would you justify diverting resources from the private sector to this project?

If it could be argued that this particular project yielded longer-term benefits, and that the market rate of discount is too high in the sense that it gives undue weight to present benefits and costs over future benefits and costs than is socially desirable, then there would be some justification. The justification for a lower social rate of discount is that the market only reflects the wishes of the current generation, whereas governments ought to take a longer-term perspective.

An alternative justification may be in terms of the distribution of costs and benefits. If the project was an effective means of targeting help to the poor (say a new hospital in an area where there is a lot of poverty) then the government may want to apply a lower rate of discount.

Assume that a project has an initial construction cost of Rs10 000, takes a year to come into operation and then has a life of 3 years. Assume that it yields Rs5000 per year in each of these 3 years. Is the NPV positive at (a) a 10 per cent discount rate; (b) a 15 per cent discount rate?

(a) NPV = –Rs.10 000 + Rs.5000/1.1 + Rs.5000/1.21 + Rs.5000/1.331
= –Rs.10 000 + Rs.4545 + Rs.4132 + Rs.3757
= +Rs.2434

The NPV is therefore positive at a 10 per cent discount rate.

(b) NPV = –Rs.10 000 + Rs.5000/1.15 + Rs.5000/1.3225 + Rs.5000/1.5209
= –Rs.10 000 + Rs.4348 + Rs.3781 + Rs.3288
= +Rs.1417

The NPV is therefore still positive at a 15 per cent discount rate.

Why is this type of cost–benefit analysis (CBA) simpler to conduct that ones assessing the desirability of a new road or airport?

There are specific scenarios, with very precise assumptions. The main purpose was not to demonstrate the positive NPV (which depends on the assumptions about the value of the environmental benefits of reducing emissions), but to compare the relative advantages of the alternative scenarios.

With a CBA of a new road or airport, there have to be assumptions, not only about the environmental impact, but about the value of time saved or additional time incurred by travellers, about the value of lives saved or lost and about the costs to local people disturbed by the road or airport. Thus the measurement of externalities is likely to be more problematic than in the case of lowering the sulphur content of road fuel.

How do merit goods differ from public goods?

They could be provided by the market (albeit with too little consumed). The problem of non-excludability does not apply.

Note: The concepts in this question will not be tested in the exam.

Summarize the economic policies of the major political parties. (If it is near an election you could refer to their manifestos.) How far can an economist go in assessing these policies?

You will need to look at the current policies.

Economists in their role as economists cannot challenge fundamental normative issues – such as whether it is desirable to have a much more substantial redistribution of income from the rich to the poor. They can, however, examine whether the factual claims of the parties are correct, and whether the policies they advocate will bring the effects they claim.

Note: The concepts in this question will not be tested in the exam.

Give some examples of how correcting problems in one part of the economy will create problems elsewhere.

Two examples are:

- A local authority reduces street parking in the centre of a town in order to reduce congestion on the streets, but succeeds in encouraging commuters and shoppers to park outside the town centre in residential areas, thus reducing the quality of life for those living in those areas.
- The government taxes the consumption of electricity in order to encourage people to become more fuel efficient and thus to reduce power station emissions. Some people respond by switching to burning coal, with the result that emissions from this source increase.

THE MARKET FOR FACTORS OF PRODUCTION

Why might the labour supply curve be backward bending?

Because after a point the income effect of the higher wage dominates the substitution effect. Note that when we talk of substitution, it is substitution between work and leisure. The labourer must decide how to utilize his time, i.e. how much hours to spend working and how much to spend relaxing. If the wage rate (which also represents the worker's income) becomes very high, given that leisure is a normal good, the demand for relaxation increases. This is the income effect and it is positive for leisure and negative for hours worked. However, as the wage rate increases, the opportunity cost of relaxing increases and prompts the worker to work more. This is the substitution effect and is negative for leisure and positive for hours worked. The labour supply curve becomes backward bending when the income effect of higher wages dominates the substitution effect.

What is the distinction between MRP_L and VMP_L? When is this distinction important?

Labour's marginal revenue product (MRP_L) = MR * MPP_L
Labour's value of marginal product (VMP_L) = P * MPP_L
The two terms mean the same when the product market is perfectly competitive, because then: MR = MC = P. However, when the product market is not perfectly competitive, for e.g., if it is a monopoly, then P > MR at the profit maximizing output, and therefore $VMP_L > MRP_L$. The important implication of this is that a monopolist would pay labourers less than the "value of the output that their labour helps produce", i.e. the VMP_L. To that effect this can be viewed as exploitation of labour.

Do any of the following contradict marginal productivity theory:

a) **Nationally negotiated wage rates;**
b) **Discrimination;**

Even if marginal productivity theory were not relevant in these cases, the theory would still be accurate in the sense that if firms wanted to maximise profits then they should employ workers to the point where $MC_L = MRP_L$.

But do any of the above four cases necessarily contradict marginal productivity theory?

a) No, not necessarily. Firms may find it convenient (in terms of the costs of negotiating and the avoidance of disputes) to pay nationally agreed wage rates. They then could employ workers up to the point where their MRP was equal to this wage rate (which, given that the firm was a 'wage taker' would be equal to the MC_L).

b) Yes. Discrimination would lead to firms employing those workers against whom they were discriminating below the level where their MRP was equal to their MC_L.

Which of the following are stocks and which are flows?

a) **Unemployment.**
b) **Redundancies (job lay-offs).**
c) **Profits.**
d) **A firm's stock market valuation (share price).**
e) **The value of property after a period of inflation.**

Stocks: (a), (d) and (e). They are measurements at a point in time.

Flows: (b) and (c). They are measurements over a period of time. (Note that (e) would only be a flow if it were measuring the rise in the value of property over a period of time.)

What is the present value of a machine that lasts three years, earns Rs.100m in year 1, Rs.200m in year 2 and Rs.200m in year 3, and then has a scrap value of Rs.100m? Assume that the rate of discount is 5 per cent. If the machine costs Rs.500m, is the investment worthwhile? Would it be worthwhile if the rate of discount were 10 per cent?

Using the formula given in the lectures:

$$\begin{aligned} PV &= Rs.100/1.05 + Rs.200/(1.05)^2 + Rs.300/(1.05)^3 \\ &= Rs.95.24 + Rs.181.41 + Rs.259.15 \\ &= Rs.535.80 \end{aligned}$$

Thus the investment is profitable at a discount rate of 5 per cent given that the machine costs Rs.500.

If the rate of discount is 10 per cent, then the present value this time is given by:

$$\begin{aligned} PV &= Rs.100/1.1 + Rs.200/(1.1)^2 + Rs.300/(1.1)^3 \\ &= Rs.90.91 + Rs.165.29 + Rs.225.40 \\ &= Rs.481.59 \end{aligned}$$

With a 10 per cent discount rate, therefore, the investment would not be profitable.

What market price would a piece of land sell for if it earned Rs.10,000 rent per year, and if the rate of interest were 5 per cent?

According to the formula, market price of land = [annual rental value (in Rs.) / interest rate (in % p.a.)]
Rs.10,000/0.05 = Rs.200,000.

What does this tell us about the relationship between the price of an asset (like land) and the rate of interest?

The higher the rate of interest, the lower the market price of the asset. The asset would have to be cheaper (i.e. yield a higher rate of return) to persuade the purchaser to sacrifice the alternative of earning the market rate of interest.

INFORMATION ECONOMICS

There are many dimensions to information economics. The first of these was touched upon in the lectures on background on demand under "demand under uncertainty." We touched concepts like moral hazard and adverse selection. Below we touch upon some of the remaining aspects that were not covered there. PART A contains questions related to the economics of information products on the web that were covered in the lecture, while PART B contains other "general" information economics issues.

PART A:

What has the internet done to the cost of distributing information?

It has virtually brought down this cost to zero. Distributors can place information on their websites and consumers can access it or download it from there without any additional cost to the distributor. There are however still the initial costs of preparing the information in a format that can be placed on the web, and also the costs of maintaining the website and marketing the website to interested consumers of that information. However, in terms of the marginal cost of distributing information to one more consumer, that cost has now been brought down to virtually zero by the internet.

What are experience goods?

These are goods whose value can be ascertained by the consumer only after experiencing or consuming it. For example if a university wanted to deliver an online course in say, finance, it might allow potential students to take part of the course free and then if they are satisfied they could proceed with a charge for the remaining or whole of the course. The same principle applies to buying software on the web. You are often given a free trial version before you asked to commit to buy the software.

How is it possible for companies like MSN or yahoo to provide free e-mail or chatting facilities or "web space" to internet users the world over?

Everything has a cost. Providing free web space or e-mail account to millions of users also must have a cost. Now if the users are not paying this cost, someone else should. This someone else is the ever-growing pool of businesses (travel companies, financial advisors, supermarkets, fashion outlets – you name it) who wish to advertise their products to millions of potential customers worldwide. Yahoo and MSN, because they have such a large captive clientele can serve as the ideal vehicle media for their advertisements. Thus you will see that whenever you log on to the yahoo page there is some advert that appears without you asking. This is to ensure that "all" users do actually see the advert, so that "some of them" buy the product being advertised.

PART B
(YOU WILL NOT BE TESTED ON THIS IN THE EXAM)

Assume that you wanted the following information. In which cases could you (i) buy perfect information, (ii) buy imperfect information, (iii) be able to obtain information without paying for it, (iv) not be able to obtain information?

- **a) Which washing machine is the most reliable?**
- **b) Which of two jobs that are vacant is the most satisfying?**
- **c) Which builder will repair my roof most cheaply?**
- **d) Which builder will make the best job of repairing my roof?**
- **e) Which builder is best value for money?**
- **f) How big a mortgage would it be wise for me to take out?**
- **g) What course of higher education should I follow?**
- **h) What brand of washing powder washes whiter?**

In which cases are there non-monetary costs to you of finding out the information? How can you know whether the information you acquire is accurate or not?

(a) (i) or (ii);
(b) (iii) (by asking people currently doing the job) or (iv);
(c) (iii) (by obtaining estimates);
(d) (iii) (albeit imperfect, by inspecting other work that the different builders have done) or (iv);
(e) As (d);
(f) (iii);
(g) (iii);
(h) (i) or (ii) (as in (a)) or (iii) by experimenting.

All could involve the non-monetary costs of the time involved in finding out.

If the information is purely factual (as in (c) above), and you can trust the source of your information, there is no problem. If you cannot trust the source, or if the information is subjective (such as other people's experiences in (b) above), then you will only have imperfect information of the costs and/or benefits until you actually experience them.

Make a list of pieces of information a firm might want to know, and consider whether it could buy the information and how reliable that information might be.

- Some examples include:

The position and elasticity of the demand curve: Market research can provide some information, but it is very unreliable, especially in an oligopolistic environment, where the actions of rival are unpredictable.

- Next year's wages bill:

The information cannot be purchased, but it could use its own past experiences to predict (albeit imperfectly) the outcome of wage negotiations.

- The costs of alternative inputs:

This information is probably available free from suppliers.

- Ways of saving taxes:

Employing accountants can help the firm save money here.

What type of information can be a public good? (Clue: do not confuse a public good with something merely provided by the government, which could also be provided by the private sector.)

An example of information that is nearer to being a public good, is information that is simple enough not to require being presented as a set of tables or as a report. It can be told from one person to another: as such it would be difficult to enforce copyright. An example would be the current rate of inflation or the size of the balance of trade deficit or surplus.

Lesson 25

INTRODUCTION TO MACROECONOMICS

MACROECONOMICS

As a subject, macroeconomics only began to be taught in colleges and universities in the 1940s after the influence of a very influential British economist, John Maynard Keynes who believed the macro economy (with its associated variables) deserved to be understood and analyzed in its own right, and not just as an aggregation of the various micro-markets, as was believed earlier.
Macroeconomics is a branch of economics that deals with the performance, structure, and behavior of a national economy as a whole. The variables of interest change from the price, demand or supply of a particular product to the economy-wide price level, aggregate demand and aggregate supply.

AGGREGATE DEMAND (AD)

Aggregate demand (AD) is the total planned or desired spending (expenditure) in the economy during a given period. AD is the sum of consumption, investment, government spending and net exports (i.e. exports minus imports), and is inversely related to the aggregate price level through the wealth, interest rate and international purchasing power effects.

AGGREGATE SUPPLY (AS)

Aggregate supply (AS) is the total value of goods and services that all the firms in the economy would and can willingly produce in a given time period. Aggregate supply is a function of available inputs, technology and the price level. It slopes upward in P-Output space but the exact slope depends whether the economy is operating at below full employment (flat) or full employment (steep).

CLASSICAL ECONOMICS

Classical economics is widely regarded as the first modern school of economic thought. Its major developers include Adam Smith, David Ricardo, Thomas Malthus and John Stuart Mill. Sometimes the definition of classical economics is expanded to include William Petty, Johann Heinrich von Thünen, and Karl Marx. Classical economists were the earliest brand of economists the world knew. They were essentially micro-economists who believed the macro economy was an uninteresting aggregation of individual (or micro) markets, and any problem at the macro level was necessarily a symptom of some micro level problem.

OPTIMAL ROLE OF GOVERNMENT UNDER CLASSICAL ECONOMICS

The optimal role for the government under classical economics was one of laissez-faire. They believed that if the prices of goods, services and factors were allowed to be determined by the free operation of the forces of demand and supply (i.e. the price mechanism) the best possible outcome for resource allocation would obtain. In other words the economy would be at the full employment level, and it would not be possible to improve that situation through government intervention.
Before 1930, there was no concept of macroeconomics. There were number of events happened during 1920-30s that necessitated the need of macroeconomics. Until the 1930s, most economic analysis did not separate out individual behavior from aggregate behavior. With the Great Depression of the 1930s and the development of the concept of national income and product statistics, the field of macroeconomics began to expand.

THE CONCEPT OF INVISIBLE HAND

Invisible hand was a concept introduced by Adam Smith in 1776 to describe the paradox of laissez-faire market economy. The invisible hand doctrine holds that, with each participant pursuing his or her own private interest, a market system nevertheless works to the benefits of all as though a benevolent invisible hand were directing the whole process. According to Adam smith, invisible hand of the market operates therefore the market mechanism is the best model if the economy wants to operate at a high level of efficiency. Since the classical economists believed on perfectly competitive markets, so according to them shortages & surpluses are temporary phenomena when markets decide about the price, market would automatically clear these shortages & surpluses through price mechanism. According to say's law, "supply creates its own demand". When there is surplus labor in the economy that situation could not persist so long because the excess supply of labor push the prices of labor go down, wage rate goes down, firms will demand more labor due to lower wage rates, so in this way, supply of labor creates its own demand of labor.

Invisible hand is the term used by Adam Smith to describe the natural force that guides free market capitalism through competition for scarce resources. According to Adam Smith, in a free market each participant will try to maximize self-interest, and the interaction of market participants, leading to exchange of goods and services, enables each participant to be better of than when simply producing for him/her. He further said that in a free market, no regulation of any type would be needed to ensure that the mutually beneficial exchange of goods and services took place, since this "invisible hand" would guide market participants to trade in the most mutually beneficial manner.

FULL EMPLOYMENT

The Classical economists assumed that if the economy was left to itself, then it would tend to full employment equilibrium. This would happen if the labour market worked properly.

Full employment is a state of the economy in which the productive resources of the economy are fully employed. Output may be expanded from this full employment level by asking laborers to work overtime or renting capital from outside. An alternative (historical) definition of full employment was: that level of employment at which no (or minimal) involuntary unemployment exists.

An important law the Classical subscribed to, which assumed particular importance in the context of the Great Depression, was Say's law: "supply creates its own demand." The implication of this was that involuntary unemployment (people being unemployed against their wishes) was a temporary phenomenon as the excess supply of labour would cause wages to fall thereby prompting firms to demand more labour. If there was persistent unemployment, it was voluntary, i.e. workers themselves preferred to remain unemployed.

THE CLASSICAL VIEWS ABOUT GREAT DEPRESSION

The Classical' reading of the three problems of the Great Depression, i.e. low investment, high unemployment and low output, was as follows:

a. Investment was low because the interest rate was too high in the loanable funds market. Policy recommendation: savings be increased to lower the interest rate and boost investment
b. Unemployment was high because of obstructions to the free market mechanism in the labour market which were preventing wages from falling to the market clearing level. Policy recommendation: these obstructions: benefit payments to unemployed, taxes on income and trade unions be eliminated.

FAILURE OF THE CLASSICAL MODEL

After 1930, the classical model failed. Solution of classical economist was not found reasonable to solve the world crises prevailed at that time. The Great Depression was the longest and severest recession the world has ever seen. It struck North America and Europe in the late 1920s after the Wall Street crash of 1929 (and following the earlier hyperinflation in Germany, and the formation of the Soviet Union) and lasted till the mid 1930s. It was characterized by persistent high unemployment, low investment by firms and falling prices of goods, services and factors. Hyperinflation is inflation at extremely high rates (say 1000, 1 million, or even 1 billion percent a year).

At that time, unemployment rate went upto 25% in 1933 in USA and Western Europe. The classical model failed because they did not give satisfactory solution to all these problem.

They kept on insisting on old doctrine. They focused on the removal of impediments of free market economy.

KEYNES AND THE ORIGINS OF MODERN MACRO ECONOMICS

Keynesian economics also called Keynesianism or Keynesian Theory is an economic theory based on the ideas of the 20th-century British economist John Maynard Keynes. Keynesian economics promotes a mixed economy, in which both the state and the private sector are considered to play an important role. He argued that government policies could be used to promote demand at a macro level, to fight high unemployment and deflation of the sort seen during the 1930s. Keynes gave the reasons of great depression and also suggested the policy advice on how Govt can rectify the situation.

THE KEYNESIANS' VIEWS ABOUT GREAT DEPRESSION

Keynes' view on the causes of the Great Depression and what needed to be done was very different. He believed that there were overarching problems of low demand and static pessimistic expectations that needed to be addressed rather than disequilibria in the loanable funds, labour and goods markets. In particular, he maintained that:

- **a.** Low investment was because of firms' bearish expectations about their ability to sell the products they produced. Firms needed to see that the potential buyers of their goods had the money and the willingness to buy goods before they could be convinced to undertake more production thereof. Therefore, higher savings, which would lead to low consumption expenditure on goods and services would not increase but decrease investment by reinforcing firms' bearish expectations about their ability to sell their products. Policy recommendation: households should be convinced to increase consumption and reduce saving.
- **b.** Unemployment was high and rising because the labour market equilibrium was moving further and further away from the full employment level. This was not because wages were being prevented from falling to the market clearing level, but because the market clearing level fell further with each wage decrease. This happened because a reduction in wages also lowered consumers' earning and spending power reinforcing firms' pessimistic view of their ability to sell their products. Policy recommendation: higher money payments to consumers should be given out (possibly by the state) in order to increase their ability to buy the goods being produced by firms.

Lesson 28

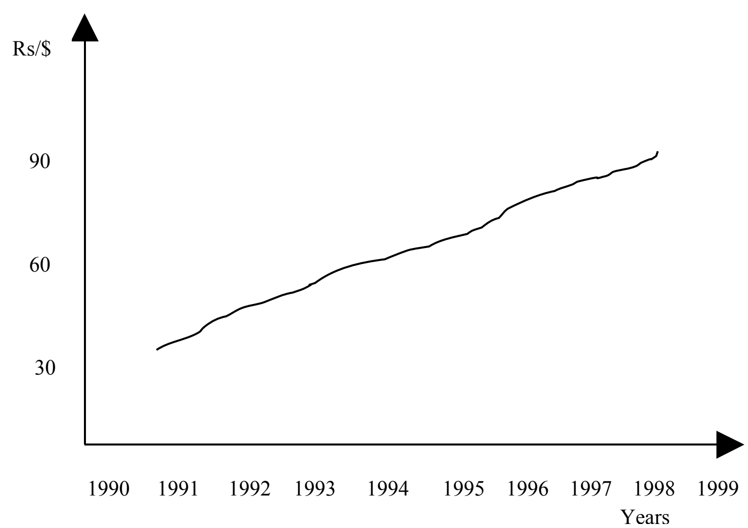
MACROECONOMIC DATA & NATIONAL INCOME ACCOUNTING

THE USE OF MACROECONOMIC DATA

As said: "there are lies, damned lies and statistics." Likewise, macroeconomic statistics are also susceptible of both manipulation and misinterpretation. In order to ensure that you understand what a particular number or data representation really means, the following need to be considered:

- **i.** Data might be used selectively. Certain information might be excluded: e.g. inflation as a whole might have increased but food inflation might have fallen.
- **ii.** In graphs, the vertical and horizontal scales used might be such in a way so as to paint a very dramatic (or totally benign) picture of things.
- **iii.** Values used might be absolute, not proportionate. People might be paying higher taxes, but as a proportion of income, the same may have fallen, as incomes might have risen even higher.
- **iv.** Questions of distribution might be ignored. For e.g., while the economy might have become richer overall, the ownership of the higher wealth might be highly skewed so that the richer have become richer and poor poorer.
- **v.** Data might be nominal or real. Nominal data is recorded in money terms, unadjusted for inflation. Real data is nominal data adjusted for changes in prices. Most macroeconomic data is presented and analyzed in real terms so as to permit meaningful intertemporal and cross-country comparisons.
- **vi.** Certain time periods might be excluded: e.g. economic growth might be 4.5% over the 1990-95 timeframe, but only 3.5% over the 1988-97 horizon.
- **vii.** Data might be aggregate, ignoring per capita considerations: e.g. a country's national income goes up but per capital income goes down due to a bigger population.
- **viii.** Vii also applies in the context of growth rates. So if national income is growing at 5% p.a. but the population at 6% p.a., per capital income would be falling at the rate of 1% p.a.

Graphical presentation of data (use of scale) Exchange rate



5 people in the economy	**Income in (000)s**				
	A	**B**	**C**	**D**	**E**
Old government	100	110	120	130	140
New government	60	70	75	300	450

Mean (old government) = $\frac{610}{5}$ = 122 Median (old government) = 120

Mean (new government) = $\frac{955}{5}$ = 191 Median (new government) = 75

Nominal data is the data which is expressed in monetary terms while real data is the data after adjusting nominal data for inflation.

Year	**1979**	**1980**	**1981**	**1982**	**1983**	**1984**	**1985**	**1986**
Real GDP Growth Rate	1%	-5%	1%	3%	4%	5%	2%	1%

Between 1982 & 1985, average growth rate = 3.5%
Between 1979 & 1986, average growth rate = 1.5%

NATIONAL INCOME ACCOUNTING

Gross Domestic Product (GDP):

Gross domestic product (GDP) is the value of the total final output produced inside a country, during a given year. GDP, like all measures of national income, is a flow (as opposed to stock) figure accruing over the period of one year. Gross Domestic Product is the total market value of all goods and services produced within the political boundaries of an economy during a given period of time, usually one year. This is the government's official measure of how much output our economy produces.

Concept of Flow and Stock:

A flow figure refers to a certain period of time. A stock figure implies a particular point in time and therefore changes instantaneously. Flows accumulate into stocks. Changes in stocks equal flows. In accounting terminology, stocks are balance sheet items, while flows are income statement items.

The change in stocks measures changes in the value of unsold inventories in a given time period. When aggregate demand is high and running ahead of current production, the value of stocks held by businesses tends to fall. This is known as de-stocking. Conversely when demand falls, business might be left with an unplanned increase in unsold output leading to a rise in stocks. Changes in stocks are normally a good leading indicator of where the economy is likely to head in the next six months.

Stock: A variable or measurement that is defined for an instant in time (as opposed to a period of time). A stock can only be measured at a specific point in time. For example, money is the stock of production that exists right now. Other important stock measures are population, employment, capital, and business inventories.

Flow: A variable or measurement that is defined for a period of time (as opposed to an instant in time). A flow can only be measured over a period. For example, GDP is the flow of production during a given year. Income is another flow measures important to the study of economics.

METHODS OF MEASURING GDP

There are three equivalent ways of measuring GDP:

i. **The product or value added method** which sums the value added by all the productive entities in the economy;

ii. **The expenditure method** which sums up the value of all the “final goods” transactions taking place in the economy;

iii. **The factor income method** which sums up all the incomes earned by all the factors of production in the economy (rent for land, wages for labour, interest for capital, and equity returns for entrepreneurship).

The three methods are equivalent. One way to see why this must be so is because in an ex-post sense, aggregate supply (i) = aggregate demand (ii) = national income (iii).

Value added is the difference between the value of goods produced and the cost of materials and supplies used in producing them. Value added consists of the wages, interest and profit components added to the output by a firm. Value added is the difference between inputs and outputs e-g if a firm spends Rs.500 making a good (inputs) and sells its product (output) for Rs.750, then value added is Rs.250.

OR

Value added is the increase in the value of a good at each stage of the production process. The value that's being increased is specifically the ability of a good to satisfy wants and needs either directly as consumption good or indirectly as a capital good. A good that provides greater satisfaction has greater value. In essence, the whole purpose of production is to transform raw materials and natural resources that have relatively little value into goods and services that have greater value.

Final and Intermediate Goods:

Final goods are meant for direct use by the end consumer rather than for further processing. Intermediate goods are those that are intended for further processing. So an iron rod, if purchased by a household as a weapon against infiltrating thieves would categorize as a final good, but if purchased by a firm for use in the making of an automobile would categorize as an intermediate good.

GDP might be calculated at market prices (includes sales tax paid by consumer as part of the final price) or at factor cost (excludes sales tax). If there is no sales tax, the two measures collapse to the same thing.

Final good: A good (or service) that is available for purchase by the ultimate or intended user with no plans for further physical transformation or as an input in the production of other goods that will be resold. Gross domestic product seeks to measure the market value of final goods. Final goods are purchased through product markets by the four basic macroeconomic sectors (household, business, government, and foreign) as consumption expenditures, investment expenditures, government purchases, and exports. Final goods, which are closely related to the term current production, should be contrasted with intermediate goods--goods (and services) that will be further processed before reaching their ultimate user.

Intermediate good: A good (or service) that is used as an input or component in the production of another good. Intermediate goods are combined into the production of finished products, or what are termed final goods. Intermediate goods will be further processed before sold as final goods. Because gross domestic product seeks to measure the market value of final goods, and because the value of intermediate goods are included in the value of final goods, market transactions that capture the value of intermediate goods are not included separately in gross domestic product. To do so would create the problem of double counting.

Difference between total value and value added:

<u>**Firm A:**</u> Produces steel from raw iron and sells it for Rs 100,000 to firm B.

<u>**Firm B:**</u> Buys steel worth Rs 100,000 then processes it to produce a car body worth Rs 200,000.

<u>**Firm C:**</u> Buys the car body, adds the other parts etc. and sells complete car for Rs 450,000.

Value added or GDP:

Value of transactions: 100,000 + 200,000 + 450,000 = 750,000

GDP = Sum of the value added by each of the firms:

= 100,000 + (200,000 – 100,000) + (450,000 – 200,000)

= 100,000 + 100,000 + 250,000 = 450,000→ GDP.

Also total expenditure of consumer on car = 450,000

Example:

Good Col 1	Seller Col 2	Buyer Col 3	Trans Value Col 4
Steel	Steel producer	Machine producer	100,000
Steel	Steel producer	Car producer	300,000
Machine	Machine producer	Car producer	200,000
Tyres	Tyre producer	Car producer	50,000
Cars	Car producer	Consumer	500,000
Total value of transaction	-	-	1,150,000

Lesson 29

MACROECONOMIC DATA & NATIONAL INCOME ACCOUNTING (CONTINUED)

GDP at factor and market prices:
Factor price is the price at which firm sells its final output to the consumers. While market price includes factor price plus the indirect taxes imposed by the government. GDP at market price is higher than the GDP at factor price.

GDP at factor cost = GDP at market price – Indirect taxes

Net Domestic Product (NDP):
Net domestic product (NDP) is obtained by subtracting depreciation from GDP. Depreciation is the reduction in the value of a capital good due to the wear and tear caused during production. The total market value of all final goods and services produced within the political boundaries of an economy during a given period of time, usually a year, after adjusting for the depreciation of capital.

NDP = GDP – Depreciation allowance

Depreciation:
The wearing out, breaking down, or technological obsolescence of physical capital that results from use in the production of goods and services. To paraphrase an old saying, "You can't make a car without breaking a few socket wrenches." In other words, when capital is used over and over again to produce goods and services, it wears down from such use.

Gross National Product (GNP):
Gross national product (GNP) is the value, at current market prices, of all final goods and services produced during a year by the factors owned by the citizens of a country. Thus the income earned by Pakistani citizens working in the US would be included in Pakistan's GNP but excluded from Pakistan's GDP. Conversely, the income earned by a US citizen (individual or corporate) in Pakistan would be included in Pakistan's GDP but excluded from Pakistan's GNP. Generally, GNP = GDP + net factor income from abroad. The abbreviation for gross national product, which is the total market value of all goods and services produced by the citizens of an economy during a given period of time, usually one year. Gross national product often was once the federal government's official measure of how much output our economy produces.

GNP = GDP + Net factor incomes from abroad

Net National Product (NNP):
Mathematically, national income is net national product (NNP). It is GNP adjusted for depreciation. In words, it is the net output of commodities and services flowing during the year from the country's production system in the hands of ultimate consumers. The total market value of all final goods and services produced by citizens of an economy during a given period of time, usually a year, after adjusting for the depreciation of capital. Net national product, abbreviated NNP, has the same relation to net domestic product (NDP) as gross national product (GNP) has to gross domestic product (GDP). Net national product also has the same relation to gross national product that net domestic product has to gross domestic product. Like NDP, NNP is a measure of the net production in the economy.

NNP = GNP – Depreciation allowance

National income (NI):
NNP is often referred to as national income. The total income earned by the citizens of the national economy as a result of their ownership of resources used in the production of final goods and services during a given period of time, usually one year. This is the government's official measure of how much income is generated by the economy.

National Income and Gross Domestic Product:
National income (NI) is the total income earned by the citizens of the national economy resulting from their ownership of resources used in the production of final goods and services during a given period of time, usually one year. Gross domestic product (GDP) is the total market value of all final goods and services produced within the political boundaries of an

economy during a given period of time, usually a year. Although national income is generated by the production of gross domestic product, the value of production does not entirely result in earned income. In other words, national income can be derived from gross domestic product after a few adjustments.

Real GDP: The total market value, measured in constant prices, of all goods and services produced within the political boundaries of an economy during a given period of time, usually one year. The key is that real gross domestic product is measured in constant prices, the prices for a specific base year. Real gross domestic product, also termed constant gross domestic product, adjusts gross domestic product for inflation. You might want to compare real gross domestic product with the related term nominal GDP.

Nominal GDP: The total market value, measured in current prices, of all goods and services produced within the political boundaries of an economy during a given period of time, usually one year. The key is that nominal gross domestic product is measured in current, or actual prices; the prices buyers actually pay for goods and services purchased. Nominal gross domestic product is also termed current gross domestic product.

Real flow includes services of land labor and capital going from households to firms, and products of firms as physical goods of services flowing to households. Real GDP therefore excludes the effect of prices and focuses entirely on the volume (or quantity) of goods and services produced.

Money (or nominal) flow includes the payments firms make to households for factor services and also it includes the household spending money to buy goods from firms. Nominal GDP would therefore include the effect of changes in the price level, as it is a measure of the money value of goods and services produced.

PRICE DEFLATOR

The process of converting nominal GDP into real GDP is known as deflation. A price index calculated as the ratio nominal gross domestic product to real gross domestic product. Also commonly referred to as the implicit price deflator, the GDP price deflator is used as an indicator of the economy's average price level.

$$\text{GDP Deflator} = \text{Nominal GDP} / \text{Real GDP}$$

It is the price deflator (see price\ratio expression in brackets below) which enables us to move from nominal to real GDP. It provides a measure of the change in prices from the base (or benchmark) year to year ‘a’, given values for some aggregate price index for the two years:

$$\text{Real GDP}_{\text{year a}} = \text{Nominal GDP}_{\text{year a}} \text{ X } (\text{Price Index}_{\text{base year}} / \text{Price Index}_{\text{year a}})$$

Using a similar formula and the same base year, Real GDP year b can be calculated and then be compared with Real GDP year a to get an idea of real GDP growth over the ‘a’ to ‘b’ period.

Example:

- Nominal GDP was $150 billion in 1985
  & $300 billion in 1994
- ASSUME that prices have risen by 50% over the period.
- Real GDP in 1994 measured in 1985 prices:
  = $300 billion x 100/150 = $200billion

Hypothetical economy

	Year 1	**Year 2**
Apples produced	100	150
Chicken produced	100	140
Cost per apple Rs	2	4
Cost per chicken Rs	4	6

- Nominal GDP in year1 = (100 x 2) + (100 x 4) = 600
- Nominal GDP in year2 = (150 x 4) + (140 x 6) = 1440

- Growth rate in nominal GDP = $\frac{1440 - 600}{600}$ = 140%
- Year2 price index at year 1 prices = $\frac{150 \times 2 + 140 \times 4}{290}$ = 2.966
- Year2 price index at year 2 prices = $\frac{150 \times 4 + 140 \times 6}{290}$ = 4.966

→ Current year price index

GDP Deflator:

2.966X = 4.966 x100

X = 167.4%

This is the price level in percentage terms prevailing in the current year (Year2) relative to the price level of Year1.

Real GDP = $\frac{1440 \times 100}{167.4}$ = 860

Growth rate = $\frac{860 - 600}{600}$ = 43%

Per Capita GDP:

Per capita GDP is simply the total GDP of the economy divided by the no. of people in the economy. The GDP of China might be bigger than the GDP of Switzerland but in average per capita terms, Switzerland's income might be several times that of China's; the figure given in the lectures was 160.

THE PURCHASING PARITY (PPP)

The purchasing power parity (PPP) measure of GDP recognizes the fact that a given amount of income in one country might not be able to purchase the same quantity of goods and services in another country. So, for e.g., if China' per capita income is 1/160th of Switzerland's per capital income, it might be that goods and services in China are much cheaper and therefore China's per capita income does not need to grow 160 times in order to deliver the same standard of living as in Switzerland. The PPP GDP per capita is therefore a more sensible measure to use for comparison across countries at different levels of development. This is indeed the reason why many international development organizations prefer this over simple GDP per capital.

Per Capita Income, Personal Income and Disposable Income:

Per capita income is obtained by dividing the national income by the total number of population. It is the average annual income per head for all the inhabitants of the country; it is used to represent the standard of living of the people.

Personal income of an individual is the total amount of income s/he receives form deploying all the different factors of production s/he owns. Aggregate personal income is just the above definition aggregated for the whole of the economy. The total income received by the members of the domestic household sector, which may or may not be earned from productive activities during a given period of time, usually one year. The primary use of personal income is to measure the income actually paid out to the household sector. After adjusting for income taxes, personal income forms the basis for consumption expenditures on gross domestic product.

Disposable income is obtained by subtracting the amount of direct taxes from the personal income of the person. Aggregate holds as above as well. The total income that can be used by the household sector for either consumption or saving during a given period of time, usually one year. This is the income left over after income taxes and social security taxes are removed and government transfer payments, like welfare, social security benefits, or unemployment compensation are added.

DRAWBACK OF GDP BASED MEASURES

There are many caveats with a GDP based measure of national income.

- **i.** GDP by definition excludes productive activities in the informal economy. Thus, activities such as a person painting a wall in his own house, or a woman cooking food in her house, would be excluded by a GDP-based measure. In countries where a large part of economic activity goes unreported and undocumented (like in lower income countries), the GDP might seriously understate the level of national income and production.
- **ii.** GDP cannot include the black or illegal economy. So, for example, if a banned good is produced illegally and exported outside the country, and foreign money is received in exchange for it, then that “export revenue” will not be included as part of the GDP. Lower income countries often confront a large black economy which cannot be documented. Living standards in such countries are therefore often higher than what a per capital income measure based on GDP would suggest.
- **iii.** A GDP-based measure of welfare or living standards also needs to be corrected for externalities. For e.g., if a country’s GDP is growing at a very fast rate but this is at the cost of rising environmental pollution (which might cause serious future health hazards) or non-renewable natural resource depletion, then a simple GDP measure of income will overstate the performance of the economy and ignore the serious long-term risks it faces.

Lesson 30

MACROECONOMIC EQUILIBRIUM; THE DETERMINATION OF EQUILIBRIUM INCOME

THE KEY VARIABLES OF THE MACROECONOMIC MODEL

The circular flow of money in the economy helps illustrate the Classical and Keynesian notions of macroeconomic equilibrium. The circular flow depicts incomes flowing from firms to households in return for factor services supplied by households to firms, and subsequently these household incomes being expended on goods and services supplied by firms to households.

The key variables for a macroeconomic model are:

Y = Income
C = Consumption
S = Savings
I = Investment
T = Taxes
G = Government Expenditures
M = Imports
X = Exports

CIRCULAR FLOW

The continuous movement of production, income, and resources between producers and consumers. This flow moves through product markets as the gross domestic product of our economy and is then the revenue received by the business sector in payment for this production. This stream of revenue then flows through resource markets as payments by businesses for the resources employed in production. The payments received by resource owners, however, is nothing more than the income of the household sector. The resource owners of the household sector use this income to purchase goods and services through the product markets, coming full circle to where we began.

THE CONCEPT OF LEAKAGES AND INJECTIONS

A leakage or withdrawal is any use of the income received by households that does not return as revenue to domestic firms. Savings, taxes and imports are examples of leakages as this money does not fall as expenditure on goods and firms produced by domestic firms.

Injections are payments to firms not originating from households: government spending, firms' investment and exports are all examples of injections into the circular flow.

Injection:

A non-consumption expenditure on gross domestic product, including investment expenditures, government purchases, and exports. Injections are combined with leakages in the injection-leakage model used to identify equilibrium aggregate output in Keynesian economics. The notion of injection is best viewed through the circular flow, in which investment expenditures, government purchases, and exports are "injected" into the main flow between output, factor payments, national income, and consumption.

Leakage/Withdrawals:

A non-consumption uses of income, including saving, taxes, and imports. Leakages are combined with injections in the injection-leakage model used to identify equilibrium aggregate output in Keynesian economics. The notion of leakage is best viewed through the circular flow, in which saving, taxes, and imports are "leaked" out of the main flow between output, factor payments, national income, and consumption.

Injection-leakage model:

A model used in Keynesian economics based on the equality of non-consumption expenditures (or injections) and non-consumption uses of income (leakages). On one side of the equality is saving, taxes, and imports -- the non-consumption leakages. On the other side of the equality is investment, government purchases, and exports -- the non-consumption injections. The injection-leakage model provides an alternative to the Keynesian cross for identifying equilibrium aggregate output.

MACROECONOMIC EQUILIBRIUM: CLASSICAL VIEW

Macroeconomic equilibrium in a Classical sense refers to joint equilibrium in all the underlying sectors or markets of the economy. So S must equal I (loan able funds market; key players are banks and financial markets), G must equal T (fiscal sector; key player is government) and X = M (external sector, key players are importers and exporters). Any disequilibrium at the macro level was attributable to disequilibrium in one or more of these individual markets.

MACROECONOMIC EQUILIBRIUM: KEYNESIAN VIEW

Macroeconomic equilibrium in a Keynesian sense obtains when total injections equal total leakages (or total withdrawals), or aggregate supply equals aggregate demand. These are two equivalent notions of Keynesian equilibrium and can be expressed respectively as:

$$\mathbf{S + T + M \equiv I + G + X} \text{ and}$$
$$\mathbf{AS = Y = AD \equiv C + I + G + (X\text{-}M);}$$

where AS is aggregate supply, Y is national income, AD is aggregate demand, C is consumption, I is investment, G is government spending, X is exports, M is imports, S is saving and T is taxes.

Withdrawal = Injection

By definition → $\underline{S + T + M} = \underline{I + G + X}$

(Adding C to both sides)

$C + S + T + M = C + I + G + X$ (Now taking M on other side)

$C + S + T = C + I + G + X – M$

$M = Cf + If + Gf$

$Cd = C – Cf$ $Id = I – If$ $Gd = G – Gf$

R.H.S = $(C – Cf) + (I – If) + (G – Gf) + X$

= $Cd + Id + Gd + X$ => Total expenditure on domestic goods

$Cd + Id + Gd + X = C + I + G + X – M$

Total expenditure on domestic goods = AD

For Equilibrium

AD = AS

C + S + T = C + I + G + X – M

In L.H.S

- $C + S + T = Y$ (total income = Y)
- Therefore another way of viewing Keynesian equilibrium is:

Y = AD = AS

Income = Expenditure = Output

Keynes' major insight was that equilibrium in the individual markets was not a necessary condition for equilibrium at the macro level. Indeed it was possible for all the individual markets or sectors to be in disequilibrium but aggregate demand and supply to be equal, and therefore the overall economy to be in equilibrium. As such, he argued that in the face of macroeconomic equilibrium (situations like unemployment, high inflation etc.) policy needed to focus on aggregate demand and aggregate supply rather than individual markets.

To refresh your memories, aggregate demand is the total planned or desired spending in the economy during a given period. It is determined by the money supply, aggregate price level, consumption, domestic investment, government spending and taxes, net exports (i.e. exports minus imports). Aggregate supply is the total value of goods and services that firms would willingly produce in a given time period. Aggregate supply is a function of available inputs, technology and the price level.

Disposable income (Y_d) is that part of the total national income (Y) that is available to households for consumption or saving. So $Y_d = Y – T$.

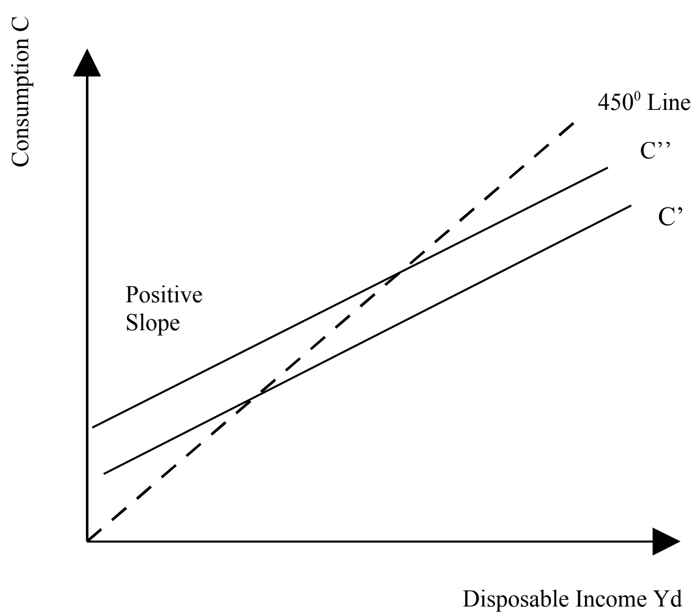
CONSUMPTION AND CONSUMPTION FUNCTION

Consumption (C) is the amount of national income that is spent on goods and services produced by domestic firms in a given period of time. Consumption is the most stable and important component of aggregate demand, accounting for about two-thirds to three-fourths of GDP in most countries.

The consumption function is a schedule relating total consumption to personal disposable income. It usually takes the form **$C = a + bY_d = a + b(Y\text{-}T)$**, where a is the minimum level of consumption that must take place even if Y_d is zero, and b is the marginal propensity to consume.

When drawn in expenditure-income space, the consumption function plots as a straight line with positive intercept, and a positive (but less than 1) slope. The slope is merely the MPC. The intercept is positive because some consumption must happen even at a zero level of income (people will borrow and spend on food for e.g.), and the slope is less than 1 because not all the income is consumed (part of it is saved).

Disposable Income (Yd)	Consumption (C`)	Saving (S = Yd – C`)
500	500	0
550	540	10
600	580	20
650	620	30
700	660	40
750	700	50
800	740	60



Marginal Propensity to Consume (MPC) and Marginal Propensity to Save (MPS):

Marginal propensity to consume (MPC) is the extra amount that people consume when they receive an extra dollar of disposable income. MPC's numerical value is usually between 0.5 and 1, but can vary considerably across different countries, population age groups, and stages of a person's life.

Marginal propensity to save (MPS) is the fraction of the additional dollar of disposable income that is saved. Thus, MPC = 1 – MPS. Average propensity to consume (APC) is the ratio of total consumption to total disposable income. Average propensity to save (APS) is the ratio of total saving to total disposable income. As before, APC = 1 – APS.

THE SAVING FUNCTION

The saving function yields the amount of saving that households of a nation will undertake at each level of income. A usual formula is $S = c + d(Y_d)$. d is MPS, positive, and usually less than 0.5.

The relationship between saving and the interest rate is also important. The relationship is positive, is plotted in i-S space, and implies that household saving increases as the interest rate goes up, i.e. the incentive to keep one's money in the bank and earn interest thereon increases as the return on that money increases.

CALCULATION OF APC, APS & MPC, MPS

- **Average propensity to consume:**
  APC = C / Yd
- **Average propensity to save:**
  APS = S / Yd
  Or
  APS = 1 – APC
- **When** **Yd** = \$500 billion **APC** = 1 **APS** = 0
- **When** **Yd** > \$500 billion **APC** < 1 **APS** > 0
- **Marginal propensity to consume:**
  MPC = ∆C / ∆Yd
- **Marginal propensity to save:**
  MPS = ∆S / ∆Yd
  Or
  MPS = 1 – MPC

APC = C/Yd	APS	Yd	C	MPC(∆C/∆Yd)
500/500=1.0	0	500	500	
540/500=0.98	0.02	550	540	40/50=0.8
580/600=0.97	0.03	600	580	40/50=0.8
620/650=0.95	0.05	650	620	40/50=0.8
660/700=0.94	0.06	700	660	40/50=0.8
700/750=0.93	0.07	750	700	40/50=0.8
740/800=0.92	0.08	800	740	40/50=0.8

Lesson 31

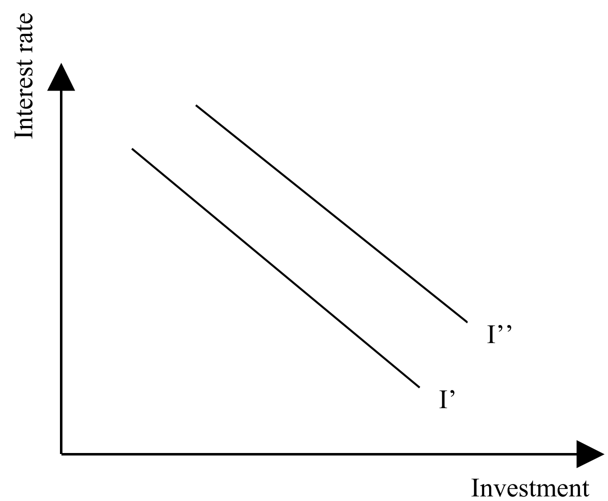
MACROECONOMIC EQUILIBRIUM; THE DETERMINATION OF EQUILIBRIUM INCOME (CONTINUED)

INVESTMENT AND INVESTMENT DEMAND CURVE

Investment (I) or gross capital formation is any economic activity (usually undertaken by firms) that forgoes consumption today with an eye to increase output in future. Investment is by far the most volatile component of aggregate demand.

The investment demand curve shows the relationship between the level of investment and the cost of borrowing for the firm (i.e. the interest rate), plotted in i-I space. The cost of borrowing is important because most investments are financed using borrowed resources (e.g. loans from banks). The relationship between the interest rate on such borrowing and investment demand is obviously negative, i.e. as the interest rate goes up, investment demand decreases.

Investment demand curve



TYPES OF INVESTMENT

Investment can be of various types: residential and non-residential construction, purchases of producer durables (i.e., capital equipment, machinery etc.) and buildup of business inventories. While all these different types are affected t some extent by the interest rate, there are other important determinants as well.

- **i.** Residential construction depends upon the number of willing house-buying households, their wealth and indebtedness levels, their ability to obtain a house-building loan from financial institutions and the cost of housing units.
- **ii.** Non-residential construction depends upon the willingness and ability of firms to buy commercial property, the vacancy rate of existing units, the needs of business units for additional commercial space, and firms' ability to meet increased rental costs which are directly linked to their current and expected costs and sales.
- **iii.** The demand for producers' durable purchases depends on utilization of existing productive capacity, the availability of advanced (more efficient) technology, current and expected sales and existing and future competition.
- **iv.** Changes in business inventories depend on current and expected sales, current and expected inventory prices, and certainty of inventory deliveries.

IMPORTS, EXPORTS AND TRADE BALANCE

- Imports are goods and services that are produced in another country and consumed in the home country. Thus a refrigerator produced in Korea brought into Pakistan to be sold here locally would characterize as an import.
- Exports are goods and services that are produced in the home country and consumed in another country. Thus a communications satellite produced in Pakistan but sold to neighboring Iran would categories as a Pakistani export.
- A country's imports are related to its level of income, exchange rate, domestic prices relative to prices in foreign countries, import tariffs (taxes and customs duties levied on imported goods), and quantitative restrictions (quotas) on imported goods. Imports are influenced by the same variables except that they are affected by foreign, not home, country income levels.
- Trade balance is the excess of exports over imports. A negative trade balance is called a trade deficit. Because the determinants of a country's exports and imports change with time, it is reasonable to expect a country's trade balance to change over time.
- Fiscal Policy is a government program with respect to i) expenditure (G): the purchase of goods and services and spending in the form of subsidies, unemployment benefits etc. and ii) tax revenue (T): the amount and type of taxes.
- T-G is referred to as the fiscal balance. If G>T, there is a fiscal or budget deficit; if G<T, there is a fiscal or budget surplus. If G=T, there is a balanced budget.

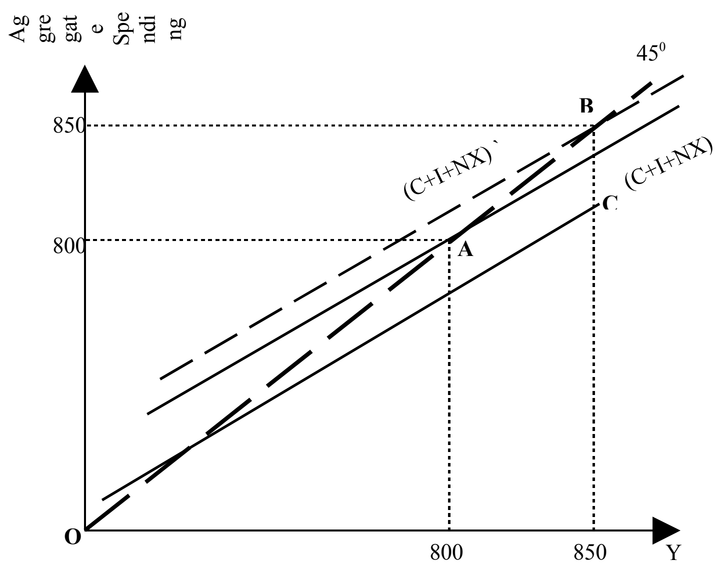
DETERMINATION OF NATIONAL INCOME

The 45^0 Line approach to equilibrium:

The 45^0 line drawn in Y-X space has the feature that at any point on the line, the horizontal and the vertical distances are the same. Thus, if the units used to measure X and Y are the same (i.e. same scales), the values of the two variables at any point along the line are equal.

The 45^0 line drawn in expenditure-income (or AD-Y) space captures macroeconomic equilibrium in the economy (recall, macroeconomic equilibrium obtains when AS = AD = Y). At all points along this line, expenditure and income are equal.

Superimposing the aggregate demand (AD) line or expenditure function on the 45^0 line diagram helps get the level at which a particular economy's equilibrium is struck; i.e. the point at which the AD line intersects the 45^0 line.



The AD or expenditure function is given by **AD = C + I + G + (X-M)**. We saw earlier that the C function was upward sloping with positive intercept but slope between 0 and 1 (i.e. less steep than the 45^0 line). Adding the I, G and X-M functions to the C function simply involves

moving the C line vertically upwards (parallel shift). So C+I will be higher than C by the amount of I; C+I+G will be higher than C+I by the amount of G; and C+I+G + (X-M) will be higher or lower than C+I+G depending on whether the country is running a trade surplus or deficit, respectively.
Starting from a certain equilibrium level, any increase in G, I and (X-M) will cause a multiplied increase in income. Thus depending on the slope of the AD line, it is possible for a $10mn increase in G (shown by an upward vertical shift of the AD line) to lead to a $50mn increase in equilibrium income and expenditure.
In the above example, if the economy had started from the full-employment equilibrium, then the $10mn increase in G would lead to an inflationary gap. An inflationary gap refers to a situation where there is pressure on prices to rise. The size of the inflationary gap is $10mn, i.e. the amount by which the C+I+G+NX line must shift down to bring equilibrium income back to the full employment level. Likewise, a deflationary gap could result if, starting from the full employment level, there was a reduction in G.

In Billions of $					
Output (Y) Col 1	**Consumer spending (C) Col 2**	**Investment spending (I) Col 3**	**Net Exports (NX) Col 4**	**Aggregate spending (C+I+NX) Col 5**	**Surplus / Shortage (C+I+NX – Y) Col 6**
650	570	100	10	680	30
700	610	100	10	720	20
750	650	100	10	760	10
800	690	100	10	800	0
850	730	100	10	840	-10
900	770	100	10	880	-20
950	810	100	10	920	-30

ALGEBRAIC DETERMINATION OF EQUILIBRIUM
Algebraic determination of equilibrium can be done by inserting the plugging the consumption function in place of C in the equation

$$\mathbf{AD = C + I + G + (X\text{-}M)}$$

In the absence of taxes, a consumption function simply collapses to C = a + bY
In equilibrium,

$$\mathbf{Y = AD}$$

Therefore,

$$\mathbf{Y^* = a + bY^* + G + I + (X\text{-}M)}$$

This leads to:

$$\mathbf{Y^* = [1/(1\text{-}b)]\ .\ [\text{“}a\text{”} + I + G + X\text{-}M]}$$

a is the autonomous part of consumption, i.e. the level of consumption that is independent of income.
Equilibrium analysis can also be done using the injections-leakages approach, i.e. by identifying the point where the upward sloping leakage function (S+M+T) intersects the horizontal injections line (I+G+X).
Example:
Consumption function
C = A + b Y
Where
A = Level of consumption
b = MPC = ΔY / ΔC
Y = Income
Consumption function can be formed by:

650 – 610 = C – 610
750 – 700 = Y – 700
C = 0.8Y + 50
Y = C + I + NX
Putting C = 0.8Y + 50
Y = 0.8Y + 50 + 100 + 10
0.2Y = 160

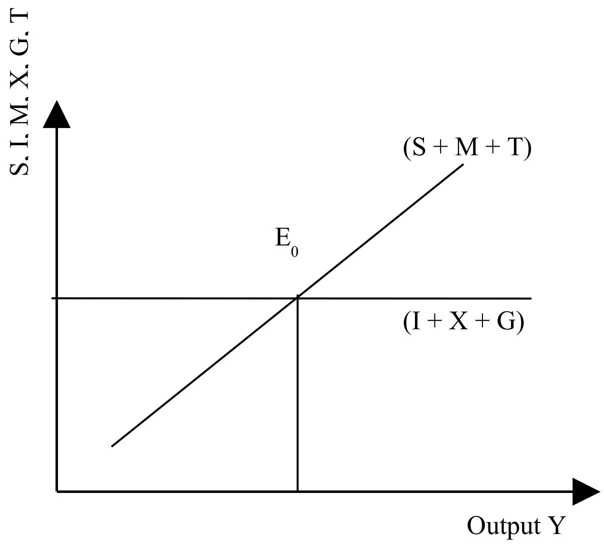
$$Y = \frac{160}{0.2} = 800$$

THE WITHDRAWALS - INJECTIONS APPROACH TO EQUILIBRIUM

W = J
S + T + M = I + G + X

Output (Y) Col 1	Saving (S) Col 2	Gross Imports (M) Col 3	Gross investment (I) Col 4	Gross exports (X) Col 5
650	80	80	100	90
700	90	80	100	90
750	100	80	100	90
800	110	80	100	90
850	120	80	100	90
900	130	80	100	90
950	140	80	100	90

G Col 6	T Col 7	Withdrawals Leakages (S+M) + T Col 8	Injections (I+X) + G Col 9	Difference (Leakages - Injections) Col 10
10	10	170	200	-30
10	10	180	200	-20
10	10	190	200	-10
10	10	200	200	0
10	10	210	200	10
10	10	220	200	20
10	10	230	200	30



How responsive this equilibrium level of income or output to changes in AD or Aggregate expenditures: The Keynesian Multiplier:
Reverting to the 45^0 line approach, the term **[1/(1-b)]** is called the Keynesian multiplier ("k") and is the factor by which equilibrium output, income or expenditure increase in response to an increase in AD (caused by an increase in "a", G, I or X-M). The higher is b, the bigger is the multiplier.
Mathematical representation of Keynes multiplier is as follows:

$$Y = C + I + G + NX$$

As $C = a + bY$

Then,

$$Y = a + bY + I + G + NX$$
$$Y - bY = a + I + G + NX$$
$$Y = \frac{a + I + G + NX}{1 - b} = \frac{1}{1 - b}(a + I + G + NX)$$
$$\mathbf{k = \frac{1}{1 - b}}$$

If any item in the numerator goes up by 10, Y will go up by 10 / 1 – b
If b = 0.8 $\Delta G = 10$

$$k = \frac{1}{1 - 0.8}$$

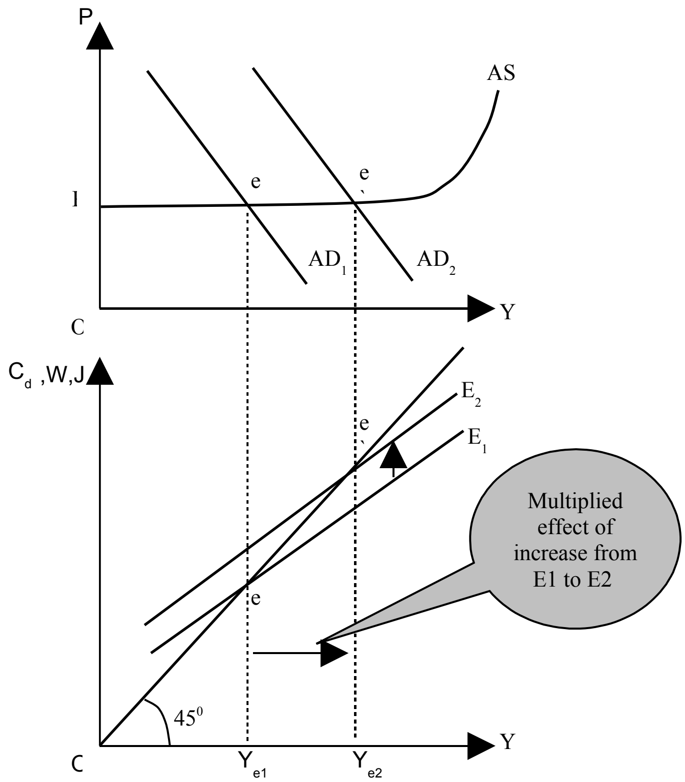
$\Delta Y = \Delta G \times b = 50$

- The higher the value of b, the bigger will be the size of the multiplier,
- The smaller the value of b, the lower will be the size of the multiplier.

Lesson 32

MACROECONOMIC EQUILIBRIUM; THE DETERMINATION OF EQUILIBRIUM INCOME (CONTINUED)

KEYNESIAN AD & AS APPROACH TO EQUILIBRIUM WITH THE 450 LINE APPROACH

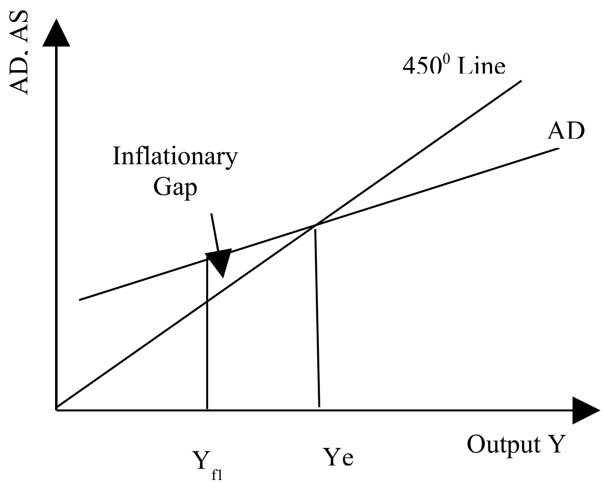


As the aggregate expenditures increases, AD also increases and output will also increase. Change in expenditures is less but change in income is higher due to the multiplier effect. This is the case of horizontal portion of AS where prices are constant.

KEYNES'S INTUITION ABOUT THE MULTIPLIER WAS AS FOLLOWS

An increase in AD caused by an injection into the circular flow, e.g. higher government spending on wages paid to government employees, would lead to higher money wages held by government servants. Higher wages would translate into higher consumption expenditure on goods and services in the economy, leading to higher money incomes of sellers of goods and services. When firms see consumers more prosperous, they are incentivised to produce more, thus their demand for labour goes up. This triggers a second rise of income increases in the hands of workers (who are also consumers) leading to a further multiplied effect on consumption, production and hiring. And so on. The multiplier effect would not be infinite as there are leakages (saving, taxes, and imports) from the circular flow of incomes each time the workers receive wages from firms. The lower the leakages and the higher the marginal propensity to consume, the higher will be the multiplier effect.

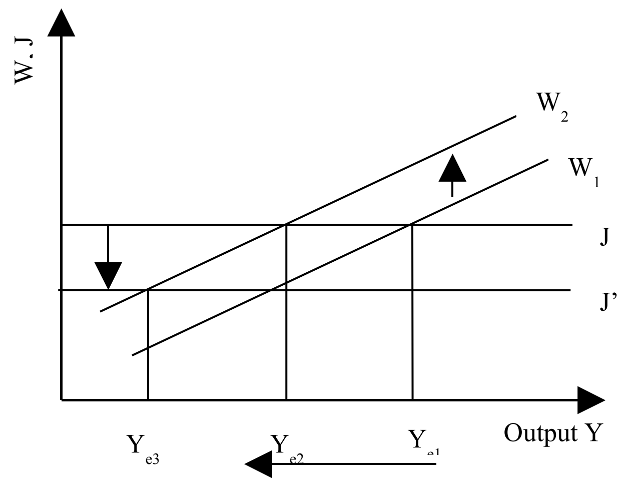
Inflationary Gap



KEYNES PARADOX OF THRIFT

The reverse multiplier effect can be illustrated in the context of Keynes's paradox of thrift, which highlights the negative impact of higher saving in an economy in recession. As noted earlier, Classical economists thought the solution to the problem of low investment during the Great Depression was high real interest rates caused by low savings. If the latter could be increased, the real interest rate would fall and investment would pick up. However, Keynes said that such thrift (or conservative saving behavior) would accentuate the recession. As people save more, they will spend less. Firms will therefore produce less, and labour hiring will, as a result, fall, leading to a decline in incomes. This decline would also happen in a multiplied fashion, causing a huge decline in national income. The paradox lies in the fact that that saving, while usually considered good for any one individual, can actually be harmful to the overall economy if everyone started saving.

Figure: How higher savings lead to lower national income



THE ACCELERATOR

The accelerator is a related concept which formalizes the investment response to output or income changes in an economy. The key observation here is that when an economy begins to recover from a slump, investment can rise very rapidly and, in percentage terms, the rise in investment may be several times the rise in income. Since investment is an injection into the circular flow of income, these changes in investment will cause multiplied changes in income and thus heighten a boom or deepen a recession. The formula for the accelerator is $\alpha = I/(\Delta Y)$, or $(\Delta K)/(\Delta Y)$, noting that $I = \Delta K$, where I is investment and K is capital (the stock of plants, buildings or machinery in the economy).

The reason why investment increases by much more than a change in income is as follows:

Suppose firms anticipate national income (and hence the demand for their products) to rise by 10% p.a. over the next 5 years. In response, firms will therefore normally look to undertake an investment (such as buying a machine) which will enable them to meet this new demand for the entire 5 year period. It is usually neither feasible nor possible to buy a machine that has a one year life! Thus, it is easy to see why a given annual change in output (10%) might prompt firms with a five year horizon to make an investment of over 50%.

Reverting back to the formula, the size of the accelerator, α, depends on the marginal capital to output ratio: $(\Delta K)/(\Delta Y)$. This is the cost of extra capital required to produce a Re.1 increase in national output. So if Rs.2 billion worth of capital is required to produce Rs.1 billion worth of output, then $(\Delta K)/(\Delta Y)$ is 2. It is easy to see that, other things being equal the marginal capital-output ratio and the accelerator are essentially the same. α is likely to be greater than 1.

INTERACTION OF ACCELERATOR AND MULTIPLIER

It is obvious that the interaction of the accelerator and multiplier can set off a chain reaction in the economy which can life output and income manifold. For e.g., if there is a rise in government expenditure, this will lead to a multiplied rise in national income. But this rise in national income will set off an accelerator effect: firms will respond to the rise in incomes (and the resulting rise in consumer demand) by investing more. But this rise in investment will constitute a further rise in injections and thus will lead to a second multiplies rise in income. And so on…

The reason why such an interaction cannot raise output infinitely is because of two reasons i) the economy runs into the full-employment constraint, i.e. there is a fixed number of workers in the economy, and ii) output must grow at an increasing rate (something which is difficult to sustain for very long) in order for investment to continue rising. This is because the accelerator links investment to changes in output, not the level of output. So for e.g., if output rises in year 1 by Rs.3bn, in year 2 by Rs.2bn, and in year 3 by Rs.1bn, then with $\alpha = 2$, investment will be Rs.6bn, Rs.4bn, and Rs.2bn in years 1, 2 and 3 respectively. As can be seen, I falls even though output is rising, leading to a reverse multiplier accelerator chain reaction to be set off effect will be reversed. The key point to remember, again, is that investment is related to “changes in income” not the “level of income”, and therefore “changes in income” have to increase in order for investment to increase. A mere increase in level is not important.

EXERCISES

If we were trying to get a 'true' measure of national production, which of the following activities would you include: (a) washing-up; (b) planting flowers in the garden; (c) playing an educational game with children in the family; (d) playing any game with children in the family; (e) cooking your own supper; (f) cooking the supper for the whole family; (g) reading a novel for pleasure; (h) reading a textbook as part of studying? Is there a measurement problem if you get pleasure from the do-it-yourself activity itself as well as from its outcome?

The difficulty stems from separating production from consumption. In the paid-employment sector of the economy, the distinction is clear. With production, money flows from firms to households (as wages, etc.), and with consumption, money flows from households to firms. Many activities in the home, however, have both a production and consumption element. Although they all lead to a benefit when complete (e.g. washing-up leads to clean dishes), several, if not all, could give pleasure while they are actually being performed. If so, should two lots of benefits be recorded (or even three)? Playing an educational game with children can give pleasure to the children, future benefits to the children from the educational element, and pleasure to the parents. Then there is the cost element. Should this be deducted? It is not deducted for marketed output. In other words, the final value of goods and services sold is what is included, not the value minus the cost of producing them: costs such as the disutility (effort, boredom, etc.) experienced by workers.

Ideally, a true measure of national welfare, as opposed to national production, should be only a net measure (i.e. benefits from consumption, minus costs of production). If this principle was used to measure welfare in the household, then all pleasurable activities should be included with a positive sign (including things such as reading a novel for pleasure) and anything causing displeasure should be recorded with a negative sign. Most of the above activities would have elements of both benefits and costs. However, when marketed national production is recorded, costs are ignored, and so for comparative purposes, household production should be recorded on the same basis, and only the benefits recorded.

All the above items bring pleasure, either directly (such as reading a novel) or indirectly (such as doing the washing-up), and in this sense they should all be included, but whether activities that give direct pleasure should count as production or merely as consumption, is a question of definition.

Review this question after the balance of payments lectures. If the Malaysian ringgit is undervalued by 47 per cent in PPP terms against the US dollar, and the Swiss franc overvalued by 53 per cent, what implications does this have for the interpretation of Malaysian, Swiss and US GDP statistics?

The GDP figures understate the purchasing value of Malaysian national income by 47 per cent relative to US national income, and overstate the purchasing value of Swiss national income by 53 per cent relative to US national income. In other words, at the exchange rates in question, Malaysian national income seems 47 per cent lower relative US national income than it really is in purchasing terms, and Swiss national income seems 53 per cent higher relative to US national income than it really is in purchasing terms.

If there are no sales taxes, no net factor income from abroad and no depreciation, will the GDP at market prices and national income measures collapse to the same thing?

Yes.

- **i.** National income is NNP at factor cost = GNP at factor cost - depreciation. Since depreciation = 0, NNP at factor cost = GNP at factor cost.
- **ii.** GNP at factor cost = GDP at factor cost + net factor income from abroad. Since net factor income from abroad = 0, GNP at factor cost = GDP at factor cost
- **iii.** GDP at factor cost = GDP at market price – sales taxes. Since sales taxes = 0, GDP at factor cost = GDP at market price
- **iv.** Therefore NNP at factor cost = GDP at market price.

If nominal GDP has increased by 10% over last year but real GDP has fallen by 2%, by what percentage must have prices risen?
12%. Real GDP growth rate = GDP growth rate – the rate of inflation. -2% = 10% +?;? = -12%.

Is population growth good or bad for a country's economic welfare?
It depends. If the energies of the growing population (and labour force) can be usefully and efficiently employed towards productive activity, then the growth impact of the larger population may dominate the negative impact of the large denominator in the per capital income formula. However, if more and more people produce less and less (law of diminishing returns) and the quality of human capital created is generally poor, then it might well be that a large population leads to a decline in per capita income and hence average living standards.
It also depends on the starting level of the population. In many African countries, centuries of slavery and migration to other countries, and decades of disease and wars, have led to the working populations in these countries to fall below the minimum threshold required for them to "take off" in an economic sense. By contrast, many South Asian and East Asian countries are quite heavily populated and could use a little cooling down of population growth rates.
How should one treat macroeconomic statistics?
With extreme caution, as such data is likely to be used and abused by different interest groups to support their respective stories. It is usually not the data which is misleading or not consistent with the truth but the manner in which it is presented which makes a certain interpretation of that data more likely. Objective analysis of data consists in stripping it off its particular dressings and looking at all the possible stories it can tell.
By what would we need to divide GDP in order to get a measure of labour productivity per hour?
The total number of hours worked in the year throughout the country.
Is the size of the underground or black economy likely to increase or decrease as the level of unemployment rises?
It could rise or fall depending on which of two effects is the larger. On the one hand, if a certain proportion of unemployed people claim unemployment benefit and work in the underground economy, then, with a higher official level of unemployed, the size of the underground economy is likely to be bigger. On the other hand, if the economy is in recession, it is likely that the size of the underground economy will shrink along with the rest of the economy.
Name some external benefits that are not included in GDP statistics?
Three examples are: the pleasure people get from seeing other people's attractive houses and gardens, aesthetically pleasing architecture, improved health from a better diet.
Are worries about the consequences of economic growth a 'luxury' that only rich countries can afford?
This is a very cynical way of looking at the issue. The point is that the marginal benefit of increased output in a poor country is likely to be much higher than in a rich country (given the diminishing marginal utility of income). Thus if a cost–benefit study were done of specific growth policies, the benefits would probably enter with a higher value per unit in a poor country than in a rich country. This does not mean that the cost should be ignored. It is just that people may be prepared to make bigger sacrifices for increased output in poor countries than in rich countries.
We must be careful with these arguments, however. They could be used to 'justify' policies that are highly damaging to the environment by governments which have little long-term interest in the welfare of the people, or by firms which are unconcerned about the environmental consequences of their activities. The point is that costs should still be taken into account: it is just that the benefits should possibly be given a higher weighting.
If a retailer buys a product from a wholesaler for £80 and sells it to a consumer for £100, then the £20 of value that has been added will go partly in wages, partly in rent and partly in profits. Thus £20 of income has been generated at the retail stage. But the

good actually contributes a total of £100 to GDP. Where then is the remaining £80 worth of income recorded?

At the wholesale stage and earlier. Each stage adds value – value that is partly in wages, partly in rent, etc. When the values added at all the stages are summed, this gives the final value of the good.

An index called the index of sustainable economic welfare (ISEW) has been developed by certain economists to measure sustainable development in different countries. The index takes account of factors like depletion of natural resources which reduces the likelihood of growth being sustained. Make out a case against using ISEW. How would an advocate of the use of ISEW reply to your points?

There are three major criticisms of ISEW. The first concerns its use as a substitute for GDP. GDP is not meant to be a true measure of living standards, both now and sustainable into the future. Instead it is primarily a measure of output that involves exchange, an important measure when attempting to understand the relationship between aggregate demand (as expressed through exchange relationships) and aggregate supply. The second criticism concerns the selection of items that should be included. Any list could be criticized for including too many or too few items. For example, it could be argued that various forms of public expenditure should be included (other than on health and education, which are already included). On the other hand, various forms of 'services of household labour' are not clearly production. They could be seen as a form of consumption. For example, does time spent gardening constitutes work or pleasure? We would not include watching television or sitting relaxing as production, so should be include gardening or any other hobbies as production which generates pleasure when consumed or merely as pure consumption? Similarly, do relationships between people constitute the 'provision of services' or is it rather mere joint consumption? The third and perhaps the most serious criticism concerns measurement. How, for example, should the depletion of non-renewable resources or long-term environmental damage be measured? Such measurement entails various value judgments about the relationship between present and future costs. In fact, the value placed on all non-marketed items is likely to be highly controversial (more so than marketed items, where corrections for market distortions could relatively easily be made).

An advocate of ISEW would reply that ISEW, as its name says, is meant to be a measure of sustainable economic welfare, and not a measure of marketed output and is thus doing something different from the conventional use of GDP. If GDP is used, not for its original purpose, but as a measure of welfare, then ISEW is superior. As far as the selection of items and their measurement is concerned, there will be inevitably be disagreement, because people have different values. But here the advocate of ISEW would reply with the last two sentences of the box.

What are the conditions for macroeconomic equilibrium in the economy.

- **i.** Injections (governemnt spending, exports, investment) into the circular flow of incomes must equal the withdrawals (saving, taxes, imports) from the circular flow; or
- **ii.** Aggregate demand must equal aggregate income must equal aggregate supply.

The two approaches are equivalent. Note, however, that the equilibrium of an economy, at least in a Keynesian world, does not imply the full-employment equilibrium. It is possible for inflationary and deflationary gaps to exist.

What are the major macroeconomic variables involved in the determination of national income?

C, I, G, X, M, T, S, prices, exchange rate, interest rate and money supply. We focus on the first seven in this part of the course, but will enrich our analysis with the remaining four later in the context of the IS-LM approach to equilibrium determination and international finance considerations.

What does the 45 degree line in expenditure-income space represent?

It represents all the points at which the economy is in equilibrium, i.e. the expenditure on domestic goods and services is equal to the supply of domestic goods and services is equal to the incomes distributed to factors used in the production of those goods and services

Are the following net injections, net withdrawals or neither? If there is uncertainty, explain your assumptions.

- **i. Firms spend money on research.**
- **ii. The government increases personal tax allowances.**
- **iii. The general public deposits more money in banks.**
- **iv. Pakistani investors earn higher dividends on overseas investments.**
- **v. The government purchases US military aircraft.**
- **vi. People draw on their savings to finance holiday trips abroad.**
- **vii. People draw on their savings to finance holidays within Pakistan.**
- **viii. The government runs a budget deficit (spends more than it receives in tax revenues) and finances it by borrowing from the general public.**
- **ix. The government runs a budget deficit and finances it by printing more money.**

- **i.** Increase in injections (investment).
- **ii.** Decrease in withdrawals (taxes).
- **iii.** Increase in withdrawals (saving).
- **iv.** Fall in withdrawals (a reduction in net outflow abroad from the household sector).
- **v.** Neither. The inner flow is unaffected. If, however, this were financed from higher taxes, it would result in an increase in withdrawals.
- **vi.** Neither. The inner flow is unaffected. The consumption of domestically produced goods and services remains the same.
- **vii.** Decrease in withdrawals (saving).
- **viii.** Neither. An increase in government expenditure (or decrease in taxes, or both) is offset by an increase in saving (i.e. people buying government securities).
- **ix.** Net injections. An increase in government expenditure (or decrease in taxes, or both) is not offset by changes elsewhere. Extra money is printed to finance the net injection.

It is possible that as people get richer they will spend a smaller and smaller fraction of each rise in income (and save a larger fraction). Why might this be so? What effect will it have on the shape of the consumption function?

It is likely that the rich will feel that they can afford to save a larger proportion of their income than the poor. The consumption function will slope upwards, but get less and less steep. This means mpc will fall as incomes rise.

What effect will the following have on the mpc:

- **a) a rise in the rate of income tax;**
- **b) people anticipate that the rate of inflation is about to rise;**
- **c) the government redistributes income from the rich to the poor?**

- **a)** The mpc will fall. Note that here we are relating consumption to gross income. For any given gross income, a rise in taxes will cause a fall in disposable income and hence a fall in consumption.
- **b)** The mpc will rise as people spend a larger fraction of any rise in income, for if they wait to consume, their incomes will be worth less in the enxt period in urchasing power terms.
- **c)** The mpc will increase, because the poor have a higher mpc than the rich.

What would be the impact of changing the determinant variables given in the first column (below) on consumption and saving. Must saving always fall if consumption falls?

Determinant	**Consumption**	**Saving**
Income (rise)	rise	rise
Assets held (increase in)	rise	fall

Taxation (fall)	rise	rise
Cost of credit (lower interest rates)	rise	fall
Expectations (that prices will rise)	rise	fall
Redistribution of income (becomes more equal)	rise	fall
Tastes and attitudes (people want to consume more)	rise	fall
The average age of durables (increases)	rise	fall

Thus there are two determinants of consumption (namely income and taxation, which will not cause saving to rise if consumption is caused to fall.

Why, if the growth in output slows down (but is still positive), is investment likely to fall (i.e. be negative)?

Because firms will require a smaller increase in capital. They will thus buy fewer extra machines and other equipment: i.e. investment will fall. The underlying concept is that of Keynes's investment accelerator which relates induced investment to changes in output rather than the level of output.

Give some other examples of changes in one injection or withdrawal that can affect others.

- A rise in government expenditure on infrastructure projects may encourage firms to invest, or, on the other hand, may replace private investment.
- A rise in taxation will reduce savings and imports as well as consumption of domestic goods and services.
- A depreciation of the exchange rate will lead to increased exports (an injection) and decreased imports (a withdrawal). This could encourage increased investment in the domestic economy.
- Higher savings will mean less total consumption, including less expenditure on imports.

Keeping Keynes's paradox of thrift arguments in mind, is an increase in saving ever desirable?

Yes.

- If there is a problem of excess demand, an increase in savings will reduce inflationary pressures.
- If investment increases over time, an increase in savings will allow these increases to be financed without problems of rising interest rates or inflation, problems which would have the effect of curtailing the investment.

The present level of a country's exports is £12 billion; investment is £2 billion; government expenditure is £4 billion; total consumer spending (including on imports) is £36 billion; imports are £12 billion and expenditure taxes are £2 billion. The economy is currently in equilibrium. It is estimated that an income of £50 billion is necessary to generate full employment. The marginal propensity to save is 0.25.

a) **Is there an inflationary or deflationary gap in this situation?**
b) **What is the size of the gap? (Don't confuse this with the difference between Y_e and Y_f.)**
c) **What would be an appropriate government policy to close this gap?**

Injections (J) = £12bn + £2bn + £4bn = £18bn
Domestic consumption (C_d) = £36bn – £12bn – £2bn = £22bn
∴ Expenditure on domestic goods, E = C_d + J = £18 + £22 = £40bn

Multiplier = 1/mps = 1/0.25
= 4

a) Deflationary gap. If the economy is in equilibrium, then Y = E. Thus Y_e = £40bn. But full employment is achieved at an income of £50bn. There is thus a deflationary gap.

b) £2.5bn. This is the amount that must be injected (given a multiplier of 4) in order to increase national income by £10bn from the current £40bn to the full-employment level of £50bn.

c) Increase government expenditure by £2.5bn.

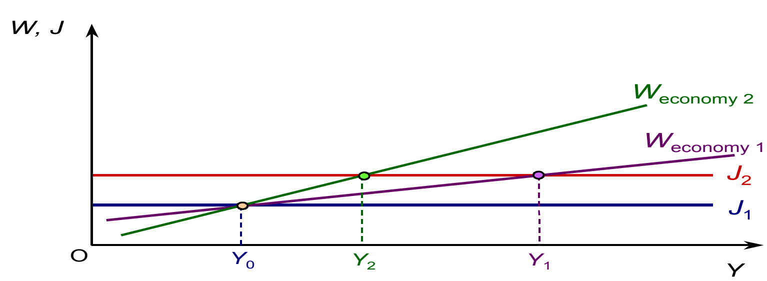
Why does investment in construction and producer goods industries tend to fluctuate more than investment in retailing and the service industries?

Because demand for the output of these industries (which are 'investment' goods industries) fluctuates much more as a result of the accelerator effect.

Give some examples of single shocks and continuing changes on the demand side. Does the existence of multiplier and accelerator effects make the distinction between single shocks and continuing effects more difficult to make on the demand side than on the supply side?

Examples of single shocks include government expenditure on a specific project, a surge in consumer spending in anticipation of a rise in taxes and a temporary movement in the exchange rate (a depreciation causing a rise in aggregate demand through increased exports and decreased imports, and an appreciation causing a fall in aggregate demand). Examples of continuing changes include a sustained increase in consumer or business confidence, which builds over time, and changes in interest rates that then remain for a period of time. The multiplier and accelerator will amplify single shocks on the demand side and the process will last for several months. Aggregate demand will not go on and on rising, however, unless there are continuing changes on the demand side, which then continue to be amplified by the multiplier and accelerator. Thus the effects are somewhat less clear cut than with changes on the supply side, but it is still possible to distinguish between single shocks on the demand side and continuing changes (even if the single shocks do cause multiplier and accelerator effects).

Draw an injections and withdrawals diagram, with a fairly shallow W curve. Mark the equilibrium level of national income. Now draw a second steeper W curve passing through the same point. This second W curve would correspond to the case where the mps is higher.



Economies with different marginal rates of taxation

Assuming now that there has been an increase in injections, draw a second J line above the first. Mark the new equilibrium level of national income with each of the two W curves. You can see that national income rises less with the steeper W curve. The higher mps has a dampening effect on the multiplier. A higher tax rate has the same dampening effect as well by reducing the size of the multiplier (by increasing the size of the term in the denominator). Multiplier with taxes = 1/[1-{mpc(1-t)}];as t increases, (1-t) falls,, therefore 1-{.} rises, causing the multiplier to fall.

What effects will government investment expenditure have on public-sector debt (a) in the short run; (b) in the long run?

a) Increase. Unless financed by extra taxation, an increase in government expenditure (for whatever purpose) will lead to an increase in public-sector debt.

b) Possibly decrease. If the investment leads to extra output and income, then the extra tax revenue from the extra incomes and expenditure could more than offset the cost of the investment, thereby leading to a fall in public-sector debt.

If cuts in interest rates are not successful in causing significant increases in investment, how can they lead to economic recovery? What, in these circumstances, determines the magnitude of the recovery?

They can lead to recovery if they cause consumers to borrow more. The increased spending causes a multiplied rise in national income. The magnitude of the recovery depends on (a) the amount of extra consumer spending; (b) the size of the multiplier; (c) whether there is any subsequent increase in investment (through the accelerator effect).

How do people's expectations influence the outcome?

People's expectations will reinforce whatever it is they expect (self-fulfilling expectations). If firms expect a rise in government expenditure to lead to a) higher interest rates, b) a reduction in private-sector investment and hence c) no expansion of the economy, they will reduce their investment plans, thus bringing about the effect (i.e. economic stagnation) that they had anticipated.

If the government increases spending by Rs.10bn and finances it totally from taxes, will there be any expansionary impact on output?

Yes. The increase in spending is an injection of Rs.10 bn. The withdrawal, however, is less than Rs.10bn, as saving (a withdrawal from the system falls). Why does saving fall? Because higher taxes reduce disposable income and therefore given a fixed mps out of disposable income, saving will fall. The concept is called balanced budget multiplier, i.e. the fact that tax-financed spending (which has no effect on the fiscal balance) can still be expected to have a multiplied (albeit much smaller) effect on equilibrium output and income.

Lesson 33

THE FOUR BIG MACROECONOMIC ISSUES AND THEIR INTER-RELATIONSHIPS

To study any major issue or problem in macroeconomics, it is important to address at least three **questions** about it:

- **i.** Why is it important (i.e. what are its costs);
- **ii.** What are its causes or the possible diagnoses of the problem; and
- **iii.** What is the policy prescription associated with each different diagnosis.

We will look at **four major problems** here:

- **i.** Unemployment
- **ii.** Inflation
- **iii.** Balance of payments problem
- **iv.** The lack of growth

Because this is a course in introductory economics, some of these problems will have to be introduced in the context of the economic history of HICs.

We'll begin with unemployment, which, as you know, peaked during the Great Depression leading to the great rift between Keynes and the Classicists, and the subsequent birth of modern macroeconomics. We shall then move to inflation which became a major problem in the 1970s in the context of the two oils price shocks. We'll then turn to balance of payments disequilibria which have especially afflicted the LICs since the early 1980s. Finally we'll look at the challenges involved in achieving sustainable and sustained economic growth.

One thing you should remember at all times is that macroeconomic problems tend to be related to each other, and it is difficult (and often misleading) to analyze them in isolation. It would be important, therefore, to see how these four problems might be interrelated.

UNEMPLOYMENT

The History of Unemployment:

The history of unemployment, which is relevant to macroeconomics, started way back in the Great Depression (1929-33), when unemployment rates reached levels of 25% in the US and western Europe. During the Second World War, the problem subsided due to higher government defense spending and war-related recruitment. Post-war, unemployment rates continued to fall as rebuilding efforts gathered pace all over the world, esp. in war-ravaged Europe and East Asia. Many colonies gained independence during this time as well (late 1940s till early 1960s) and undertook massive infrastructure and industrial initiatives which absorbed a large part of the workforce. The problem did not raise its head again until the 1970s, when two oil price shocks (1973, 1979), and the associated episodes of cost-push inflation and balance of payments deficits in many countries led to a global recession that was compounded in the early 1980s by rising interest rates (induced by tight US monetary policy). The world recovered from recession in the mid-1980s, and following a brief recession (that lasted till the early 1990s), saw the rise of the new economy (i.e., the age of information technology) and a rapid growth in jobs related thereto. By the turn of the millennium, however, the new economy bubble had burst, causing decline in growth rates in many HICs. This decline was compounded by the events of September the 11th in the New York and the subsequent "war on terror" started by the U.S. and its allies. It is worth noting that Japan, the world's 2nd largest economy, has remained in recession virtually all through the 1990s. Interestingly, there has been a general rise in the rate of unemployment in many LICs (except East Asia) over the last 2-3 decades – a rise that has not been strictly correlated with global boom-recession cycles. The unemployment problem in these LICs was seen to assume a more permanent nature due to their high population growth rates – rates that far outstripped the rate of new job-creation in these economies. Also there was a loss of jobs in many of these countries due to adoption of capital-intensive as opposed to labour-intensive technologies in industrial production.

DEFINITION OF UNEMPLOYMENT

While unemployment can be defined in terms of absolute numbers, in most cases, it is the rate of unemployment which is quoted and which enables cross-country comparisons. The unemployment rate is defined as the ratio of the no. of unemployed people divided by the sum of the employed and unemployed people. A rate of 3-4% is usually considered low, 10-15% considered high, and over 20% considered extremely high. It is worth mentioning that unemployment figures, because they are such a sensitive political issue, are often under-stated by government and over-stated by opposition groups. In most LICs, "official" unemployment rates are seriously misleading, and you can find the government quoting a figure of 5% for unemployment, when the actual rate is around 20-25%, if not higher.

Unemployment is the state in which a person is without work, available to work, and is currently seeking work.

Types of Unemployment:

There are several types of unemployment. **Frictional unemployment** occurs when a worker moves from one job to another. While he searches for a job he is experiencing frictional unemployment. **Structural unemployment** is caused by a mismatch between the location of jobs and the location of job-seekers. "Location" may be geographical, or in terms of skills. The mismatch comes because unemployed are unwilling or unable to change geography or skills. **Cyclical unemployment**, also known as demand deficient unemployment, occurs when there is not enough aggregate demand for the labor. This is caused by a business cycle recession. **Technological unemployment** is caused by the replacement of workers by machines or other advanced technology. **Classical or real-wage unemployment** occurs when real wages for a job are set above the market-clearing level. This is often as a result of government intervention, as with the minimum wage, or unions.

COSTS OF UNEMPLOYMENT

If unemployment is voluntary, i.e. people do not work because they feel they are better off being unemployed, the costs are borne essentially by society, not the individual per se. The costs are:

- **a.** Output and hence national income is lower than potential.
- **b.** Government loses tax revenues because of a.
- **c.** Firms lose revenues as they could have employed more workers and produce and sell more.
- **d.** Other workers lose additional wages that they might have otherwise been able to earn with higher national output.
- **e.** There is a general tendency for crime and violence to rise in society as unemployment levels increase.

If unemployment is involuntary, then all the above broader social costs are borne, but private individual costs for the unemployed individual must also now be added. These would include loss of personal income, mental stress due to loss in self-esteem, worsening of relationship with family or friends.

The Concept of Labor Force (LF):

A fundamental concept in relation to the above definition of unemployment is that of the labour force (LF). The labour force is essentially the denominator in the formula for unemployment rate. The labour force includes all people eligible and able to work, so excludes children, elderly people, parent(s) busy raising children, the handicapped and terminally ill etc.

Note, however that there is a difference between a person who is able to work and a person who is also willing to accept a particular job. So, for instance, a chartered accountant who is looking for a job may be offered the job of a bus driver, but he will not accept that job because it is not worthy of his qualifications, and/or offers a wage below his reservation wage. Thus, in order to be employed you need to fulfill two conditions: you have to be a member of the labour force (LF) and you have to be willing to accept a particular job ("AJ"). This distinction will be developed further shortly.

DEFINITIONAL PROBLEMS WITH UNEMPLOYMENT RATE

Keeping the AJ and LF distinction aside for a while, there can be some other definitional problems with the unemployment rate that need to be addressed. In particular, we might wish to see why the unemployment rate may be reported as:

a. Lower than it actually is (i.e. the severity of the problem of there being very few jobs compared to workers is under-stated). There are two common reasons: underemployment and disguised unemployment. Underemployment refers to the situation when a person is reported as employed but is actually only doing a part-time job. Disguised unemployment is a situation where a person gets a salary but does not really have a job to do, as is often the case for excess workers in government departments.

b. Higher than it actually is (i.e. the severity of the problem of there being very few jobs compared to workers is over-stated). The possible reasons here could be:

   i. The existence of child labour, i.e. children taking the jobs that would otherwise have been available to adults.

   ii. Incompatibility between skills and jobs, i.e. there might be a demand for workers possessing a certain skill, but the people who are seeking jobs do not have that skill and therefore remain unemployed.

   iii. People doing more that one job. If a doctor works in a hospital in the morning and runs his/her clinic at home in the evening, s/he is actually doing two jobs. Were s/he to concentrate on his/her home practice only, the hospital job would become available to some other doctor.

   iv. Unemployment benefit given by the state to the unemployed. This is usually the case for the welfare states (most HICs) where people can live off rather well on unemployment benefit and therefore choose to not work. In LICs, where the state does not quite pay unemployment benefits, the analogy would be beggary. Some people would rather beg on the streets than work honorably and earn their living.

An important concept related to unemployment is that of its **duration**. The duration of any particular unemployment spell depends on when the benefits of accepting a certain job exceed the costs of continuing to searching for a better job (see also the section on search or frictional unemployment below). In an aggregate sense, however, the duration of an unemployment episode in a country depends on the rate at which people enter the pool of the unemployed (i.e. become unemployed) and the rate at which they exit the pool of the unemployed (i.e. find an acceptable job).

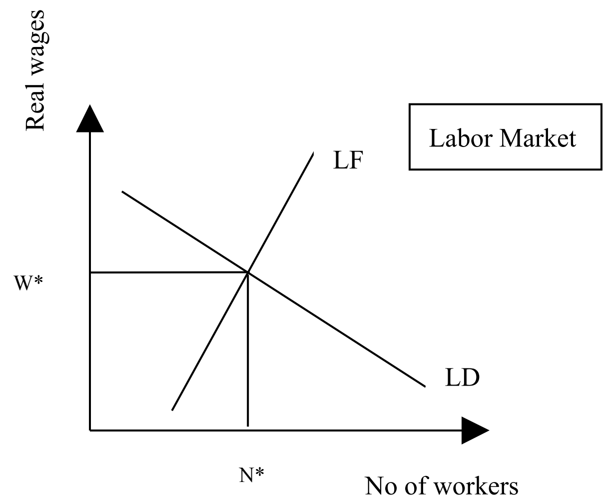
People entering the unemployed pool include those made redundant, sacked, resigning or temporarily laid off. They could also include those formerly outside the labour force, for e.g.: college leavers, women returning to the labour force after raising children.

People leaving the unemployed pool include those taking new jobs, returning to old jobs (if they had been temporarily suspended or laid off) They could also include people who have become disheartened and give up looking for a job, those who have reached retirement age, or who temporarily withdraw from the labour force (e.g. to raise a family), those who emigrate or die.

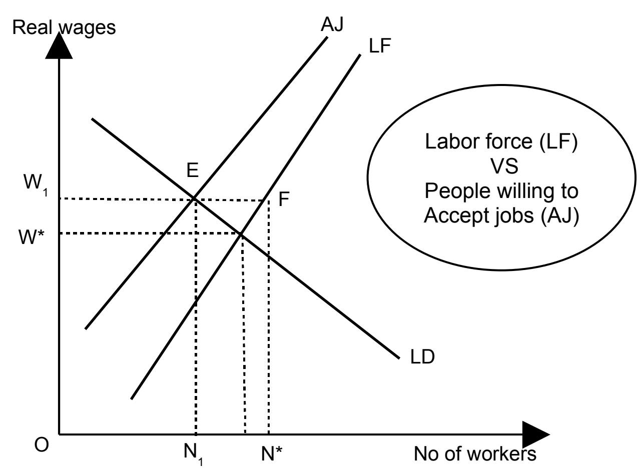
THEORIES ABOUT THE CAUSES OF UNEMPLOYMENT: THREE VIEWS ABOUT UNEMPLOYMNET

There are essentially three schools of thought regarding the causes of unemployment: **the Classical, Keynesian and Monetarist schools.** However, before we delve into the specific arguments presented thereby, we must develop an understanding of how labour market equilibrium is generally struck in an economy.

1- MONETARIST'S VIEW



There is first the supply side of labour. As mentioned earlier, however, a distinction needs to be made between LF and AJ. LF stands for the size of the labour force, and is drawn as an upward sloping fairly inelastic line in wage – no. of workers space. It is drawn as fairly inelastic because the size of labour force would be expected to be fairly unresponsive to the wage rate. A badly handicapped or terminally ill person will not suddenly decide to join the labour force just because wages went up. Some people (like parents looking after kids) however might still be inclined to become members of the labour force if wages go up; the reason why the LF curve is not perfectly inelastic (i.e. vertical).



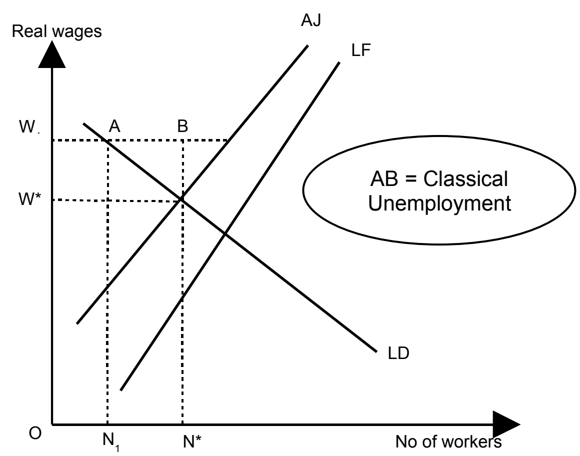
The AJ curve represents those members of the labour force which are willing to accept suitable jobs. The AJ curve is flatter than the LF curve because job seekers would be more willing to accept jobs in response to a rise in wages compared to people who are not even members of the labour force. The AJ curve is to the left of the LF curve because there will always be some people in LF who cannot accept jobs (like the terminally ill). However, given a flatter AJ and a steeper LF, it is clear why the horizontal gap between AJ and LF would narrow at higher wage rates.

The demand for labour is generated by firms, government-owned or private, which need to hire workers to produce goods and services. The lower the wage rate, the more will firms be willing to hire workers. Therefore, the demand curve for labour, LD, is downward sloping in wage – no. of workers space.

The intersection of LD and AJ determines labour market equilibrium, i.e. the no. of potential workers who will be employed (N_1), and the wage rate that they will earn (w*). The intersection of LD and LF delivers N*, the maximum possible no. of workers that can be employed at a particular wage rate. The horizontal distance between N* and N_1 is referred to as the natural level of unemployment. When the horizontal axis measure the employment rate, the same measures the natural rate of unemployment. To keep things simple, we will not differentiate between the two terms and will use them interchangeably.

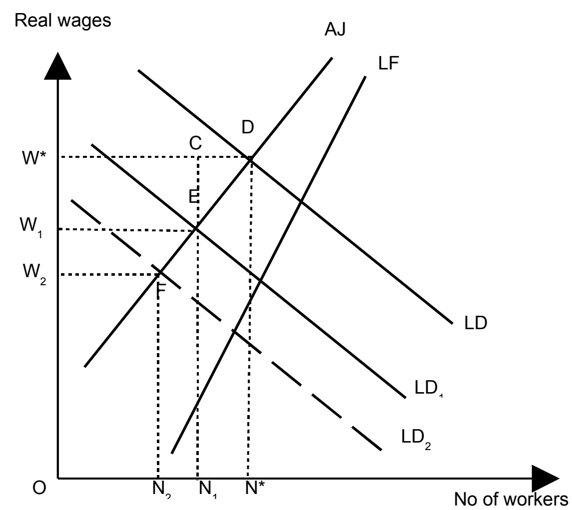
2- CLASSICAL VIEWS

The Classicists viewed unemployment as an essentially voluntary phenomenon caused by wages that were higher than the free market level. If wages were allowed to fall to the market-clearing level, firms' demand for labour would increase, removing the initial unemployment. Their main policy prescription was therefore to remove any factors (labour unions, minimum wage legislation, and unemployment benefit) that prevented wages from falling to market-clearing levels. The theoretical problems with this view of unemployment aside, there were serious difficulties that could be expected in implementing the above policy prescription. Unions were politically strong bodies and could not simply be wished away; removing the minimum wage threshold would hurt the poorest workers, whose wages would now fall below the threshold. Similarly, removing unemployment benefit was likely to be seen as an attack on the safety net for the poorest sections of society.



3- KEYNESIAN VIEWS

Keynes located the origins of unemployment in deficient aggregate demand. According to him, if aggregate demand could be boosted by pumping government expenditure, this would cause the demand for labor to increase as part of a multiplier effect. The increased demand would absorb the excess supply of laborers, thus alleviating the unemployment problem.



THE MONETARISTS' VIEWS ABOUT UNEMPLOYMENT

The Monetarists viewed unemployment essentially in its natural rate context, and thus as a problem that could not be cured through wage decreases or demand injections. According to them, the economy usually operated around the full employment level, and thus the only type of employment to worry about the natural rate kind. Attention, therefore, needed to focus on the horizontal distance between the AJ and LF curves at the market-clearing wage rate, w*. If this distance could be reduced, by shifting the AJ curve to the right (i.e. closer to the LF curve), the unemployment rate could be reduced.

According to monetarists, the reason why people on the labour force might not be willing to accept jobs, was located in **frictional (search), structural and seasonal reasons.**

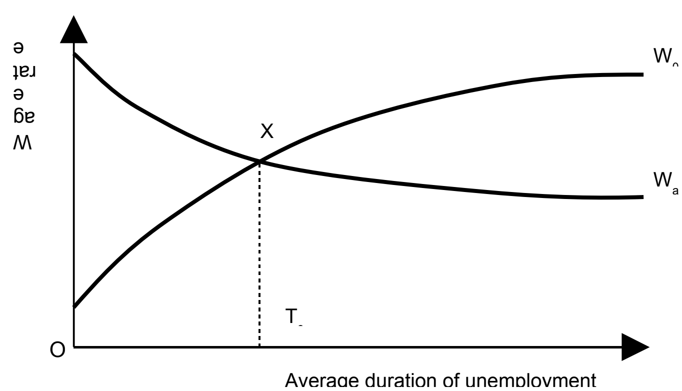
Frictional unemployment was defined as unemployment caused by delays in matching job-seekers to jobs. Delays were seen as being essentially caused by a lack of information, and thus job centers newspapers etc. with better information on jobs etc. could help.

Structural unemployment was associated with changes in the structure of an economy: changes in demand patterns (tastes, fashion etc.), changes in methods of productions (capital vs. labour intensive production techniques being adopted, the replacement of some jobs with computer-based solutions). Also, if industrial or business activity were reduced in a particular region of the economy due to environmental or law and order reasons, or if a geographical area suffers a natural calamity, the resulting unemployment would also be categorized as structural unemployment. Both market-friendly and interventionist policies have been suggested to address this type of unemployment. Among the market-friendly policies are encouraging people to i) retrain themselves (for e.g., facilitate training of mine workers for computer jobs), ii) get on their bikes and look for jobs. Interventionist policies included directed government action to match jobs with skills. Thus the government could give financial grants for training a certain group of people for a certain type of job etc.

Seasonal unemployment relates to certain types of workers going out of job due to seasonal factors. Thus crop producers in cold countries would have little to do when the fields are covered with snow from December till March. The policies that could address this problem would be to facilitate labour migration in winter months, or to develop alternative tasks for the winter (holding winter games in the region and developing it as a tourist spot for e.g.!)

Monetarists also suggested **supply side measures** which generally reduced the incentive to work. One example is lowering income tax rates and thus increasing the incentive for people to work the extra hour and earn the extra dollar.

Understanding Frictional UE, Structural UE
Seasonal UE



Lesson 34

THE FOUR BIG MACROECONOMIC ISSUES AND THEIR INTER-RELATIONSHIPS (CONTINUED)

HYSTERESIS

Hysteresis refers to the permanent effects of a temporary change. In the context of unemployment, for example, a temporary fall in demand in the economy which leads to lower economic activity and hence lay-offs by firms may have more permanent effects given the search behavior of workers (laid off workers become disheartened as time goes by and if they fail the first few job interviews; they become lazy, their skills rust, they become used to unemployment benefit and are therefore less likely to find jobs) and the recruitment behavior of firms (by the time firms need to re-hire, their own motivation and speed of recruitment has slowed; they're not putting up enough job adverts and hence the likelihood of instant recruitment decreases). The result could well be longer spells of unemployment. Generally, it has also been seen that the likelihood of a person leaving the unemployment state falls as the duration of his/her unemployment spell lengthens.

INFLATION AND DEFLATION

Inflation is a situation in which there is a continuous rise in the general price level. **Deflation** is the opposite of inflation and occurs when the general level of prices falls. The rate of inflation is the percentage annual increase in average price level.

Pure inflation is a special case of inflation in which the prices of all the goods and services in the economy are rising at the same rate. So if an economy produces three goods: apples, shirts and cars, and they cost Rs. 5, Rs. 100 and Rs. 400,000 respectively in 1992, while the prices in 1993 are Rs. 6, Rs. 120 and Rs. 480,000, and the prices in 1994 are Rs. 9, Rs. 180 and Rs. 720,000, respectively, then we can say that there was pure inflation of 20% in 1993 (over 1992) and pure inflation of 50% in 1994 (over 1993).

MEASUREMENT OF INFLATION

More generally, inflation (in % p.a.) is measured as

$$[(P_t - P_{t-1})/P_{t-1}]*100$$

Where P_t refers to the average price level prevailing in year t, and P_{t-1} is the average price level prevailing in period t-1. The term average price level usually refers to the value of an index, like consumer price index or producer price index etc., which weights the prices of goods according to their share in the total nominal GDP.

Dates	Price level	Inflation (%)
30th June 2000	100	------
30th June 2001	105	5%
30th June 2002	107	1.9%
30th June 2003	120	12.1%

Ideal Inflation Rate for an Economy:

It is difficult to say what the ideal inflation rate for an economy is. But it is not usually zero. A small positive inflation rate of about 3% is considered healthy for mature HICs while 7% is quite acceptable for fast growing emerging economies. Inflation rates above 10% are generally considered undesirable.

Some countries, esp. in Latin America, have recorded inflation rates in 100s and 1000s of percentage p.a. – these episodes were known as hyper inflation episodes. The first country to suffer hyperinflation, and perhaps in its severest form, was Germany in the 1920s. The country was burdened with very high debt – linked to obligations taken on as a result of being defeated by the allies in the 1st World War. The German government could not find the resources to pay off the debt or the interest thereon and resorted to printing money. This, however, plunged the local economy into hyperinflation.

The Choice of price Index:

There are important statistical challenges to resolve in the definition of inflation. The choice of price index often affects what one can say about inflation. Thus it might be that overall inflation is high, but food inflation is low. Similarly, there can be a price index for students, the health sector, housing and each has a different meaning attached to it. When talking about inflation, it should be clear which index is being referred to. The most commonly used index when referring to overall inflation in the economy is the retail or consumer price index (CPI).

The CPI and PPI:

CPI measures the cost of a fixed basket of consumer goods in which the weight assigned to each commodity is the share of expenditures on that commodity by consumers. The producer price index (PPI) is the price index of goods and raw materials sold at the wholesale level to producers: examples of goods in the wholesale basket are steel, wheat, cotton etc.

However, all these indices are measures of price inflation. A slightly different but related concept is that of wage inflation, which measures the rate of increase of average wages in the economy. Since real wages are nominal wages adjusted for prices, it is straightforward to see that if wage inflation is greater than price inflation, real wages are rising, and vice versa. Countries where price inflation is very high often keep track of wage inflation rates to ensure that the real wage (which measures the purchasing power of workers) remains constant. The practice of linking wages to prices is called "index-linking" and is the norm in many Latin American countries.

COSTS OF INFLATION

- **a.** Inflation redistributes income away from those on fixed incomes (or who do not have the bargaining power to renegotiate wages) towards owners of land, property or assets whose prices are generally quite sensitive to the general price level. Traders also raise prices faster when they see a rise in the inflation rate. But the salary class is often constrained from doing so. In many African and South Asian countries, the problem has been quite acute. Wage earners, esp. in the government sector, had lost (by the 1990s) up to 90% of the purchasing power of their incomes from the 1960s. The reason, a gradual but persistent decline in real wages what could not be reversed. This has led to corruption and inefficiency in many countries.
- **b.** High inflation increases uncertainty for firms, thus impacting investment rather negatively. The point to note here is that inflation is most volatile at high levels (thus the fluctuation of the inflation rate around 25% will be much higher than its fluctuation around 2%). Thus high inflation translates into uncertainty about prices, which means inability to accurately forecast firm revenues and expenditures, which means lower investment ex-ante.
- **c.** Balance of payments problems are also often a result of high domestic inflation. Rising prices in a country, if not offset by equally offsetting depreciation of the exchange rate, can lead to the domestic currency becoming overvalued and therefore to a decline in exports and a rise in imports – in other words to a deterioration of the current account.
- **d.** Resource wastage is high when inflation is high. Extra resources (time and money) are dedicated, both by individuals and firms, merely to hedge against the purchasing power erosion effects of inflation. Restaurants have to change their menus frequently; price lists have to be issued more frequently.

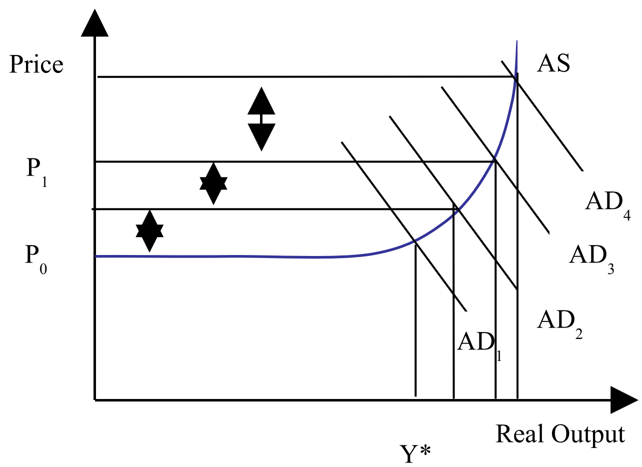
THEORIES ABOUT THE CAUSES OF INFLATION: THREE VIEWS ABOUT INFLATION

There are three views here: The Traditional Keynesian Views, The cost push inflation, the monetarist views of quantity theory of money.

1- TRADITIONAL KEYNESIAN VIEW

Keynes sees inflation and unemployment as opposite sides of the same coin, so is merely a result of excessive aggregate demand (demand-pull inflation). Assuming that increases in aggregate demand in a Keynesian world do have some output and employment impact, one can think of plotting the relationship between inflation and unemployment as trade-off: a downward sloping curve in inflation-unemployment space. This curve is called the Phillips curve names after the economist who first quantitatively documented this trade-off in the context of the UK economy in the 1950s and 60s. The Phillips curve tradeoff can be summarized as follows: Lower unemployment can be achieved only at the cost of higher inflation. The policy prescription flowing from this particular diagnosis of inflation was simple: reduce aggregate demand by contractionary fiscal and/or monetary policies.

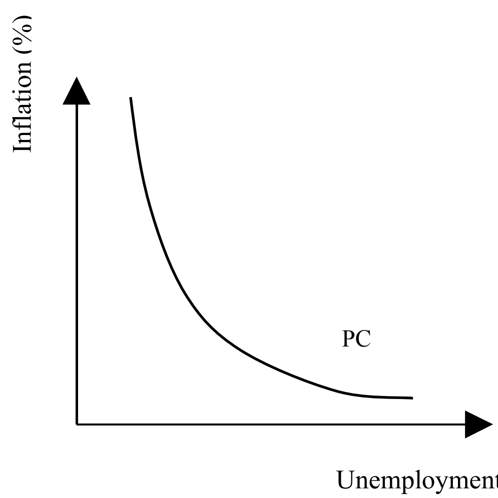
Demand Pull Inflation



Keynes looked inflation and unemployment as the opposite side of the same coin. If inflation increases, unemployment decreases and vice versa.

Philips Curve

W.A.Philips gave the idea of Keynes a formalized shape and draws the Philips curve stating inverse relationship between unemployment and inflation rate.

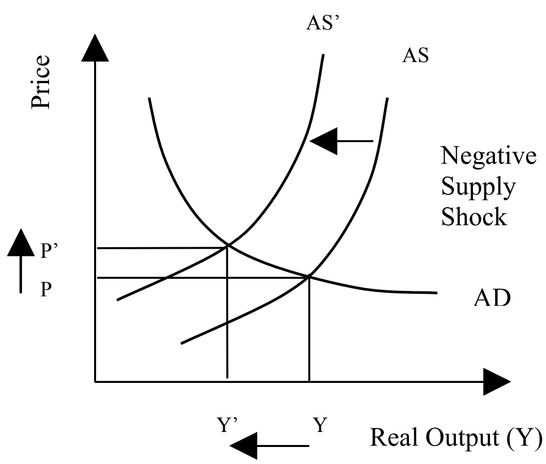


Philips's views were consistent with the Keynesian views. The cost of reducing inflation is unemployment and the cost of reducing unemployment is inflation.

2- COST-PUSH INFLATION

This view came to fore in the 1970s when the world was confronted with a situation of rising prices but high unemployment (stagflation), something that demand-pull theories could not explain. It was observed that the two oil price shocks in the 1970s, which were essentially supply side shocks (because they increased the cost of production), were capable of producing such a situation. In AD-AS space, such a supply shock would be shown by shifting the AS curve to the left (and up) causing prices to rise and output (and employment) to fall. In the context of the Phillips curve, the supply shock would be shown by shifting the Phillips curve out to the right, reflecting a structural shift in the inflation-unemployment trade-off. The name given to the resulting higher inflation (at any level of unemployment) was cost-push inflation. The policy prescription appropriate for dealing with it, included supply-side measures such as developing alternative energy sources, fuel efficient technologies, production cost reduction methods, and reducing tax distortions (that reduce the incentive to produce), increasing competition (in search of productivity gains), removing price floors etc. Keynesian demand management policies were obviously not seen as relevant in this context. It is important to note that sometimes what appears as cost-push inflation is actually driven by higher demand. For e.g., let’s say demand for property increases in an economy; this causes housing prices to rise, causing rents to rise, and causing workers to demand higher wages. Higher wages causes firms’ production costs to increase prompting them to raise goods prices which in turn cause retail prices to rise. At every point of the who chain, it is the costs that are rising (rental costs, production costs, purchase costs of retailers), but the cause of these rising costs is higher demand for property. This situation is often branded cost-push illusion.

Cost Push Inflation



Summing up in the words of Hazrat Ali (AS), “high inflation together with a deteriorating law and order position are hallmarks of the worst possible government.”

Lesson 35

THE FOUR BIG MACROECONOMIC ISSUES AND THEIR INTER-RELATIONSHIPS (CONTINUED)

3- THE MONETARISTS VIEW

In economics, the **quantity theory of money** is a theory emphasizing the positive relationship of overall prices or the nominal value of expenditures to the quantity of money.
Monetarists located the causes of inflation in the Quantity Theory of Money (QTM), which provided an explanation for inflation totally independent from that for unemployment. QTM states: MV = PQ, where M is the real money supply, V is the velocity of money (the no. of times money is circulated in the economy in a year), P is the price level and Q is the real output. Assuming a constant V and a stable (natural rate) output Q*, changes in P could be explained totally by changes in M. A stable M would imply a stable P. Thus the Monetarist key to solving the inflation problem was a stable money supply set to grow at the rate of growth of natural rate output (Q*). For Monetarists, the concern was not the government's expansionary fiscal policy per se, but the manner in which the fiscal deficit was financed. If the government financed its deficit by borrowing from the central bank (i.e. printing money, and thus expanding money supply), this would be tantamount to inflationary finance of the budget. If, however the government financed the deficit by borrowing from banks or the retail savers, then there would be little inflationary consequences.

Monetarism and Philips curve:

It is also instructive to see how Monetarist's viewed the Phillips curve, and the inflation-unemployment tradeoff. Monetarists believed that the economy generally gravitated around a full-employment or natural rate level, and any positive output or employment impact of inflationary demand policies would have be limited. The duration and extent of this limited impact would depend on how much money illusion private agents suffered from.

Money illusion is when agents base their decisions on their expectations about inflation (set in period
t-1), so that when government driven actual inflation (increase in prices and wages) in period t exceeds expected inflation, agents view the increase as real rather than nominal, and therefore erroneously spend more than they should.

Adaptive expectations: In setting expectations for period t+1, however, they learn from the previous period, raising their inflationary expectations based on the outcome in period t. Only an inflation rate higher than this new expected rate can convince them to spend more. The net effect of learn and error process is that inflation rises very steeply in response to continued demand-injections by government until the effect on spending and employment is virtually zero – a vertical Phillips curve. This is how Monetarists characterized the long-term tradeoff between output and inflation, i.e. that there was no trade-off and that expansionary demand policies (i.e. expansionary monetary policy) translated fully into higher inflation with no impact on employment whatsoever.

THE BALANCE OF PAYMENTS (BOP)

BOP is an accounting record of a country's transactions with the rest of the world. To illustrate the related concepts in a non-complicated way, we shall assume a two-country world (Pakistan and the US), and view things from the Pakistani side.
The balance of payments (or BOP) measures the payments that flow between any individual country and all other countries. It is used to summarize all international economic transactions for that country during a specific time period, usually a year. The BOP is determined by the country's exports and imports of goods, services, and financial capital, as well as financial transfers. It reflects all payments and liabilities to foreigners (debits) and all payments and obligations received from foreigners (credits). Balance of payments is one of the major indicators of a country's status in international trade, with net capital outflow.

Before we can fully grasp the BOPs, it is important to develop an understanding of the market for foreign exchange. Foreign exchange, in the Pakistani context, simply means US dollars (note that foreign exchange from the US's point of view would be Pak rupees).

THE MARKET FOR FOREIGN EXCHANGE

The market for foreign exchange (or dollars) in Pakistan works like the market for any other commodity (like apples, oranges etc.). We have an upward sloping supply curve and a downward sloping demand curve. We operate in the same price-quantity framework, noting, however, that quantity in this context means the quantity of dollars and price in this case means price per dollar, i.e. the rupee price of a dollar. However, the latter is simply the Rupee/US$ exchange rate, and hence we can label the vertical axis accordingly.

The foreign exchange (**currency** or **forex** or **FX**) market exists wherever one currency is traded for another. It is by far the largest financial market in the world, and includes trading between large banks, central banks, currency speculators, multinational corporations, governments, and other financial markets and institutions.

Exchange rate:

In finance, the exchange rate (also known as the foreign-exchange rate, forex rate or FX rate) between two currencies specifies how much one currency is worth in terms of the other. For example an exchange rate of 123 Japanese yen (JPY, ¥) to the United States dollar (USD, $) means that JPY 123 is worth the same as USD 1.

HISTORY OF EXCHANGE RATE IN PAKISTAN

During the past five decades, Pakistan's foreign exchange regime has been moving towards a deregulated and market-oriented direction:

Before the 1970s, Pakistan linked its currency, rupee, to the Pound Sterling. With the economic influence of the USA getting more apparent, in 1971, Pakistan linked rupee to the U.S. Dollar.

Pakistan fell into a budget deficit in 1982, when the strengthening U.S. Dollar made remittances abroad through official channels slumped. The plunging black market rate suggested that the rupee pegged to the U.S. Dollar largely deviated from the underlying economic realities. In this view, Pakistan put the rupee on a controlled floating basis, with the currency linked to a trade-weighted currency basket.

In 1998, to alleviate the financial crisis in Pakistan, the authorities adopted a multiple exchange rate system, which comprised of an official rate (pegged to U.S. dollar), a Floating Interbank Rate (FIBR), and a composite rate (combines the official and FIBR rates). Export proceeds, home remittances, invisible flows, and "non-essential" imports can be traded at the FIBR rate.

With the economy recovering from the crisis in 1999, the three exchange rates were unified and pegged to the U.S. within a certain band. This band was removed in 2000.

Now, Pakistan is maintaining a floating rate. Under this exchange rate system, each bank quotes its own rate depending on its short and long positions. Strong competition, however, means the exchange rates vary little among the banks. Under the prevailing Exchange Control Act, the State Bank of Pakistan on application may authorize any person or institution to deal in the foreign exchange market. By virtue of this vested authority, the SBP may determine the extent to which a Bank would be authorized to deal in various currencies.

Lesson 36

THE FOUR BIG MACROECONOMIC ISSUES AND THEIR INTER-RELATIONSHIPS (CONTINUED)

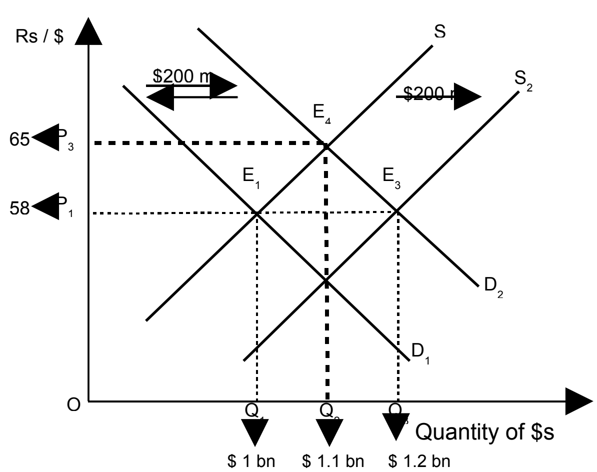
DEVALUATION
This is the act of reducing the price (exchange rate) of one nation's currency in terms of other currencies. This is usually done by a government to lower the price of the country's exports and raise the price of foreign imports, which ultimately results in greater domestic production. A government devalues its currency by actively selling it and buying foreign currencies through the foreign exchange market.

REVALUATION
This is the act of increasing the price (exchange rate) of one nation's currency in terms of other currencies. This is done by the government if it wants to raise the price of the country's exports and lower the price of foreign imports. This is an appropriate action if the country is running an undesired trade surplus with other countries. The procedure for revaluation is for the government to buy the nation's currency and/or sell foreign currencies through the foreign exchange market.

APPRECIATION
This is a more or less permanent increase in value or price. "More or less permanent" doesn't include temporary, short-term jumps in price that are common in many markets. Appreciation is only those price increases that reflect greater consumer satisfaction and thus value. While all sorts of stuff can appreciate in value, some of the more common ones are real estate, works of art, corporate stock, and money. In particular, the appreciation of a nation's money is seen by an increase in the exchange rate caused by a growing, expanding, and healthy economy.

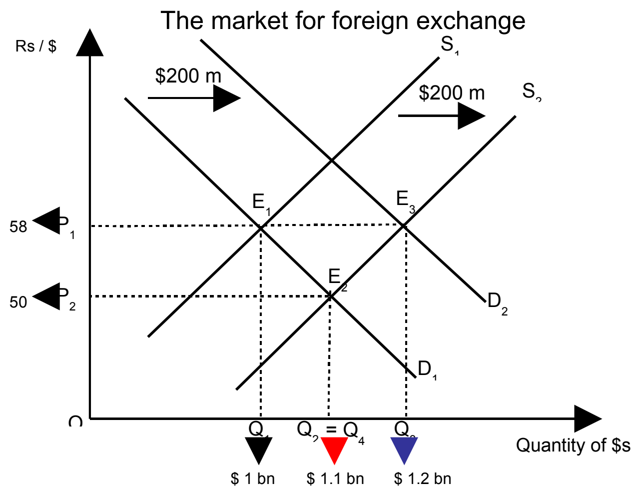
DEPRECIATION
This is a more or less permanent decrease in value or price. "More or less permanent" doesn't include temporary, short-term drops in price that are common in many markets. It's only those price declines that reflect a reduction in consumer satisfaction. While all sorts of stuff can depreciate in value, some of the more common ones are capital, real estate, corporate stock, and money. The depreciation of capital results from the rigors of production and affects our economy's ability to produce stuff. A sizable portion of our annual investment is thus needed to replace depreciated capital. The depreciation of a nation's money is seen as an increase in the exchange rate.

The market for foreign exchange



THE FORCES OF DEMAND AND SUPPLY IN FOREIGN EXCHANGE MARKET
Now, consider what the forces of demand and supply are in this market. First consider what could cause the supply curve for dollars to shift to the right, i.e. what would cause the supply of

dollars in the market to increase given a certain exchange rate. Well, any transaction which has the effect of bringing dollars into the country would have this effect. Examples of such transactions are net inflows of US investment into Pakistan, Pakistani exports to the US, remittances from Pakistanis working in the US. Now what are examples of transactions that cause the demand for dollars to increase? Pakistani imports of US goods, Pakistani travelers traveling to the US, Pakistani students paying for study in US universities, profits repatriated to US by US firms operating in Pakistan. All these will cause the demand for dollars in the market to increase.
Any transaction which causes the supply curve of dollars to shift to the right is recorded with a positive sign on the BOPs (as it corresponds to an inflow of dollars), while any transaction which causes the demand curve to shift to the right is recorded with a negative sign on the BOPs.



EQUILIBRIUM IN THE MARKET OF FOREIGN EXCHANGE

Equilibrium in the market for foreign exchange occurs at the point of intersection of the supply and demand curves. In BOP terminology, this is when all the +vs and the –ves balance; i.e. the BOPs is zero (external balance).
Given an initial equilibrium, it is useful to study note how market equilibrium responds to a shift in, say, the supply curve. A distinction has to be made between the cases when the exchange rate is fixed by the government and when it is left to float freely.
When the exchange rate is fixed, the government has to make up for any excess or shortfall in the market. Thus, if the **supply curve shifts to the right** (say due to a rise in exports), and there is an excess supply of dollars, the government must step in and purchase those excess dollars from the foreign exchange market, pumping the equivalent local currency in the process. This purchase is an example of foreign exchange market intervention by government (often implemented by the central bank on behalf of the government).
Similarly, if there is a **rightward shift in the demand curve** (due to, say, a rise in imports), and there is a situation of excess demand for dollars at the given exchange rate, the government must step in and supply those dollars from its coffers in exchange for local currency. The government's foreign exchange reserves fall and the local currency (rupee) supply contracts as a result of this kind of intervention.
To let the exchange rate float freely is to allow the price mechanism to bring about automatic equilibrium in the foreign exchange market. Thus in this case, if there is an excess supply of dollars, the price of the dollar falls (i.e. less rupees are required to purchase one dollar). In other words, the rupee appreciates vis-à-vis the dollar. Conversely if there is an excess demand for dollars in the market, then this pushes the exchange rate up (i.e. more rupees will now be needed to buy one dollar). In other words, the rupee depreciates. Note that an increase (decrease) in the price of the dollar is equivalent to a(n) depreciation (appreciation) of the rupee, not vice versa.

PARTS OF BOP

The BOPs can be divided into three parts:

i. Current account,
ii. Capital account and
iii. Changes to reserves.

(1) THE CURRENT ACCOUNT

The current account balance is essentially the trade balance (exports minus imports), but with net factor receipts from abroad added.

If the exchange rate is fixed, then changes in reserves must mirror the combined balance on the current and capital accounts in order to bring the overall BOPs to zero. If the exchange rate is floating, then changes to reserves can remain zero, as the adjustment burden is borne by the exchange rate which appreciates (depreciates) in response to a joint surplus (deficit) on the current and capital accounts.

External Transactions:

External transactions which have no long-term (or future) flow implications for the current account are recorded on the current account. Thus exports, imports, and factor payments (foreign workers' outward remittances, interest on foreign debt, and dividends on profits of foreign firms) and factor receipts (overseas Pakistanis' inwards worker remittances, interest earned on foreign assets held, dividends earned by Pakistani firms abroad) are all recorded on the current account.

Compare these transactions with the taking on of a new long-term foreign debt – recorded with a plus sign under the capital account. Here, the initial inflow of dollars does not make the transaction complete in an inter-temporal sense. The money that has come in will have to be repaid, both principal and interest over the future. Similarly, foreign investment coming into the country. The initial dollars coming in will imply a future stream of current account outflows in terms of dividend remittance abroad.

In the long-term, the current account and capital account should usually mirror each other. So if the current account is in deficit, you would expect the country to be borrowing or attracting foreign investment on the capital account to bring the overall BOPs to zero. Similarly, if the current account is in surplus, you would expect the country to be lending to the rest of the world or investing outside the country.

Current account balance (+ or -)

(i) Goods or visible balance (+ or -)

(+) Exports
(-) Imports

(ii) Services or invisible balance (+ or -)

(+) Exports
(-) Imports

(iii) Income and transfers (+ or -)

(+) Factor income from abroad, e.g. worker remittances, dividends, interest
(-) Factor payments, e.g., MNC profits, interest on debt

(2) THE CAPITAL ACCOUNT

The capital account generally provides a direct picture of the net asset position of a country vis-à-vis the rest of the world. If the capital account stays in surplus year after year, this indicates the country's increasing indebtedness to the rest of the world. If however, the capital account stays in deficit year after year, this means the country's indebtedness to the rest of the world is falling.

At the introductory level, BOP problems normally refer to a deficit on the current account, since the capital account is assumed to be passive. Thus external disequilibrium is usually associated with a situation where the trade balance (exports – imports) is in deficit.

This raises two questions:

a. **Is a current account deficit necessarily bad?** The answer is no. Recalling the condition for macroeconomic equilibrium S+T+M = I+G+X, and rearranging, we can get {M-X} = [I-S] + (G-T). The {} term is the current account or trade balance, the [] term gives the private sector resource deficit (i.e. the excess of the private sector's investment over savings) and the () term is the government fiscal deficit. It is clear that if M>X because G>T (i.e. government is spending in excess of its resources), then the current account deficit might be unsustainable (i.e. bad), especially if the government's spending is essentially of a current nature. However, a trade deficit which finances private investment that would otherwise not have been possible, is likely to be desirable, esp. if the private sector is investing in industries that will have future export potential (because this means the country will have the foreign exchange reserves in the future to pay off the debt that is being incurred today to finance the current account deficit).

b. **How can a current account, which is in deficit, be restored to balance?** Firstly it must be recognized that perennial current account deficits of the sort implied in the question only obtain under fixed exchange rates (because under floating exchange rates, the disequilibrium would self-correct through exchange rate depreciation). One quick fix solution to sort out current account deficits under fixed exchange rate regimes is to have an economic deflation. The theory here is as follows: when a country's national income rises, it spends more; part of that spending falls on imported goods; higher imports cause the current account to worsen. The reverse is also true: lower income must reduce import spending and therefore improve the current account spending. However, economic contraction is a rather painful way of restoring current account equilibrium. A less painful one suggested by economists is devaluation, the name given to exchange rate depreciation but in the context of fixed exchange rates. (The corresponding term for exchange rate appreciation is revaluation.) A devaluation attempts to bring the exchange rate in line with its long-run equilibrium level, i.e. a level consistent with international competitiveness. Competitiveness is simply defined as the real exchange rate (RER), where RER = (P_f/P_d)*NER; NER is the nominal exchange rate (in Rs/$), P_f is the price level prevailing in the foreign country (US), and P_d is the price level prevailing in the home country (Pakistan). The formula simply says that, given a fixed NER, if inflation is higher in Pakistan (relative to the US), Pakistani exports will become less attractive (or competitive) in the international market. As a result, our exports will fall, and current account will go into deficit. To rectify the situation, the NER can be devalued so as to make our goods cheaper and bring competitiveness back to its original higher level. However, there are many provisos attached to the devaluation policy prescription. Devaluation only works if the country's exports and imports are elastic, otherwise the price effect of the devaluation will dominate the volume effect and the current account will worsen. Secondly, the country must have excess productive capacity in order to meet the higher demand for exports that is created as a result of the devaluation. Thirdly, the country should not have a very high foreign debt whose burden increases so much as a result of the devaluation that the negative effects associated therewith overwhelm any positive competitiveness effects.

<u>Capital account (+ or -)</u>

(+) Incoming FDI, FPI or other private capital
(-) Outgoing FDI, FPI or other private capital
(+) Borrowing, aid inflows
(-) Payments of debt principal, aid outflows

(3) CHANGES TO OFFICIAL FOREIGN EXCHANGE RESERVES (+ OR -)

(+) Sales of foreign exchange by government, i.e. drawdown of reserves
(-) Purchase of foreign exchange by government, i.e. build-up of reserves

Balance of payments (1+2+3 =0, or net errors and omissions)

Lesson 38

THE FOUR BIG MACROECONOMIC ISSUES AND THEIR INTER-RELATIONSHIPS (CONTINUED)

THE CONCEPT OF ECONOMIC GROWTH AND GROWTH RATE

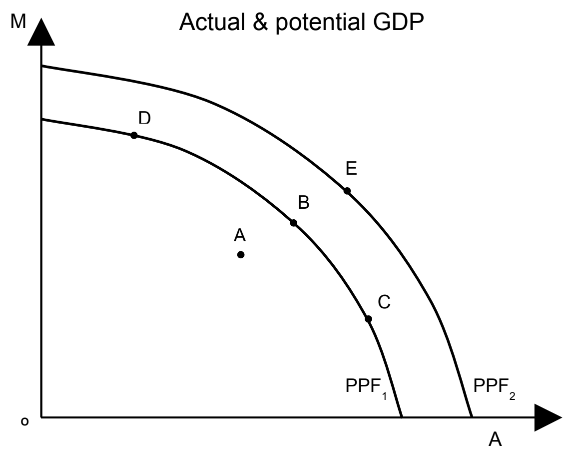
Economic growth is increase in an economy's level of production, output or income. We can talk about production or output in two broad definitional contexts. One, we can compare real GDP with some other measure of welfare (for e.g., one which adjusts for externalities, social indicators, the black market, purchasing power parity, income inequality etc.). Two, we can talk about potential vs. actual output. Potential output is the aggregate capacity output of a nation; the maximum quantity of goods and services that can be produced with available resources and a given state of technology.[1] In our discussion here, we will abstract from such complexities and take output to simply mean real GDP.

The growth rate of a country's real GDP can be negative, positive or zero. A growth rate of between 2-3% is considered normal for mature developed countries; for LICs, 5-7% is considered healthy and 7%+ excellent.

ACTUAL & POTENTIAL GDP

The GDP gap or the output gap is the difference between actual GDP and potential GDP or potential output. The calculation for the output gap is Y-Y* where Y is actual output and Y* is potential output or the natural level of output. If this calculation yields a positive number it is called an expansionary gap and indicates an economy in expansion; if the calculation yields a negative number it is called a recessionary gap and indicates an economy in recession.

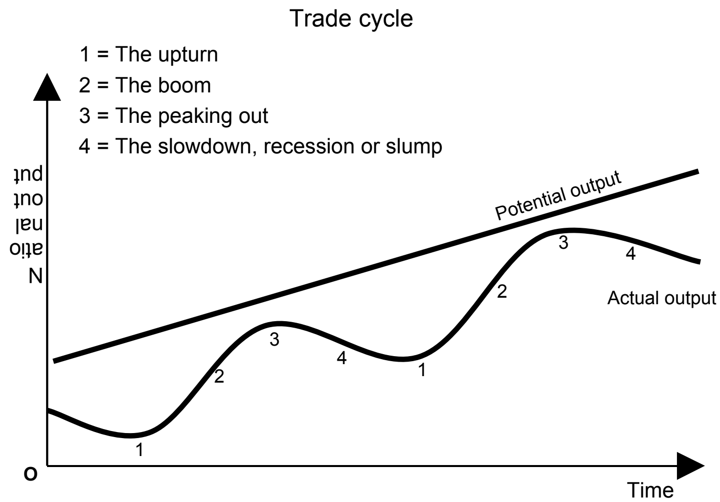
The percentage GDP gap is the actual GDP minus the potential GDP divided by the potential GDP. (*Actual GDP − potential GDP*) / *Potential GDP*.



TRADITIONAL THINKING ABOUT GROWTH

Traditional thinking on growth was that it can be driven either by an increase in factor resources (land, natural resources, labour, capital), i.e. an increase in potential GDP, or by more efficient use of the factors, i.e. a move from inside the PPF to the PPF. The policy implication attached to this line of thinking was simple. Countries must either accumulate factors of production (esp. capital), or develop more cost-efficient technologies and methods of production to utilize those resources better. In any event, factors of production were at the heart of growth theory.

[1] Thus, when the production possibilities frontier of a country shifts out, that represents an increase in potential GDP. Actual GDP can be less than or equal to potential GDP, and is usually less. The difference between potential and actual GDP is sometimes referred to as the *output gap*.



REAL VS NOMINAL GDP

Nominal GDP measures the value of output during a given year using the prices prevailing during that year. Over time, the general level of prices rises due to inflation, leading to an increase in nominal GDP even if the volume of goods and services produced is unchanged.

Real GDP measures the value of output in two or more different years by valuing the goods and services adjusted for inflation. For example, if both the "nominal GDP" and price level doubled between 1995 and 2005, the "real GDP" would remain the same. For year over year GDP growth, "real GDP" is usually used as it gives a more accurate view of the economy.

Relation between Real GDP and Nominal GDP

Nominal GDP is calculated using current prices whereas real GDP uses constant prices. The difference between the nominal GDP and real GDP is due to the inflation rate in market. The relationship between inflation, real GDP and nominal GDP is explained by Fisher Equation. *Real GDP = Nominal GDP – Inflation*

AGGREGATE GDP VS PER CAPITA REAL GDP

Economy's total income **or the** sum total of all incomes in an economy in a given period, usually a year is known as the aggregate GDP or aggregate income level.

When studying growth, it is always instructive to analyze changes in per capita real GDP along with changes in real GDP. Per capita real GDP growth adjusts GDP growth downwards by the population growth rate and gives a more accurate indication of improvements in living standards in a country. For mature HICs, Real GDP growth rate = per capita real GDP growth rate, since the population size in these countries is quite stable.

It is also important to note that even a small per capital real GDP growth rate (say around 2% p.a.), if sustained for a very long very of time (say 100 years) can deliver huge improvements in living standards. The U.S. and Japan in the 19^{th} and 20^{th} centuries and East Asian tiger economies in the last four decades are a neat example of this.

Why Growth is an Important Macroeconomic Issue:

It is obvious why growth is an important macroeconomic issue. Every government aspires to deliver a higher growth rate for the country. High growth rates means higher national income which means better living standards on average, which in democracies, means happier electorates and therefore increased chances of re-election for another term in office. However, while all countries might wish to achieve high growth rates, in practice, only a handful have been able to convert the wish into reality.

HOW PER CAPITA GROWTH RATES RELATED TO THE AGGREGATE GROWTH RATE IN AN ECONOMY? DEFINING GDP GROWTH RATE

y = Y / L

Where,

- Y = Total GDP

- L = Population
- y = Per capita GDP

Taking log of both sides

- ln y = ln Y – ln L

Taking derivative *w.r.t.* time

- $\frac{1}{y}(dy/dt) = \frac{(dY/dt)}{Y} - \frac{(dL/dt)}{L}$

$$\mathbf{gy = gY - gL}$$

Growth rate of per capita income = Growth rate of total output - Growth rate of population

Real Gross Domestic Production figures

Countries	Ratio of 1999 / 1870	Annual growth
Japan	100	3.7
US	66	3.4
Australia	43	3.1
Sweden	33	2.8
France	15	2.2
UK	10	1.9

RGDP per capita figures

Countries	Ratio of 1999 / 1870	Annual growth
Japan	27	2.7
US	10	1.8
Australia	4	1.2
Sweden	14	2.2
France	10	1.9
UK	5	1.3

Pakistan's growth rate statistics since independence

Era	Aggregate Real GDP
60s	6.7
70s	4.8
80s	6.4
90s	4.7

Per Capita RGDP	Population
4	2.7
1.7	3.1
3.3	3.1
2	2.7

Reference: Zaidi.A, Issues in Pakistan's Economy

LINK BETWEEN GROWTH AND THE VARIOUS FACTORS OF PRODUCTION

Capital:

Any increase in capital should cause an increase in growth rate of output. **Capital deepening** is a term used in economics to describe an economy where capital per worker is increasing. It is an increase in the capital intensity. Capital deepening is often measured by the capital stock per labour hour. Overall, the economy will expand, and productivity per worker will increase. However, economic expansion will not continue indefinitely through capital deepening alone. This is partly due to diminishing returns and wear & tear. **Capital widening** is a term used to describe the situation where capital stock is increasing at the same rate as the labour force, thus capital per worker remains constant. The economy will expand in terms of aggregate output, but productivity per worker will remain constant.

Labor:

Human capital also matters for economic growth. Quantity as well as the quality of labor should also be considered. This is also an engine of growth.

Land:

Pakistan is an agrarian country in which land matters much. Japan and Korea has been grown rapidly because they have used their scarce land very efficiently.

Raw materials:

If stock of raw materials increases economy will produce more output which will increases growth rate of output.

Technical knowledge:

If there are technical advancements then production will increase, growth rate of output will also increase. The factors of technological advancements are learning by doing, invention, innovation etc.